

5th Applied Active Inference Symposium (Part 1, Nov 12, 2025) ~ LIVE

LIVE | 10:37:38

Greetings, everyone. Welcome to the 5th Applied Active Inference Symposium. It's November 12th, 2025. Some brief introduction slides and then to Carl Friston for a keynote address and discussion. So welcome to the Active Inference Institute. We are a non-profit, volunteer-run organization dedicated to improving the rigor, accessibility, and applicability of the Active Inference Framework. Since 2021, we have hosted an annual Applied Active Inference Symposium with Carl as the first presenter back in 2021, which serves as an end-of-year celebration and convening for our ecosystem. At the link symposium.activeinference.institute, you can look at all the previous programs and proceedings. We're here to highlight applied and translational work in Active Inference across domains and from groups around the world. Meanwhile, we service opportunities and needs of our ecosystem, get feedback, and facilitate connections among participants. That's you. You can join the Institute Discord, donate to support the work, stay tuned via newsletter, and actively infer on. About the program, there are three days of programs, and the bulk of the proceedings will be live-streamed to YouTube at the three links that are available at the program. And you can watch it live and re-watch it at the same link. And there may be some adjustment at the fine scale with time zones and pre-recorded and live and scheduling and all fun kinds of things. But big picture, there's going to be three live streams that will consist the bulk of the program. In the participant registration, we asked if people wanted their responses to be public. And for those who did, we integrated their responses. So you can read the raw responses to people's questions that people were asked, like what would be interesting for them? What would be useful? How can Active Inference have a big impact in 2026? And we also used LLM to integrate the responses and do embeddings on the responses, all kinds of cool stuff. So you can check out that open source code. A big thank you to our team of co-organizers. Alexandra Mikhailova, Andrew Pachea, Bradley Licia, Christian Martin, Corey Slater, Daniel Friedman, Maria Garcia, Maria Luisa Iannico, Pablo FM, Sylvia Zhang, and Zora Ramanijad. It was a great and streamlined process with these co-organizers. So absolutely was a team effort. And I thank them all again. If there's some kind of session presenter or development that you'd like to see in next year's symposium, you would be welcome to provide that feedback or even to step up and be a co-organizer. It's a lot of fun. Email us if you're interested. Also, thank you to our sponsors. Our first sponsor is Navigating Uncertainty, a book bringing the science of active inference and the free energy principle to life through story, art, and human experience by Anna Pereira with a foreword by Carl Friston. And Anna will have a presentation later in the program. And also our sponsor for this symposium, Upcycle Club, an independent social initiative that actively supports a select group of noteworthy institutions and innovative projects dedicated to fostering research, education, and original thinking. So now, without further ado, to Carl's keynote. And I thought, how to introduce this keynote? And on this slide, I have Carl thinking of the 2012 paper, The History of the Future of the Bayesian Brain, where

there's a quote, which was about the neuroimaging community, but I believe it applies to the active inference community today as well. And he wrote, in short, the future of imaging neuroscience is to refine models of the brain to minimize free energy, where the brain refines models of the world to minimize free energy. This endeavor itself minimizes free energy because our community is itself a self-organizing system. And so here, from the past to the predictive future, we have generative AI and all these different domains of application for active inference. So hopefully a fittingly aesthetic introduction to what should be a very fascinating keynote, followed by a discussion. So thank you, Carl, for joining. And you can share your screen and looking forward to your keynote here. Thank you for that lovely introduction. And just to say, it's a great pleasure to kick off this symposium. And I'd like also just to reiterate my thanks on behalf of the audience and the viewers to the co-organizers, and particularly you, Daniel, for stewarding us all in this community direction. It is a great endeavor. And I just want to again, thank you for that. So I'm going to start the slightly retrospective feel. I'm mindful that now we're in almost the sort of second generation of people applying or interested in the free energy principle and in particular active inference. So I'm going to stand back a bit. And this presentation is going to have a certain retrospective global take on the foundations and the principles upon which active inference is based. And in particular, it's based upon a presentation given to the Neuro AI workshop at last year's NeurIPS and also the International Active Inference workshop earlier this year in Montreal. However, this presentation has got a twist because I know people always like to know what the latest thing is. So at the end of this presentation, I'm hoping just to foreground a domain of questions and challenges that face artificial and general intelligence research, the kinds of problem solving abstractions and reasoning that we that are characteristic of natural intelligence that now have become a goal or a target for certain sections of the machine learning community. So what I'm going to do in order to sort of look at how one could deploy or apply active inference to understand, formalise and simulate logical reasoning and abstraction, I'm going to refer back to a rather old paradigm now, the three ball paradigm, just to revisit that and look at how we can use active inference in its most general sense to underpin the sample efficiency that we can use. So that's where we can use the world. So that's where we're going to start. In fact, I'm going to cheat and we'll get Geoffrey Hinton to actually start for us. Now, I'm not sure whether this is going to be audible on the live stream. So, Daniel, if I could ask you, I'm just going to play a few seconds and if you can hear it, then we'll just continue. And if you can't, I'll actually read out what Geoff actually said. So let me just start this.

I would like to read Calum Bolts.

Was that audible? Excellent. OK, so let me just put this in context.

Geoff was being presented with the UCD Ulysses Medal a year or so ago.

And right at the end of his presentation and indeed right at the end of the question and answer session from the audience,

he was asked the following question. If you could go back in time and meet your hero, academic hero, who would you like to meet?

And his response was, I'd like to meet Helmholtz.

So let's just listen to his explanation for that.

And I'd like to meet Helmholtz. Helmholtz believed in unconscious perceptual inference.

He basically thought that when you're doing vision, you're doing lots of inference, but it's not kind of conscious, deliberate inference.

You're inferring from the distal, from the proximal stimulus, you're inferring what the state of the world is that gave rise to that.

And he was clearly right about that. So that was one branch of Helmholtz's work.

A different branch of Helmholtz's work was free energy.

There's something called Helmholtz free energy.

And what Helmholtz didn't realize was that if you're trying to do inference in a complicated model with complicated latent variables,

a statistician would tell you to do inference by given the current parameters of your model, take the data and figure out the best way for your model to explain the data.

And there might be several ways for your model to explain the data,

but you have to figure out the probability of each of those ways in which your model could explain the data in order to adjust the parameters of your model.

And what variational inference says is actually you don't have to get that probability right.

You can approximate that probability and still guarantee you'll learn a better model, or rather there's some bang that'll improve.

And so the two completely different bits of Helmholtz's career, the work on free energy and the work on perceptual inference, were actually very closely related.

And free energy is the key to being able to do perceptual inference tractably.

And I'd love to tell Helmholtz that.

I'm not going to say that's that.

Isn't that lovely?

So I'm basically going to just try and unpack what Geoff was intimating there, and do so with reference to three of the great intellectual architects of the physics, from my point of view at least, of artificial intelligence.

Clearly Helmholtz whose contributions come at many, many different levels.

Also a nod to Feynman in the sense it wasn't quite Helmholtz's free energy

that underwrites variational inference.

The free energy in question is actually due to Richard Feynman.

But I also want to acknowledge the sort of physics of measurement and observation

and certainly the perspective afforded by Edwin James in the context of things like the maximum entropy or cannibar principles.

So Helmholtz, where do we start with Helmholtz's contributions?

My favourite point is a slide which I've shown many, many times summarised by this beautiful quote.

Objects are always imagined as being present in the fetal division as would have to be there in order to produce the same impression on the nervous system.

So this is his notion of unconscious inference.

This is his notion that the brain is a constructive organ generating explanations in an inside out kind of way that provide an accurate and as we will see simple account of sensory impressions on the nervous system.

And clearly this is very closely related to the notions of perception as hypothesis testing by people like Richard Gregory.

Ideas that were taken with great effect and deployed in machine learning by people like Jeffrey Hinton and Peter Dine,

who actually built a Helmholtz machine as a metaphor for the Bayesian brain.

Lending or borrowing from probability theory in the sense of Bayesian statistics and of course the work of Richard Feynman on his work on the pathological formulation of quantum electrodynamics.

So what's this notion of unconscious inference?

One could argue it goes right back to the students of Plato.

I'm bringing it back to the 16th century.

This, I think, beautifully illustrates the notion of unconscious inference and perhaps even conscious inference in this instance.

So this is a painting by a 16th century oil painter famed for painting still lives that when viewed from another perspective,

give you a very different kind of visual impression.

So if previously you saw a bowl of fruit, but now you see a face, the point being made here is that you made that face.

This is something that you brought to the table as an explanation, a hypothesis, a model, a fantasy, literally making the brain a fantastic organ, a purveyor of explanations and fantasies that best explain our sensory input.

I think that perspective lends a certain gravity to Feynman's meme, which we all quote regularly.

What I cannot create, I do not understand.

And that requires a very concrete meaning when read in light of Helmholtz's notion of unconscious inference.

If you can't create it on the inside, you have no understanding, knowledge, belief, no real ability to recognise what it is.

So I'm here reading the creation again in this sort of constructive perspective on the brain as a statistical organ creating explanations for its sensorium.

How can one formalise that?

Well, we can again appeal to Feynman's pathological formulation, free energy, or use of variational, calculus of variations to elaborate the right kind of objective functions that allow you to solve very difficult inference problems.

They are difficult because they entail marginalisation or impossible integration problems, but they can be rendered tractable and converted into optimisation problems through these variational bounds.

In particular, the variational free energy as a bound on log evidence.

So what do I mean by that?

Well, if we just think carefully about the nature of things that could possibly hold beliefs or have beliefs or make inferences about other things,

then immediately we have to introduce a distinction between the thing that is observing, inferring, measuring, and the thing that is observed, measured, or we are making inferences about.

And statistically, this is the Markov blanket due to Judea Pearl,

and simply separates the world into the self or the observer, and the non-self or the observed,

and the bi-directional exchange between these two systems here cast in terms of a Markov blanket comprising sensory states and active states.

So the sensory states are mediating the effects of the world on the inside,

whereas the active states are reciprocally mediating the effects of the inside on the world through some kind of action,

autonomic reflexes or actually acting with my, say, muscles or motor system.

And this construction, this sort of partition, separation, disambiguation between the observed and the observer,

and enables one to apply Feynman's path integral formulation to the dynamics of any system that yields to this kind of separation between self and non-self.

And I've just summarised this here in terms of the free energy principle,

where we can associate now or we can describe and demonstrate that the dynamics,

the rate of change in my internal states and my neural activity and my active states together,

sometimes referred to as autonomous states. Why?

Well, because they don't depend upon the external states by construction,

by the separation of the self from the non-self.

And if we can do that, we can now just write down the dynamics of those autonomous states.

I repeat my internal states and my active states as a gradient flow or a process that looks as if it is minimising,

in the sense of optimisation, this variational function of my sensory input and the beliefs about the outside that are parametrized by my internal states.

And here's the familiar expression or one way of writing down that expression.

This has been used to great effect in terms of understanding things like predictive coding with reflexes,

which we'll cover briefly later, and sort of simple from the point of view, for example, of control theory, sort of control as inference.

I just wanted to draw a bright line between two kinds of application of active inference that rests upon the notion of beliefs about what I am doing.

It takes us from control as inference into planning as inference, where I now have in mind a probabilistic representation of what I am going to do,

and certainly what I am going to do into the future.

Interestingly, if one traces back the origins of that particular move, it just rests upon considering big things, possibly with a sort of deeper hierarchical structure on the inside, where the active states no longer have direct access to the internal states deep inside some internal structure.

So if I just remove that, something really interesting happens.

What happens is we can now regard the active states, which were previously real variables that had direct access to the internal states as vicarious causes of the sensory states.

But if we're now reading the dynamics of my internal states as making inferences about the causes of my sensations, having beliefs about what is out there,

then now I am licensed to describe my internal states as including beliefs about my own actions.

And this now leads to the notion of planning as inference and introduces something called expected free energy.

Why expected? Well, if I've got beliefs about my future and in particular the consequences of my actions in the future,

those observations, those sensations, those data are now no longer real variables.

They now become random variables because they haven't yet occurred.

And now that means if I want to evaluate free energy, I now have to take an expectation over the posterior predictive density,

over the sensations that I would cause in the future if I committed to this particular action.

So you could say that this is a more, if you like, authentic kind of active inference in the sense that it could be applied to things like you and me,

basically using the path integral formulation to link the Markov blanket to full active inference,

where we have paths of least action that are determined by the free energy for my internal states, my belief updating,

and for my action, the expected free energy.

So let's just visit those two sort of aspects of free energy minimisation,

sometimes cast out repeat as a path of least, a principle of least action.

Let's just look at the sort of the internal states minimising variational free energy.

And this is where the second foundational contribution of Helmholtz comes into play.

What we can do is if we now interpret the internal states as parameterising some beliefs about the external states,

in this example, parameterising terms of a Gaussian distribution or Gaussian belief,

then because we know the expression for the dynamics of the internal states,

we can use the Helmholtz decomposition to split the dynamics into two parts.

And what I've done here is associate those two parts.

For those people who are interested in that those two parts are effectively a dissipative flow that descends free energy gradients and a non-dissipative or conservative flow that circulates in a certain angle of fashion on the isocontours of the free energy.

I can associate this non-dissipative and dissipative part with the prediction and update of a Kalman filter, which would be one very simple example of inference under a very simple generative model and the basis of controllers inference, things like linear quadratic control.

But the story I want to tell here really is that we can associate, if I knew the state of the world or had an expectation about the state of the world, I also know how it's going to change.

So I can predict how it's going to change the μ dot here.

But I also, and that would be associated with the solenoidal, the non-dissipative part, I have not been informed by any sensory information at this stage, but when I do have access to sensory information, I can now evaluate the free energy gradient and update my expectation in the light of the sensory evidence, and in particular, the prediction error that is effectively the free energy gradient.

So what's a prediction error?

Well, imagine we have this impression on our sensory epithelia, and I had in mind an internal state that represented the belief that this particular impression was caused by a howling dog.

And if I had a generative model that could generate the consequences, the sensory consequences of a howling dog causing my sensory input, I could take the prediction, subtract it from the actual sensation and produce a prediction error, a mismatch.

And then all this equation is saying, all the Helmholtz decomposition is saying, is it will look as if this prediction error is driving my beliefs in order to minimise the prediction error, where when the prediction error is minimised, I'm minimising my free energy gradients, and I've found a free energy minima.

So this provides, I think, a really nice perspective on filtering, on data evidence accumulation, that can be interpreted directly in terms of Helmholtz's decomposition.

Notice that all that this is saying is that you're minimising your prediction error as a proxy for a variational free energy.

You're never going to actually know what's out there.

This is, let's say, was actually a cat.

So what about James?

What's another way of looking at this sort of free energy minimising description of self-organisation, where the self is distinguished by a Markov-Banket from non-self?

I put this up here because I think that, you know, it just provided a really interesting dual perspective on the problem of influence, or the description afforded by inference, measurement and observation in general.

And I just express the free energy now in terms of an energy and an entropy,
an expected energy or potential and an entropy here.

And this entropy here, we can see if we want to minimise the free energy,
we need to maximise the entropy of our beliefs.

And of course, this is exactly what James was saying in terms of the measurements that we do in physics,
but crucially under constraints.

So what are the constraints?

Well, they're just the constraints that are supplied by a gerundered model.

For us, you know, affording an accurate explanation of some data under a gerundered model,
where the gerundered model is usually expressed in terms of some likelihood and prior,
the probability of some outcomes or observations,
given the hidden states that are hidden behind the Markov blanket,
and some prior beliefs that characterise my model,
namely the probability of those latent or hidden states.

I put this up because there's often a degree of confusion when talking about the thermodynamics of
inference.

So, you know, if I want to maximise the entropy of my beliefs,
why is it that often the free energy principle is cast as an account or a description of things that sustain
themselves

by minimising the dispersion or the dissipation of their, say, sensory states or observational states?

The paradox is very easily resolved in this instance by noting that when people talk about entropy,
you have to specify the particular probability density or distribution to which that entropy applies.

And in this particular instance, the maximum entropy principle applies to the entropy of the beliefs.

And this entails a minimum entropy production principle.

And that, I think, can be rendered very, very obvious in the sense,

if I now just take the time average of the free energy,
we can see immediately that as the path entropy goes down,
it goes in one direction, the entropy goes in the other direction.

So if I maximise my path integral by minimising my free energy or my expected free energy,
then I'm implicitly minimising the dispersion or dissipation of my exchanges with the environment
through the outcomes or observations supplied by that environment.

So maximum path entropy is basically equal to or equivalent to a minimum entropy production principle,
very much in the spirit of the theory of dissipative structures due to Prigogine.

Another way of re-understanding or intuiting the characteristic or the implications of free energy is just to
express it to somebody,

machine learning would express it or possibly even basic statistics.

So I've just rearranged the terms and I've just arranged them in terms of the accuracy,
the expected probability of some outcomes under some beliefs about what caused those outcomes,

the latent states minus the complexity.

So what's the complexity?

Well, it's just the divergence between my posterior beliefs, the beliefs encoded by my internal states and prior beliefs.

So that scores the divergence between what I believe now, a priori, before seeing some data, and after seeing some observations, a posteriori, it literally scores the degree to which I change my mind in order to provide an accurate account of the data. And of course, that's just an expression of principles of indifference or Occam's principle.

But interestingly, it also has thermodynamic correlates by appeal to things like the Jaszinski equality and Landauer's principle.

You can now say that if I want to change my mind, if I want to pay a complexity cost in accounting for my observations,

my measurement in a way that provides an accurate account of that measurement or observation, then I'm going to have to spend energy.

It costs some work, some thermodynamic work to erase bits and to move on some statistical manifold or to induce a divergence between my prior and my posterior, which means that in minimising free energy, equivalently in machine learning, in maximising the negative free energy or the evidence lower bound, I'm implicitly trying to minimise my computational cost.

And that also impacts in a thermodynamic context that I'm going to do it as thermodynamically, if as efficiently as possible.

And of course, this is just one aspect of biomimetic and biotic self-organisation, right through to the kind of inference that Helmholtz had in mind.

So that perspective allows us to sort of look at the sort of what some people call merely reflexive active inference

because we haven't as yet got to planning as inference through a number of different lenses.

And I've just summarised these here in the slide, which I'm sure many of you have seen before.

So at the moment, we're just talking about the notion of describing ourselves as minimising free energy or trying to maximise this evidence bound or maximise the model evidence, the evidence through our generative models, through updating our beliefs and noting that our actions should also minimise exactly this quantity.

So our actions are just going to fulfil our predictions.

Formally then, what does this mean or how could one interpret this?

Well, we know that first of all, the free energy is a bound approximation to the log evidence and the log probability of some outcomes given me as the self in question.

And what that basically means is I can interpret this log probability as the kind of outcomes that are characteristic of the kind of thing that I am.

So they have meaning for me. They characterise me.

You know, I have a certain temperature. I have a certain status.

I have all sorts of exchanges with the environment.

And all of those environments are characteristic or typical or define the kind of thing that I am.

And from this, that interpretation of free energy leads to things like reinforcement learning.

If I was a control theorist, it could be cast in terms of control theory.

Or if I was an economist, it would be expected utility theory.

If I was an information theorist, the negative of this log marginal likelihood or log model evidence is known as self information.

Also known as surprise or more simply surprise.

And it just scores the improbability, how uncharacteristic this exchange with the environment would be if I am me.

And that reading of this quantity here, the free energy in and of itself can then forge links with information theory,

the Infixmax principle, the principle of maximum mutual information or equivalently minimum redundancy principles due to Barlow.

And of course, the free energy principle itself.

That's good if I now take the time expectation or average of self information or surprise, I now get entropy.

So now we come back to the minimum entropy production principle that James was considering from the point of view of Maxent or Maxcalibur principles.

And what does that mean?

Well, it just means that it will look as if my autonomous states, my sensory states and my control states or active states are those that on average,

try to minimize the dispersion of my exchanges with the environment.

Of course, this is the Holy Grail of self organization is exemplified in synergetics due to Haken.

And if I was a physiologist, it's just a statement of homeostasis that I am the kind of thing that keeps my essential variables within viable bounds.

And ensures that they do not dissipate beyond those characteristic values.

The final interpretation, which of course is the one we're familiar with, is that this probability of some outcomes given me can also be described as model evidence.

If I associate me as a generative model of the world in which I am immersed.

And this basically says that free energy minimizing is just maximizing model evidence, sometimes particularly referred to as self evidencing.

And this provides a sort of formal account to the Bayesian brain evidence accumulation and particular implementations such as predictive coding that rest upon the isomorphism between free energy gradients and precision weighted prediction error.

So that's from free energy to self evidencing.

What about this sort of second kind of active inference where we now instead of just picking those actions that minimize free variation of free energy?

What if we now make inferences and now describe our internal states as holding beliefs about what we are doing and possibly what we are doing into the future, sometimes referred to as policies, π here.

And that now will minimize the expected free energy under some, I repeat, posterior predicted density of the consequences, the outcomes that ensue if I committed to a particular sequence of actions or combination of actions known as a policy. And this is where we move from the free energy to the expected free energy.

And I've just written this out in full because, you know, if you don't like equations, I think there's a lovely iconic symmetry here, which is, tells you a lot about ways in which you can understand the motivations for behavior and the implicit intentional aspects of the behaviors that I infer myself committing to.

So I've just written out the free energy in terms of a mixture of complexity and accuracy.

And we just move terms around to foreground the fact that we can also express this as a KL divergence or a bound on the log evidence.

And it's a bound simply because the KL divergence between my beliefs about latent states and the true posterior can never be less than zero by construction.

So if I now take the expectation of these quantities under some consequences conditioned upon a policy, what happens?

Well, the complexity becomes risk. The inaccuracy becomes ambiguity and the divergence and log evidence become expected information gain and expected value.

So just to unpack that, this is just a look at these terms and see how we can interpret them very much in the spirit of the previous slide looking at merely reflexive active inference.

So let's assume that first of all, I don't have any probabilistic description of the characteristic outcomes that I would expect given me or given my generative model.

What are we left with? Well, we're left with another KL divergence, but this one is an interesting one in the sense that it's it is going to be maximized simply because now the outcomes are in the future and they now become random variables.

And what is it an information gain or a kind of divergence or on average, a mutual information? What is it between?

Well, it's basically between the beliefs I would have about the states of affairs in the world out there in the future if I were able to see the consequences or observe the consequences of those states relative to if I did not.

In other words, it's scoring the information that is obtained by acting in this way.

It basically scores the epistemic value or the epistemic affordance.

It scores the degree to which I'm going to resolve my uncertainty about states of affairs in the future if I did this.

In the visual search literature, this is often referred to as Bayesian surprise.

It is also mathematically equivalent to maximizing mutual information between cause and consequence where cause are hidden or latent states and the consequence of the observable outcomes or consequences of those states commissioned upon some policy or active intervention on the world under my generative model.

Let's take a particular kind of uncertainty off the table.

Let's assume for simplicity that we can see every state of the world directly.

So now my observations become the states of the world and the states of the world become the observations.

So there's no ambiguity about the states or the causes of the observed consequences.

If I remove the ambiguity, I'm left with another KL divergence.

This time we want to minimize it.

And what are we minimizing?

Well, effectively, we're just minimizing the divergence between what I believe will happen if I commit to this policy relative to my prior preferences that characterize the outcomes that are characteristic of me.

And that basically scores my policies in terms of how close can I keep myself to my prior preferences.

In engineering, this would be known as pathological KL control.

You could also substitute model predictive control and some kinds of Bayesian RL under this under this formalism.

If you're an economist, this would be risk sensitive control.

Let's now make a final move and remove all reducible uncertainty from the problem.

So now there is no more information to be gained from this environment.

I'm now completely familiar with this environment and I cannot resolve my uncertainty anymore.

What am I left with?

Well, I'm just left with this quantity which we have previously associated with value.

The expected value.

So from the point of view of economics, this would just be expected utility theory.

The value I expect given my beliefs about the future under a particular policy.

So notice what we've done here.

We've actually derived a number of really important characterizations or descriptions of motivated behavior or optimal behavior in some sort.

Simply by selectively removing different sources of uncertainty.

But the most general case is this decision making under uncertainty that entails a principled approach to information gathering or uncertainty reduction.

I'll try to illustrate that here just by appealing to the dual aspects of Bayes optimal action or decision making.

So it has two aspects. The first is nicely captured by the ability of the expected value to score the probability that I'm likely to pursue this policy.

And of course, that's exactly what you'll find if you read books on Bayes' indecision theory.

So this would be, if you like, the part of the Bayes optimality that rests upon optimal Bayes' indecision theory.

But there's another way of being Bayes optimal.

And that's often cast in terms of the principles of Bayes' optimal experimental design.

Designing your experiment to elicit those data that resolve the greatest amount of uncertainty about your hypotheses pertaining to the way that those data were generated or caused.

And this goes right back to the 1950s with people like Dennis Lindley looking at the information theoretic criteria, which of course adjusts the expected information gain or the epistemic affordance that we've just described.

That underwrites exploratory behaviour, designing the right kind of experiments in order to resolve uncertainty so that you can become exploitative in accord with Bayes' optimal decision theory.

Together, expected value and information gain just mean the negative expected free energy.

And one could argue this is the essence of active inference beyond a merely reflexive active inference.

What I want to do now is just sort of return to these three perspectives or contributions to the free energy, but from now a much more technical perspective or a formal perspective.

And what I've done here is write out the free energy as an accuracy term and then all the different bits of complexity that are contributed by the various sets of unknowns in my generative model.

And every set usually brings along a mean field factor or partition of the unknowns, the random variables of different kinds.

So I picked out here the unknown states, the random variables that are the latent or the hidden states behind the Markov blanket that change from moment to moment.

We don't know our policies. This is another random variable that I'm making inferences over and then selecting the most likely policy.

But also I have model parameters. I have parameters of a model here.

I'm summarising the parameters in terms of a likelihood tensor, which we'll come back to later.

A, that has in terms of the deployment of the parameters within say a tensor. It also has a particular structure.

So we're not only do we have beliefs about states of affairs, the ways in which we're going to interrogate or act upon the world, but also we have beliefs about the contingencies, the laws, the parameterisations, the associations that underwrite the cause effect structure that are installed in our generative models.

Now, the key observation here is that if I can be described as a free energy minimising thing, then there are a number of ways in which I can minimise my free energy.

I can change my beliefs. I can change my mind about states of the world.

We could call that perception by changing Q through my internal states to minimise the variational free energy or the negative free energy in this instance.

But I can also change my posterior beliefs about the parameters of my generative model.

So we have a nice distinction now between sort of inference and learning.

And I'm using active learning in the sort of traditional sense of certainly promoted by people like David McKay in the 1980s and 90s.

And the difference here is exactly the same maths. It's just it usually unfolds over progressively longer timescales.

So I have to do lots of evidence accumulation or information gathering to resolve uncertainty about states of the world and how they change.

And then on the basis of those beliefs, I accumulate those beliefs and assimilate that into optimising my beliefs about the parameters,

say the connection weights in a neural network or the Dirichlet counts in a tensor network that would correspond to the slow process of free energy minimisation.

That basically is minimising the complexity of my model of the world maximising its compressibility, maximising the mutual information,

all of these being equivalent perspectives on the same imperative to minimise, to maximise the marginal likelihood or the model evidence.

The final thing though, is that not only can we change our posterior beliefs encoded by the, you know, our

internal states,

we can also minimise the free energy by changing our prior beliefs.

Now we're familiar with that in terms of predicating our prior over what to do next, our policies on expected free energy.

But we can also do the same with respect to a prior over the latent states and crucially, the very model itself.

And that's what I want to pursue in the final part of this presentation.

So what we're talking about now is that there are lots of opportunities to minimise free energy.

And if we're all free energy minimising things, it's likely that we will show evidence of these kinds of minimisation processes,

such that minimising, changing our priors to minimise free energy could be regarded as a description of attentional selection.

What we're talking about now is that there are lots of opportunities.

Sorry, continue, Carl.

Right, I'm talking to myself.

And the priors, sorry, policy selection would be a neat description of optimising my prior beliefs about what I'm going to do next.

And that leads to, with this sort of use of a selection word, appropriate to the kinds of unknowns that constitute my JANJ model,

leads to the notion that we can also minimise free energy through model selection, through a process akin to structure learning.

So I'm going to close with an example and then just show you sort of the kind of things that we've been, certainly colleagues at Versys recently, focusing on in terms of generalising this notion of optimising our priors over different aspects of the generative models.

This is a paradigm which I'm sure most of you have seen before.

And if you haven't, the purpose of showing it again is just to enable you to think about the way that you abstract,

the way that you reason about the cause-effect structure.

And in particular, how do you test hypotheses about the mappings from states of play, states of affairs in the world and some correct response,

the rule that underlies a reward or some feedback that you are correct.

So what we're talking about here is sort of a calculus of abstraction, a calculus of reasoning and a calculus of rule discovery,

where the rule can stand in for some kind of symmetry, some kind of equivariance,

some structure that explains when you are right, when you've got the right answer.

And this is the three ball paradigm.

Daniel, do you want to play this? Have you forgotten the rule?

Go for it.

Okay. If you haven't forgotten the rule, that's good because we can get through this more quickly.

Normally I would play this game with the audience just to get them into the spirit of introspecting.

How do we solve these kinds of abstraction problems or rule discovery problems?

So the game normally is presented as follows.

You're going to be presented with three coloured balls and you're going to choose, or your game or your job is to choose the correct colour based upon the arrangement of the three large circles.

And the location of the correct colour depends upon the colour of the upper circle, which in this instance is blue.

And then what I'm going to do is give you the correct answer and just leave that on the screen so that you can remember past instances, past trials, so that you can start accumulating evidence for or against different hypotheses or rules that could be brought to the table to

explain what's the correct colour. So that's all you know, that's your prior beliefs.

And then we normally play with the audience a game. And here there are a series of games starting with the one that we saw previously, and then a series of examples.

And what we're going to do, what I'm going to ask Daniel to do is to guess until he thinks he's discovered the rule and then tell us if he's got the right rule.

And I'm interested in how sample efficient he is. I'm interested in how many exposures does he have to entertain before he suddenly gains insight into the rule, before he has that aha moment.

And indeed, if you're watching this now or later, you know, just think about the experience of this kind of rule discovery.

You know, is it a gradual thing? Does it happen all of a sudden? Are you able to mentalise it? Is it a propositional thing? Are you arguing in words?

Or does it just come into, pop into mind? These are all really interesting aspects of this, you know, how one would formulate or formalise this kind of abstract reasoning or logical induction.

Right. So let's start, Daniel, what would start with this, this little game here, this epoch here. What colour do you think is the correct colour?

Green. Excellent. And for those of you who might not have noticed, this is exactly the same as the example of the previous slide. So I'm assuming that Daniel was...

That was training data. That was training data.

Ah, right. So what are you going to go for this time?

Red.

I'm sorry, it was green. And of course, there is not sufficient evidence at this stage for you to possibly guess any kind of rule, but have another go. Let's go to this one.

Blue.

Right. Well, think again, if the correct colour is in a particular location, can it be blue if there's no blue on the screen?

I'm open to some very creative rules.

Okay, we'll go with red.

Okay, let's do that. By Joe, you're correct, sir. And how about this one?

Blue.

You're going for blue. I'm afraid it was green.

Okay, this is, this is, this is desperate unfair. So when I normally do this with an audience, we give people

about 20 or 30 seconds between each example, so they can mull through alternative hypotheses.

And suddenly spot the one that's explained the data before.

My, at this stage, I, I spent no one in the audience will have caught it.

Daniel, let's give a bit more information. What do you think is the correct answer here?

Green.

Well done. Do you have an idea as to what might be the rule?

Always green unless otherwise absent.

That's close. It's not quite the rule, but that's, that's certainly close.

Do you want to think about it just a little bit longer?

This is relevant because when I'm going to simulate, when I simulate this game, there is a process of offline introspection that requires you to spend some time rifling through different hypotheses or explanations for this particular rule or cause effect structure.

So I'm deliberately wittering to give Daniel some time to spot any rules that he might bring to the table to make the next prediction.

What are you going to predict for this one?

You, you've already got this one, so this is easy.

Blue.

You've just told me the green.

Or green on top.

Oh, it depends on the colour of the upper circle. It doesn't say it doesn't depend on other things.

Yeah, the, all the information you need is in the, the arrangement of, of these, of these three balls. And what you've told me so far is when, um, the, uh, and all, all you, the sort of the, the, the queue, if you like, is the colour of the central ball.

So when you've already told it, when the central ball is green, uh, the correct answer is green.

Yeah. You've already told me that.

And when the central ball is blue, it's also green.

Yeah. Well, you don't know that. Oh, uh, well, that was certainly one instance, but it could have been this one here.

Fair enough.

Perhaps have another choice.

Blue.

Blue.

No, it's red.

Okay.

Then red.

Correct.

Then green.

Correct.

Red.

Green.

Green.

Green.

Green.

Blue.

Green.

Blue.

Blue.

Blue.

Blue.

Blue.

Blue.

Yeah, it can't, it can't, it can't be green because there's no, there's no green on the, uh, on the, on the, I'm teasing you and it's very unfair to, I'm sorry, I should, uh, the correct answer was in this instance, blue.

And so the, the solution now, unfortunately we don't have access to the audience, but some of you, um, what normally happens is, um, after a couple of, um, trials, following the point at which Daniel spotted the sort of first element of the rule.

Um, um, uh, people suddenly get the insight with very, very sparse information, but so usually by about sort of, uh, uh, uh, between five and eight, uh, trials.

And the, the correct answer was exactly as Daniel, um, realized even by the fifth one, I think, um, that when the center is green, the correct color is always green.

When the center is red, the correct color is on the left.

And when the center is blue, the correct color is on the right.

Um, so discounting the first one, you would not have known that until you now got to the one, two, three, four, five.

Six, seven.

Um, so this illustrates, you know, not a desperately complicated rule, but certainly has a certain context sensitivity.

Um, and the question I want to ask now is why is it that Daniel partially was able to do this, um, um, or was able to infer this on the basis of, um, well, certainly part of the rule after one, two, three, four, five, um, given, um,

he hadn't even seen a blue, a queue apart from the training, uh, trial, um, before.

So in principle, this is quite remarkable that, you know, he was able to make some inference to have a hypothesis in the absence of any definitive data,

which is, um, which, you know, is the kind of abstraction and our remarkable efficiency to discover latent structure in the world, even before we've actually seen it.

Um, but let's just see if we can actually, um, get a base optimal agent to, um, match that kind of performance.

Um, there's something which I have to implement in the simulations, uh, which was not made explicit in the task instructions.

Um, the, the simulated agent has to actually look at each of the locations from a start location and thereby has to choose where to look as well as whether to select a particular color, um, prior to it, uh, um, um, um, um, after it's decided what color to choose.

Um, so there are two kinds of actions in this particular, um, example.

It's where am I going to look and what am I going to choose?

What would I need?

What latent states would I need to generate, um, the outcomes?

Well, the outcomes are basically what color am I seeing?

Um, and where am I looking?

Um, and what kind of feedback am I getting?

Is it sort of, uh, have I yet to make a choice?

So if I have made a choice, is it right?

Or is it, uh, is it wrong?

So to generate these what, where feedback outcomes, I need to, um, I need to know what the latent states of the world are.

And the latent states are basically a sort of context that establishes the rules.

So for example, if the correct color is on the right, um, then, uh, well, what is the correct color?

Let's say it's red.

Um, and if I know that the correct color is on the, on the right, and I know that the correct color is red, and I know where I am looking, and I know whether I have chosen or not, I can generate all of these things.

Um, how do I generate these things?

Well, I, I do so through a likelihood mapping, uh, technically a likelihood tensor that goes from these, um, latent states, um, probabilistically maps to the outcomes.

So what I've done here is just show the rule that we used in the piece example in the form of an unfolded, um, likelihood tensor, um, which is fairly simple to read.

All it's saying is that, you know, if, um, the rule, uh, the current rule context is that the correct color is on the left,

then there's a lawful mapping between what I am seeing, um, and what the correct color is, uh, provided I am looking at the left.

But if I'm looking at the center, then I'll just see red.

And hence that was the rule that whenever the center was red, I would, should look to the left.

When I look to the left, I can now identify the correct color because of this isomorphic mapping between the two.

If I'm looking anywhere else, looking to right, that doesn't count.

Um, so what we're doing is basically visualizing the, the nature of a rule in terms of these sparse isomorphisms,

these sparse mappings in a likelihood, um, mapping, um, between cause and consequence.

And of course, you didn't know that.

You did know that if, um, the, the, the color, um, of the, uh, center square.

So I'm looking at the center, there's a unique color to one of three rules because there are three colors, but you don't know what the rule is.

So you started off with this, uh, and, you know, you were hoping to gather enough, uh, information or evidence

to eventually disclose this structured, um, likelihood mapping.

So we can actually simulate that.

And this is what's, um, been done here.

I'm just showing, um, the results of simulating a Bayes Optimal Agent,

um, with those ignorant priors as naive priors, um, and looking at its performance,

um, over 32 games, each game comprising sort of, uh, four, sorry, four moves, five exposures or, or, uh, to the same stimuli.

Um, we don't need to go into any detail here.

What's going on.

This just shows you the, the policies about where to look and whether to select something,

um, that are selected given all the cues, um, um, the central cues at the top here.

Um, what we, what we will notice though, is that after about, um, um, 13, uh, or 14, um, observations,

the agent is continually correct, makes no mistakes, um, which is, um, roughly the same as, uh, as, uh, Daniel's performance,

even though he half got the rule.

Um, at that point, the free energy reduces on average and the confidence, namely the, uh, the precision or the neg entropy of the belief distribution over what to do next,

what policies to engage in, uh, suddenly, um, remains quite high.

So there seems to be a sort of a reasonably abrupt change, um, after about the 15th, um, uh, the 15th exposure, uh, to a low free energy, confident, correct kind of behaviour.

Um, and, um, the question is, um, you know, what, what underlines, what underwrites this, this qualitative, um, uh, sorry, quantitative, uh, change in, in behaviour.

Um, what we can get an idea just by contrasting the belief updating, um, before and after learning sufficient amount of contingencies to have an insight into the rule.

Um, and what I'm showing here is just the simulated belief updating and image format, but also the simulated local field potentials,

or if you do electrophysiology, uh, event related potentials, um, associated with looking at the different cues before and after acquiring some knowledge through parameter learning.

So updating my beliefs about the likelihood mapping, um, uh, denoted by A.

And the thing I just wanted to highlight, it's really interesting that, you know, when I've learned what the meaning of these instructional clues are,

when I have some understanding of the course effect structure, then that meaning is actually reflected in belief updating.

So suddenly certain cues, um, suddenly evoke belief updating and simulated neural responses in those representations of the context.

So suddenly I can recognise I'm in one context simply because I, and it's me and it's, uh, implications, simply because I've now learned the implications and I've learned the rules.

Um, but what else to ask, well, why look anywhere? There's no reward in this paradigm.

Um, you know, what, why would you, um, what, what, what drives, what motivates, um, scanning around in the first place?

Well, of course we know what the answer to that is. It's, it's basically, um, the expected information gain.

But in this instance, not over the normal kind that is associated with Bayes' surprise, namely resolving uncertainty about states of the world,

but, um, using exactly the same construct, but resolving uncertainty about the actual contingencies, the rules, the parameters of my generative model.

And in some literatures, um, between information gain over states and information gain over parameters is known as novelty and salience.

And all we're saying is that generalising expected information gain to cover the parameters of the generative model, um, equips you with the kind of curiosity that can be described in terms of what happens if I do that.

Uh, what will happen? So it's not what is behind this door, but can I open this door? Um, and we are motivated to be curious in this kind of way.

So our intentional behaviour is now motivated by this, um, by responding to epistemic affordances in the environment that allows us to slowly unearth the contingencies and the parameterisations in our generative model.

So, um, and so is that a sufficient explanation of what's been going on? Well, not quite.

Remember that Daniel was able to, um, partially discover the rule after, um, six, um, uh, present one, two, three, four, five, five presentations.

How did he do that? Because he'd never seen some of the information necessary to be confident that this was a rule in play.

Yeah. Um, at this point I would, you know, I, I, I, I'm tempted to ask Daniel, but I think, I think that would be unfair.

Um, so I, I, I, I asked the audience, I said, think about how enough, what did, what, what did you know if you had discovered that rule before the, uh, the full 32, uh, exposures?

How did you, how, what, what, what did, what did you know that the model didn't know?

And of course the normal response, um, which is completely correct as well.

I knew there was a rule. I knew there was some kind of symmetry.

There was some kind of constraint, some lawful relationship, um, which constrained my hypothesis space, um, which meant that I could actually rifle through different explanations that all had a rule like, uh, structure.

Um, I may have done that serially or I may have done that, but, well, I don't know how I could cause it's likely that I, uh, I was doing this unconsciously.

Um, uh, from, but from the point of view of, um, how one might simulate the fact that you knew that there was a rule.

How could one install that prior into this kind of active inference?

And the idea is really just to focus again on optimising the prior, optimising the prior via a process of structural learning.

And basically what we're saying is that if this agent was equipped with an a priori set of rules, hypotheses, um, uh, laws, um, that are going to be embodied by the rule,

that are going to be embodied or represented by a whole range, a whole set of different kind of likelihood mappings or, uh, likelihood tensors here, different models.

Then, um, this puts constraints on the belief updating and it enables the agent to accumulate evidence for each of these models.

And when the agent is sufficiently, um, confident that their one model is the best expression of what's going on,

she can then commit to that particular model. So this is a process known as Bayesian model selection.

Um, and we can formalise that just by looking at the sort of the maths of Bayesian model selection.

Uh, and the picture that emerges is that, you know, we spend a lot of time, um, gathering information, foraging the environment in a way that allows us to build up associations and contingencies and correlations, um, as parameters in our generative model.

And then the idea now is that we're going to go back and find a simpler model that has a greater evidence, that still complies with a particular model by removing certain parameters and scoring the, whether the, uh, the free energy bound on the model evidence goes up or down.

So if I remove something from my model, if I reduce my model in some way, uh, and the evidence goes up, that means that that model was just contributing to the complexity, not the accuracy.

So I'm fully licensed to remove it and to reduce that model.

And furthermore, if I have a space of hypotheses, all of which are reduced forms of the model,

I can go through evaluating the reduced free energy or the reduced evidence and score all my models in terms of how,

uh, how much, um, uh, how likely they are in relation to each other.

And if there's a winning one, I can pick out that and I can, uh, use this base of model reduction in order to, um, uh, very quickly and efficiently assess,

now this model is the best expression for the data I have seen so far.

Of course, this is very reminiscent of what we're doing in sleep,

um, in terms of synaptic regression just by removing, um, redundant parameters, crucially, um, minimising free energy, but in the absence of any sensory data.

So there's no accuracy at all.

So the free energy just reduces to complexity.

I minimise my complexity just by removing, um, removing any, um, parameters, um, that have a substantive change in the free energy.

And that's relatively trivial to, um, evaluate, um, under these discrete space-based models or likelihood tensors.

So what we're saying is basically we go around with our eyes open, accumulating all sorts of potentially, um,

redundant associations.

And then we, we just think about things.

We close our eyes, we introspect offline, um, just to, uh, make sure that we just retain those, uh, parameters that are necessary to explain the data that we've just seen.

So that we can now maximise the model evidence and then open our eyes again, uh, and start, uh, start, start again.

And when we implement that in this agent, um, we can massively increase the, um, the sample efficiency, um, through this period of periodic introspection,

or having little naps between each of the trials, which is why I was trying to give Daniel a break, but didn't actually allow him a proper break, uh, to think about things.

And, um, increase, so increase the efficiency with which we can build our beliefs about the structure and the parameterization of our joint and model.

So here's the true rule. This is what, um, Daniel started with.

After about 12 trials, this is probably what he was representing if he didn't know that there was a rule there.

So if he'd had time or indeed did have time to a certain extent, um, to now look at this, uh, through the lens of all possible rules,

he could have identified this particular combination of subensors or, uh, this rule here after the 12th observation.

And indeed, um, when we simulate this, um, um, um, we can show a marked reduction in the discovery time and the associated, uh, uh, performance that can be scored in terms of the average confidence, um, and the, uh, free energy bound and model evidence.

Uh, this is probably the most interesting, uh, kind of result.

It basically shows how long it was until the model had a nahar moment, how long it was until one of its hypotheses, one of its rules, uh, became substantially more likely than any other.

To the extent that we could then confidently commit to that particular rule.

Um, and in some instances, it can, it can actually be with the right kind of information.

Um, it can actually be, uh, very, very, uh, uh, you know, be done after just three exposures.

Um, but you have to have the right kind of information.

Uh, this is the, the top line is the first one, um, showing that, uh, to be, uh, to be confident about the best drill, you, you, you, you should have been about sort of eight or eight or eight.

Uh, observations, which is roughly what, um, uh, which is roughly what general people do, uh, generally.

Um, um, uh, however, notice the blue squares are when people jump to the wrong conclusion.

So there is a possibility of prematurely committing to a particular model on the basis of what you've seen so far, but, um, um, it's not going to be the best explanation for what you're going to see in the future.

So they occasionally get this sort of, um, jumping to conclusion effect.

But on average for this particular game with random, um, ordering of stimuli, we're looking at, uh, you know, the, the aha moment emerging after about five to six, um, exposures, which is, uh, exactly the same that, um, uh, Daniel, um, evinced when, uh, getting the partial rule.

So the final, the final, the final thing I just wanted to, um, um, um, speak to is the last generalization, um, from expected information gain over parameters to expected information gain over the model in and of itself.

So this is new, this has never been written up or, um, um, or described anywhere, but it, it, it, it's a natural extension that rests upon, um, Bayesian model reduction and takes us now into the world of true epistemic foraging as a scientist would do when she is gathering data for or against or to disambiguate between a null and alternate hypotheses, but generalized to potentially hundreds of hypotheses.

So now we're addressing the question, um, let's assume we are going to do some Bayesian model reduction. What's the best way to gather data that will disambiguate, maximize the information gain, maximally resolve the uncertainty about this hypothesis space so that I can remove one set of hypotheses successively and winnow down my hypothesis space until the last man is standing and I can commit to this particular explanation.

And I would submit that this is, if you like, um, active reasoning, it's, it's sort of, um, actively having in mind explanations for your world and then designing experiments in a way that is the most efficient in terms of resolving uncertainty about your hypotheses.

And that just basically requires, um, and that just basically requires, um, adding an extra term to the expected free energy, um, that just scores the difference between my posterior over models, uh, with the outcomes relative to without the outcomes.

And when we do that, um, we get sort of, you know, very interesting, uh, behaviors, um, and you can actually score or plot the posterior probability over different models or hypotheses here.

This is the same paradigm that made slightly different.

So we didn't tell the agent which of the, um, locations of the colored ball, uh, colored balls signified, um, with the, the, um, the criterion ball that was the, uh, contained the correct color.

Uh, so, um, in the sense in the, in the game we just played, that was a center ball, but the, this agent didn't know that.

So it, it, it, it has to acquire a lot more information before it can commit to a particular hypothesis.

That hypothesis, um, well, the correct, um, the correct rule is the one that we just, um, um, entertained, which the center is red and the left state is the correct choice and so forth.

Um, that's rule number 65, but there are 81 other rules that all have to be, um, eliminated before we, before we can get to the correct rule.

And this is a process of elimination here, starting off with rather uninformative posterior beliefs about the particular rule or model in play.

And when we, um, are confident by applying, uh, Occam's razor, um, to this distribution of posterior hypotheses or beliefs or models, um, we can suddenly commit to this model.

And then we can identify the true likelihood mapping, um, for subsequently and basically become very confident, um, in terms of what we're, what we're doing, uh, and perform perfectly, um, under the constraints of the game, uh, with a minimum divergence between the posterior and the true, uh, true likelihoods.

Um, just for those people who are interested, um, it's interesting to think about, well, how much does this, um, expected information gain over models?

Contribute to the expected information gain over other things and how do they interact with each other?

And just to speak to that, it's, um, what we've done here is to simulate that game again, um, but using all the different kinds of, um, components and expected free energy, the expected information gain over models, states, um, parameters and states and the expected value, which in this instance we can be ignored.

Um, and then remove the various sources of all kinds of expected information gained to see what would

happen.

And interestingly, what happens if we remove this, um, this scientific or, um, um, information gain over hypotheses or models?

Um, we can still do quite a lot of, a lot of, um, resolving uncertainty about, um, models just by, um, resolving uncertainty about the parameters that constitute those models.

But we have real difficulties when it comes to two equally plausible explanations, because, um, this agent is actually deliberately sampling the world in order to maximally disambiguate between the two, uh, the two last hypotheses.

And when we remove everything, we, we, we, we, we still, there is information there, but it can take a very long time before we actually commit to any particular hypothesis.

So in summary, um, uncertainty can be, uh, you know, thought of in terms of expected surprise, namely entropy.

Um, and we can think of different sources of uncertainty and their resolution in the following way that latent states parameters and structure all, um, offer different sources of uncertainty that can be resolved through free energy, minimizing processes of inference, learning and selection of different kinds.

Uh, leading to, um, uh, uh, leading to, um, uh, one way of reading artificial curiosity in terms of succumbing to or finding affordances that can be described as salience, novelty and, uh, knowledge, um, and just to reiterate the importance of sleep.

Um, in the words of Einstein, the nearly reflexive act of inference is just basically appealing to the best explanations of the world, put very simply, everything should be made as simple as possible, but not simpler.

Um, when we move to, um, planning as inference and the importance of expected information gain that is constrained by expected value, uh, the important thing is to not stop, is to not, to stop questioning curiosity has its own reason for existing.

And on this analysis, you probably invert that, if you exist, you have to be curious in the special condition that you don't have direct access to your own actions.

And with that, it just remains for me to thank those people whose ideas I've been talking about, of course, to thank you for your attention.

Thank you very much.

Thank you, Carl.

Awesome.

All right.

Great.

I will now pass to Michael Garfield, who will facilitate this next section.

There will be some round table discussion and thank you, Michael.

Take it away.

Yeah.

Wonderful.

Uh, that was a bracing way to wake up immediately jump into your lecture and, and, uh, begin one's day.

Um, I guess the first question I have for you, and then I think this touches directly on Hongju's work and, uh, I would love to solicit input from other people on this as well.

But Carl, let's, let's, let's have you go first.

Has to do with the surprise of advanced meditators or psychonauts in a, you know, non-ordinary states of consciousness.

And like the, the, the, the loss of the distinction between self and other.

And, um, there's a, what Eric Davis calls a kind of a metabolic ontology that, um, there's a threshold beyond which the, uh, these like dyadic distinctions no longer, uh, seem to pertain.

And you end up with like, uh, kind of co-implication of self and environment or a, uh, like an end dimensional figuration of, of these relationships.

So I'm, I'm, I'm curious to hear folks on this panel wax on, uh, phenomenology of non-ordinary self other relationships in the, in light of this framing that you just laid out for us.

I'll start that because I'm the oldest.

Um, and I get the impression it's very early in the morning for you.

So, so I'm assuming you're in California, which of course, Santa Fe.

Yeah.

Santa Fe.

Okay.

That's still, that's still about sort of 11 hours, I think, or eight hours.

Uh, so it must be very early in the morning.

My apologies for that.

For me, it's getting, getting into mid afternoon.

Uh, in relation to your question, I'll try and set up the, the, the, the other, uh, panelists for this.

Uh, a, a, a very sort of, uh, technical view of this.

I guess what you're talking about is sort of non-dual states.

The, the, the distinction between self and other others.

So the, the narrative self or indeed the sort of somatic self just disappears.

Um, I think that, that again, I'm answering, uh, you know, if I had to simulate this, how would I, what kind of gerative model would I have to put in play in order to plausibly simulate this kind of phenomenology?

And the first thing that I would have to do is to actually write down a gerative model that actually had a distinction between self and non-self.

Um, and, um, or self in different, uh, in different contexts.

So, uh, that would be quite a sophisticated gerative model, very expressive gerative model that naturally now have discovered the rule or the hypothesis that it is actually an entity.

And that many things can be explained by this particular entity.

That is me being in one of the small number of states, sort of carving subjective selfhood into, into, um, some plausible.

set of states that can now very efficiently recognize I'm in this state and therefore should attend to this and should attend to that.

And then what I would do to dissolve, um, that confident commitment.

I am in this state and the, and implicitly I am me, um, would be to look at those situations under which there's a loss of, um, precision over the self, other or self in different, other contexts, um, distribution.

How would I do that?

How would I do that?

Well, operationally that would require, um, decreasing this deep precision, um, via increasing the precision of lower level processing.

Um, what would that look like?

What would that look like?

Well, it would look like I was attending to something that was, um, uh, closer to the sensorium.

Um, so you, you, you, you could sort of say this was some aspect of meditation, for example, that, you know, I am deliberately attending to assigning much greater precision to the, um, to, you know, to the sensorium, um, possibly a very limited part of the sensorium.

And that implicitly dissolves any conviction.

It doesn't, I don't need to explain the world now in terms of I cause this and I cause that.

There is no need for any ironies.

Uh, uh, you know, uh, it is not necessary in this particular instance, simply because I've, I've been able to, uh, and of course we aren't generally able to do this.

Uh, but if you become a skilled practitioner doing this, some people can, um, you know, re-deploy their precision, their entropy, their, their, their, um, their attention, however you want to articulate this away from this very high level.

So, uh, kind of model of, uh, minimal self at least.

That would be my, that's how I would approach this.

Angzhu?

Angzhu?

No, I'm unmuted.

All right.

Thanks.

Um, thank you for, thank you for inviting me in this panel talk is really great and a great honor, um, before.

So, um, well, I would say from, you know, from a more, I would say, phlamnological perspective and phlamnologist ish perspective, I would say.

Um, so it is also important to distinguish, uh, intentional, intentional, uh, quality versus intentional the entities, uh, wasn't the, um, so.

So, if we say conscious, conscious, conscious, uh, conscious lana or qualia.

The qualia, uh, comes to, to me.

Like if we, uh, if I am.

Observing a red apple.

Then the red apple can be also the qualia.

But, but in the, but in terms of perspective or the orientation, subjective orientation that.

If I would like, like, um, be, be, um, open to, um, approach, I would say open to approach to the redness of the, of the apple.

Or am I close to like, am I deliberately, uh, thinking that red apple is something bad or something, something, um, harmful to, to, um, my individual.

Um, so, so those, that kind of, uh, that kind of, uh, uh, type of, I would say type of consciousness is.

Well, the questions are referred in general as an intentional quality.

And, um, I would say much the easier, easier way to describe this would be, um, having certain perspective toward some, some kind of matter, some kind of, um, cautious matter.

So, um, I would say also modeling, not just only the red apple, the redness of the apple and the roundness of apple, but also.

What, uh, with, what exact perspective does the, um, and then the visual.

Is oriented, oriented toward it.

Um, having, having which, which perspective.

Is very important.

I would say when we try to model the consciousness.

And, um, in terms of, in terms of how, so how can the active inference framework can, um, complete in, in, in terms of modeling this perspective, I would say.

Um, let's start from, uh, let's start from the very, uh, very basics of.

Let's say, um, free entry principles, uh, the physics principle, um, that.

The individual, uh, individual turns out to have certain, certain kinds of, um, valence or certain kind of feeling or orientation towards the.

Uh, toward the certain objects.

And that also.

Might be, uh, might be self organized within the, within the, um, agent, uh, within the agent or conscious being.

Um, so what I would suggest would be, um, if we can, if we can deliver this perspective system as something, um, something.

In, in the active inference model, uh, something that.

Um, makes, makes the agent to infer or.

Um, so as a, as a different inference layer, um, or.

So, sorry.

Um, as a, as a inference, different inference layer, different dimension of inference layer.

That is, uh, that can be, uh, uh, described in as.

A, um, a certain informed form of information.

I would say, um, so that would be one, uh, one meaningful path that we could model the perspective structure.

Or the consciousness.

Certain parts of consciousness, I would say.

Adam and Alex, I don't know if either of you.

One of.

Alex seems to be getting the.

The ball from Adam here.

Uh, well, my computer crashed in the middle of this.

I think I'm not a hundred percent sure what we're talking about.

Hi, Carl.

Uh, so, uh, uh, I'm not exactly sure what we're discussing.

I think I got a little bit of what Hongju was saying, but, uh, maybe remind me of the prompt.

Yeah, the, the, the, the prompt has to do with, uh, like this, like the self as a, uh, something that is, is not necessarily persistent in, you know, it's an output of inference.

And there, there are, you know, uh, many well-documented states in which, uh, self other relationships are not, uh, in which we have a different syntax.

You know, we're like non dual consciousness being one of them.

Uh, how do we, you know, how do we think about all of that within an active inference framework?

So are you saying the output of inference is oneself?

Just make sure I got it.

Yeah, it was like, yeah, it's like the, the, the idea of like a, like there being a, a self other relationship.

Uh huh.

Uh huh.

Is it like with other as in like other entities outside of like outside.

Yes.

I get right.

Yeah.

Well, this might not be the most appropriate answer, but I, I tend to think of it as maybe a different direction.

So this is something Carl didn't also mention in his talk, uh, but he and I have worked on this concept.

So another corollary of the free energy principle is mortal computation.

Um, and he and I have written at length about that.

Uh, the audience can just type Google and it'll come up.

And, um, so then that's a little bit about talking about embodiment and inactivism and embeddedness in extended theory of mind.

And then Carl added a fifth E to the five, four, eight cognitive theory.

So it's five E cognitive theory, elementary or basal cognition in reference to our friend Michael Levin.

Um, and so like these, this is like a whole framework for discussing like sentience.

Uh, I tend to avoid discussions of consciousness a little bit.

Carl's far more braver and more equipped to address these things.

So I usually stay away from that.

I'd like to look at it like a low level, but when you say the self, um, I think at least I haven't thought of active inferences outputting something is rather active inferences, like, uh, uh, a reference to like the dynamics and mechanics of something that is embodied.

So you need to talk about like a substrate and so we need to talk about, well, what platform are we building on?

Like, is it a robot?

Um, and that is the self.

So we have kind of this like entity and again, it could be like nested Markov blankets and we could talk about lots of jargon, but the idea is that you have some kind of object and it's, uh, it's primary directive is it doesn't want to die or rather it is trying to preserve its non-equilibrium steady state, its nest.

Um, cause it doesn't want to disintegrate its heat baths.

So the whole point is it like moves through, uh, the universe and trying to collect resources.

It does the epistemic forging that active inferences all about, uh, because again, that's critical to maintaining its nest.

It's prime directive.

Um, and so like the self, the identity, you have to like touch on the notion of embodiment and you need to have like a body to talk about.

And so active inferences performed within the context of that body again, can be described as a Markov blanket.

And so that is the self.

I mean, of course it can be destroyed.

It can disappear.

Uh, if of course it fails to maintain its nest.

Um, but the idea is like, that's how I think of the self.

And then you can use the idea that that's a mortal computer.

I mean, underneath all of this is the FEP is saying we're all mortal computers and we live in a collective of mortal computers.

And so that's the other, right?

So there's other mortal computers.

So I think I kind of touch on that.

And that's kind of the way I think of it.

Rather the self is kind of like inherent to the fact we have to start talking about an entity, a substrate, something physical.

And we also need an environment of course, because inactivism is critical.

So you need some way to describe the body, its interactions, its thermodynamic coupling to some niche.

Uh, and then of course we need other, other bodies that also interact with that nation, interact with that body.

Uh, and then of course you get embeddedness, which is then how you get society and culture.

And then we can start working our way up the, the big hierarchies that Carl often likes to talk about.

Not to speak about you when you're here, but as we work up to bigger and bigger self organizing systems.

So that's my perspective.

I might be misunderstanding, but that's how I see self is it's kind of like, well, you've got to be able to talk about.

Now a platform, a substrate, a body, whatever it is, it could be E. coli.

Uh, and that could be fine.

That's a wonderful example of the most element, one of the more simpler mortal computers.

So it rather than an output.

So I don't know if maybe I misunderstood, but that's how I think of it.

And so I kind of like to touch on that is sentience.

I don't know about the consciousness part.

I very careful with that.

So sure, sure, sure.

Sure, sure, sure.

I, I, I think that, uh, you know, just to qualify that, uh, a lot of meditative traditions.

I see, uh, like laughter as like, uh, uh, an observation of surprise at realizing that, you know, one is not other than, uh, figures into this.

That I'm Adam, I want to hear you speak to this.

Um, the part about laughter in particular or.

Well, no, just broadly.

Um, well, what just came to mind with a part about, um, laughter also would be, um, some of the work Carl's done with, um, artificial curiosity and insights and some of, um, uh, Schmidt Huber's, uh, uh, framings of, uh, humor in terms of like rapid compression.

Uh, of, um, uh, sudden, uh, uh, uh, I guess, uh, sun arrival at an increasingly elegant models.

But, um, in terms of, uh, I guess what's come up is, um, as, uh, wondering about, um, in the process of, um, self evidencing, whether the most robust self evidencing involves, um, having a certain porousness.

And, uh, there's, uh, there's, um, uh, some work that's been, uh, called under, uh, different names, such as, uh, uh, psychotic Markov blankets.

Or, like, you can have Markov blankets that aren't well tuned enough that might be, like, overly rigid.

And, um, um, they don't, um, uh, adjust to the multi-scale competences they ought to be a part of.

Um, and the co-construct, they don't, uh, co-construct niches well.

And they can become, like, um, pathological or, um, like, parasitical or, like, cancerous in their, in their nature.

So, but anyway, but the, uh, uh, I guess the thing I'm thinking of is, um, uh, the importance of maintaining meta-stability, uh, or, or, or for, um, ongoing, I guess, um, uh, chaotic, uh, itinerancy or ability to adapt or evolve.

Like, uh, in, in the, in the process of, uh, preserving the self in a given form, uh, the best way to do that would be to have a grip that's not too tight or not too loose, um, on the self.

So those are some things that came to mind.

I don't know if that's made any sense whatsoever, but I put some, uh, uh, some of the thoughts in the chat and some papers.

Um, uh, there's, uh, interesting one by, uh, uh, my friend, uh, Rutger, um, Goku, um, um, um, how higher goals are constructing collapse under stress, or he'll actually talk about how systems, um, uh, uh, when they,

they're put under allostatic load, they can basically, um, uh, their practical structure can unravel a bit as, like, the deeper, um, uh, nodes or predictions, the ones related to things like self and world potentially become overloaded and destabilized and this is having a certain adaptive function so those are things that came to mind when from from your prompt but all right yeah i had another question on uh this this carl's talking about sleep and scaling you know i was i was uh just watching uh a stream of octopus documentaries talking about the the development of the octopus brain and how rapidly these animals have to grow and learn and and uh how therefore so much of their life is spent in sleep and uh i'm trying to like think about that in terms of um like what we have going on with uh ai which you know by contrast seems to uh you know we don't we don't have uh at least in any way that we think about it like a like a dreaming like a sleeper dreaming for the models that were that were put that we're using in ai systems although i i have seen some research just sort of simulated dreaming in in an effort to like uh create geospatial world models in in ai um so i'm curious about like the relationship between these things and and uh how uh energy inefficient our ai systems are and uh the possibility that like human input might be able to function as a sort of proxy of dreaming or sleep in ai systems um that's a big kind of a kind of wide ambit question but i'll kick it off there that like you know how how do we how do we let the machines sleep and dream um you know what what might that look like technically uh what role does uh uh yeah i'll just i'll stop there carl now that's a great question i've been making notes about sort of key words that people have brought to the table which i think some deserve deserve a little bit of a comment but they're just picking up on um um that particular question um it is interesting that um from the point of view of people like jeffrey hinton um building the helmholtz machine um sleep played a really important part um and if you go back and google the week at the wake sleep algorithm um you'll see the first forerunners of the kind of mortal computation that alex was talking about and interestingly what this basically entails is that you're um you're basically building a machine that can learn to infer and the way that you do it is you expose it to data um and then you learn the parameters of um a neural network that maps to posterior beliefs and then you let it sleep you take away the data and you let it have a little little period to sleep and during the sleep you actually generate from the learned representations fictive data dreaming data and that allows you now to learn the parameters of the mapping from the latent representations of the latent causes to the actual content or the outcome so this is very very similar to um in fact almost isomorphic to the original introduction the free energy principle that was cast really in terms of learning to infer um but um the ultimate objective is an inference machine exactly along the lines of the helmer's machine or or um the you know the way that i introduced the sort of hell helps in perspective

so i think your notion that you know sleep um is important is absolutely right it was there at the foundation of all um certainly the useful um um engineering endeavors in machine learning and artificial intelligence research it's lost its way a little bit with uh current transformer architectures and um um deep rl however i think it's coming back again in one sort of um muted sense if you just look at the um what reasoning means in the journey very high literature reasoning means trying to suppress hallucinations which you could call a kind of dreaming um but it's online reading that the user could actually see but trying to resolve um mistakes of your false predictions or you know implausible predictions uh by um recursively um you know in the background um generating its own prompts and or multiple prompts and then selecting so reasoning as read in the generative AI literature which has nothing to do with reasoning in the vernacular or concrete science or indeed in what i was talking about uh or indeed in terms of probabilistic induction uh you know logical induction of the course kind of josh tenenbaum might be uh you know might be interested in uh but reasoning in this particular popular literature in machine learning um large language model literature it does actually uh come back to this notion of just taking you know stopping worrying about the content and then just turning things over in your mind which has some similarity to the kind of um um um basic model comparison and uh uh uh and and reduction that um was um in my presentation so i think that yeah it's a really important thing to think about and you know again i'll come back to the fact that you're you should be asleep now if you're in california so i'm sorry about that sleep's really important uh but can i just comment upon some of the really just very quickly on on some words that came what came up during during the panel discussion which i think um would be really usefully formalized within an active inference uh setting so i i think about you know um initially we talked about valence uh i think those is really important um so the most technical and simplest explanation of valence i think would be come under the rubric of affective inference i mean this very simple view valence just is the precision or it's the negative uncertainty so things are negatively valenced when you're uncertain specifically when you're uncertain about what to do and that's sometimes associated with anxiety um sometimes in a biological context uh stress uh and chronic stress so just knowing not knowing what to do because you've got a very low precision belief belief over your policies uh can be read as and of course you have to recognize that and now you have if you like a gerative model that can recognize oh i am uh in a state of imprecision uh pertaining to my policies and therefore i must feel bad um uh and you can now um sort of unpack the implications and the virtual and or possibly vicious circles in terms of interacting with the world um that um that i think is one part of the story it's not emotional inference though and i'm now speaking to alex's focus on mortal computation of course as soon as you talk about mortal computation and things like you and me you now have to move to interceptive inference that the actual body matters so the vastness the biggest sensory organ i have is my body uh and i have to make sense of my body uh and you know i have to infer the state of my body and of course as soon as you do that you've got some sort of minimal kind of self-modeling inferring my state of self because i have to instead infer it could be completely pre or

subconscious or sub-personal you know in the sense of homeostasis but when we move to allostasis then i think

the the lines kind of bit you know the the the lines are not so distinct and you could argue that the interception in the spirit of mortal computation has to be an aspect of emotional inference and of course only you can own an emotion so i think that's important um i've also got down here self versus other um yeah i think that's a really important observation so there's a whole other argument uh which actually came under adam's uh nod to co-construction of our niches um so one thing if we co-construct a niche together is that we our world now becomes populated by other things like us which suddenly calls upon gerative models to disambiguate did i cause that or did you cause that because we are co-constructing and that may be if you like a really simple minded explanation as to why we need a notion of self simply to disambiguate between self or the consequences of the action of self the versus other and of course that's a you know a deep issue when it comes to um attachment theory or emergence of theory of mind in a developmental context for the baby to recognize that it's a self and it may actually have to first recognize that the mother is an autonomous self and then recognize well perhaps i am similar to mum and therefore perhaps i'm a thing as well and suddenly you get this this this sort of distinction between self and other so i just wanted to really endorse that that point uh i think that mark was making that you know the other hood uh is probably more important for selfhood than anything else you don't need it uh if you didn't actually co-construct your niches um the dimension um the dimensionality in terms of quality of experience and qualia um that you'll be searching for in um active inference schemes formally i would read as the order so first order inference is basically inferring the content of what's going on second order inference is inferring the uncertainty of the precision of that so it's not about the content but it's the precision you're inferring as a psychologist you would read that as attention for example um and i think that that the um some people would argue to have a quality of experience you have to have some um agentic control over the attention to the redness of the apple um which just means you've now got to have a sort of deep gerative model of a particular kind where um the depth is not now in terms of the sort of composition of content but the uh the depth in terms of parameterizing the quality of your beliefs about the content sometimes you can describe this as a sort of metacognitive aspect so i think that that that that would be a really interesting line of inquiry to explain and people do explain this um you know the distinction say between something that's normally transparent and opaque uh being defined by the ability to mentally act upon that representation on the inside through affording it greater or lesser precision but of course to do that you have to have a representation that i want to afford this to so again it it induces a dependency on a self-representation and there are all sorts of um ways that you you know one can impact that in terms of i repeat phenomenal transparency and mental action so i just wanted to say all all of those perspectives i think speak to really exciting opportunities to formalize them in terms of relatively straightforward gerative models but perhaps not too straightforward because they do involve some higher order aspects of of of inference that uh that we are particularly skilled at uh scheme solving so then i want to add to carl if you don't mind i want to extend because carl brought up

really interesting the holmholtz machine and the wake sleep algorithm uh we also you were to michael's point about generative models like again that's been around the dreaming concept like boltzman machines also used to be called by terry uh so janowski and jeff uh harmoniums uh the idea with that learning like wake sleep got generalized to contrastive divergence so another way that dreaming kind of looked like was you have what carl was mentioning like data dependent inference and then you had uh independent or let's say decoupled kind of like exploring trajectories that are confabulations this jeff would call them and then you kind of like push them away uh from what's reality which is data driven so that's kind of been i just kind of wanted to highlight that sleep has kind of been at least through the view of like oh we have negative and positive data and then today another kind of branch of research that i think dreaming has sort of kind of lived uh is in that contrastive paradigm we see a lot of self-supervised learning in the neuro ai machine learning communities um and again that's kind of like pushing away out of distribution data from indistribution data uh and again i carl and i've also done some of that too with like meta representational predictive coding um so i kind of and i think the dreaming another way that it kind of lives on is in the concept of catastrophic forgetting uh because no matter how much transformers are heralded as wonderful in the solution to everything uh they still forget deep neural networks have not really addressed this issue uh and so uh you can see even like predictive coding literature from my groups like my own and others where you use like the generative model that's being learned in this neural circuitry you sample from it and then you kind of use that as a refresh mechanism to try to improve your longer term memory and that could be viewed as like a dreaming

state or little uh let's say daydreaming phases uh and that goes back to also classical literature like robert french and those that tried to work on catastrophic forgetting like the 80s and the 90s uh and another place too i just want to highlight is let's not forget that active inference which shares a lot of connections to reinforcement learning we have experience replay which is again another way where we kind of

replay a lot of data in a buffer um and trying to kind of uh improve the learning mechanism of a policy uh so we that kind of mimics episodic memory kind of sort of so that's another place where i think it lives so i just wanted to

highlight that and then you asked michael very briefly you snuck in there about like energy efficient ai and how energy efficient it is obviously it is completely opposite of what we want transformers are again carl and i have argued this they have carbon footprints of literally cities uh to run uh you know chat gpt and llms so i think like to get to that energy efficiency that we want and i think also carl highlighted in his uh mention of interoceptive inference uh i think the answer is again we go back to the idea of a body uh you need an embodiment system they need a mortal substrate uh and then i think that kind of begins to answer where dreaming uh and this consolidation of information that again carl mentioned in his talk which is the structural selection i think you need a body because then it kind of motivates things like rest uh because again if your body is fatiguing and you need to kind of go into a state of like oh i need to repair because you guys have mentioned we've mentioned homeostasis you

guys have mentioned allostasis we should not forget autopoiesis uh the idea that we also need to repair and construct and deal with damage and so i think that that's where you need a motivation for the dream state and the rest state and i think that that's where the body sort of answers it because again that comes from the interoceptive inference i need to understand my own internal states and then i am driven to preserve them and again go back to our directive so i think that's where dreaming the future of dreaming comes into play because i think with a body and its own limitations and its own needs and requirements and then how you interact with the niche to get those resources uh motivates things like rest repair uh and again so we and not to keep saying the same thing but carl and i highlighted all the levels right you have to start with like hormiohesis or homeostasis allostasis on top of that you need to you need to describe autopoiesis uh you kind of need to go all the way you have to explain all the levels to build like a complete mortal computer that you know is again an embodiment of the free energy principle and i think then that starts to motivate things like why you would dream uh and so i think that's kind of where the future might lie even though we see it embedded like you know carl and i pointed out some examples uh where dreaming is kind of used as a mechanism to repair learning uh or to for example learn to contrast confabulations from real world data so it's just like a little thing i wanted to make a comment or long comment on and then if i'd i'll just add quickly i know we don't have too much much a question for carl i just wanted to ask him based on the talk and he mentions like a bunch of these different things about active inference and i love the formulation but something that he knows that i'm obsessed with is the role and carl you mentioned neuro ai neurobiological learning and i know from your group and a lot of those that work on it back propagation of errors is used often to train the deep neural network forms of active inference or you do things like probabilistic graphical models that are discrete which are kind of hard to scale up to very big systems what do you think is the role of getting the learning rules right like heavy and plasticity and you know because again predictive coding has local learning rules uh i often see like my group tries to show you how you can do active inference with that but i don't see a lot of what do you think is the role and the importance of getting that right that neuro-atomical description like synaptic plasticity correct and not using like the chain rule of calculus which is convenient but isn't really what our brains are doing i mean the neuroscience still has not confirmed that we are just a chain rule system back propagating along the same synapses that we do forward transmission so it's curious to know your thoughts on the role of that if you don't mind commenting if we have some time daniel do we have time yeah go for it carl and then any last comments from others right i will yeah really excellent point excellent i think at two levels just that sort of um the importance of mortal computation for the way that our tech industry is developing and the impact upon our lived world i think that's a really important uh consideration which speaks to your quite substantive issues uh in the way that we deal with big tech and the direction of travel in machine learning research and investment and building power stations in virginia for an example um so you know much of the motivation for biometric or neuromorphic uh mortal computation um inherits from that and from the the physics of information uh sometimes cast in terms of say thermodynamic computing if you like but also the importance of neuromorphic so that's i think a really important

issue beyond academia um the second issue the specific question about the particular algorithms that are used and when they diverge from a neuromorphic or um biomimetic um principle or when they diverge from

the principles of least action um evidenced in biological and real world self-organization you know that's travel ahead um and one example of that is back propagation of errors and of course the whole point of the free energy principle is it is a local free energy is an extensive property which means if you minimize it here and just here you're minimizing it everywhere if you're repeating that local um update run so in answer to your specific question about the update rules i think they're vitally important um is there a right one or a wrong one according to the complete class theorem no um there isn't what you would say is what is the implicit generative model that explains this particular functional form for the uh the um belief update uh rule for the parameters or the learning rule in machine learning talk so there will be a generative model implicit in the functional form of your learning rule um and that has to have the form exactly of that helmholtz um decomposition so there has to be a sort of a serenoidal term there has to be you know an up to an update term and a prediction term in it and there's also

going to be some sort of volatility terms there's going to be a decay term in it um so but the key point is that it has to be local

because we know that the paths of least action um are defined by action which is a time integral of an energy potential and that energy potential is an extensive function so it has to be local so you know from the start that back propagation of errors is not the right way to do it um that would be my brief answer to your question thanks carl appreciate it all the panelists now have to

to say something very wise in their concluding comments i think that's that's the pressure upon you yeah any short closing comments and then we'll go to the next session well then i'll just knock myself out quickly uh thank you carl for again answering my question i mean i obviously have argued for like a better part of a decade that backprop is not the right way to go so uh and uh i think again through the pass of least action that kind of motivates why we should care about like local biological plasticity as kind of our guiding way of designing for example predictive coding active inference systems so uh maybe to try to say something wise in the spirit of what you were just saying to the audience i think that uh yeah we should probably if you are interested in this like idea of neuro ai carl and i call it biomimetic intelligence uh but they're all kind of the same thing i think then we should be put making a bigger push uh towards biological plausible forms of learning and rules uh and i think that that's kind of a key idea and then just remember that when you're constructing these as carl has aptly pointed out there is like an implicit generative model uh what you might not be able to do with an implicit generative model i'll just briefly comment is you might not be easily able to sample from it sometimes it's better to have the explicit model uh but again uh it's kind of like a trade-off and what kind of scientific question you want to answer and what type of model you want to build so uh yeah that's my minor comment but i do think that there should be a lot a bigger push uh towards uh biologically plausible credit assignment learning uh in support of things like an active

inference framework and so maybe that's a call to the community to the active inference symposium i know my students who will be presenting later these coming days and today they know that rule but i think more people need to do it so we need a bigger push for that but maybe that's over like the next 10 years or so so so thanks for that

does anyone else

you'll have to hand the baton by calling upon yeah okay okay then i'll give it to uh michael since it look like you uh opened up your mic your turn say something wise in in the spirit i have no wisdom here i'll just say that uh this has been a uh delight i think you know it's it has prompted me to think about the the way that uh like why is it that the production of

uh like deep fake video has become so prominent for both open ai and for meta

in their in you know public facing products

and that it may be that that like

the insane uh finance like economics of these organizations

uh is actually like leveraging humans trying to

like uh dissipate

out into

like all of the possible

fantasy worlds that we can imagine with these tools and and that's somehow gonna like hold this

thing up i mean i'm sitting here thinking about the ai bubble that we're in right now and uh

whether whether it's a bubble that's going to pop or whether whether it's actually like a

a cell membrane now being held together by all of this this sort of uh collective fantasy

i'll leave it there all right thank you everyone we will be back soon

yeah thank you good talking to you carl

i will be talking with you i will be talking with you

i will be talking with you

i will be talking with you

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

All right.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

All right.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

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here a bio-inspired 3D neural compass based on head direction cells this is an engineering take on this where we're looking at this model of bat head direction cells and then this more biological paper where they look at three-dimensional head direction coding in the bat ring and it's based on this model of the bat so the bat uses echolocation and vision to sort of tune their spatial navigation they have this flight room set up where the bat is flying around the room they're taking pictures of the bat and they're looking at their brain they're looking at

the HD cells they're tuning with and their firing rate and they're looking at it for different behaviors they're looking at the tuning width in percent and then they're looking at the peak firing rate in hertz so this shows that there's this responsive head direction cells for different types of behaviors and in flight there are different types of neural responses and so if we put this into a phylogenetic perspective we have primates chiroptera and rodents these are all mammals but they're of course have different ways in which they treat optic flow so in primates we have optic flow contributing to a view representation and this results in visual specializations and primates but it has this view this is why i showed the image of that network with the place cells and the visual information so that's in primates but in chiroptera optic flow contributes to a view representation as well but this is due to visual and auditory specialization so they have this view representation and chiroptera is a sister outgroup of primates so that makes sense but they have this extra set of specializations and then rodents being kind of distant from chiroptera and primates optic flow contributes to a place representation where somatosensory and olfactory specializations dominate and so you can see that there's this relationship between how information like optic flow goes into the brain and depending on how the brain is adapted to different types of environments different types of lifestyles spatial cognition can occur in different ways but kind of rely on the same substrate and so this is just a little uh at the lower right there's just this little visualization of how in flies visual motor transformations um basically stabilize flight through this optic flow mechanism so this is actually quite a bit different than the mammalian last common ancestor and this is quite different than the last common ancestor for mammals but you see that uh you see that flies are also using uh optic flow to stabilize their movements and then that translates into how they can navigate through space o'keefe and nato actually had a lot to say about gibson and the quote is spatial cognition is a set of transformations acting on sensory input and variants in the form of things like ambient optic arrays are extracted from multiple sets of alternative viewpoints or experiences and so this paper below on the ambient optic array in eye perception kind of shows what this looks like so basically uh you have all of these possible sensory transformations from the sensory world we have this physical set of transformations and this visual set of transformations and spatial cognition is basically acting upon these transformations finding specific pathways forward so you have all the sensory information which gives you a lot of possible places to explore brain and spatial cognition filter out and leave you with a small set of alternatives that you can then explore so sensory transformations can disambiguate objective or allocentric and subject ego subjective egocentric movement these trajectories are the consequence of the sensory environment but also the way the agent moves through the sensory environment and it can kind of put things into different frames and so of course then observers or agents being observers move to different locations relative to their target and they have to be able to track that target as they move even though the perspective changes and the object changes it's kind of size and shape in the allocentric space so this little set of pipes here is you know if you have this agent here and it's moving in sort of a an arc it's going to take different viewpoints of this set of pipes and it's going to look at it in

different ways and it's going to get different information but it can always tell that it's looking at an arc furthermore it can remember all those positions and it can use those to navigate through these pipes knowing relative information relative allocentric information about the object in the environment and so this paper by rushton and warren is moving observers relative retinal motion and the detection of object movement talks a little bit about this in terms of your agent moving over time the object movement is moving over time at a certain angle and then having these points of reference where their frame of reference shifts and then they have to use their memory system to sort of align everything later when they go back to navigate it and of course we can look at very small connectomes to think about different types of plausible visual pathways so what i showed you in the last slide was this image of all these plausible visual pathways where we kind of the brain filters out all of the implausible ones and leaves us with the most likely visual pathways but in a in an anoid connectome which is very small it's less than 100 neurons we get this we can do these experiments where we can look at all the plausible to visual pathways available to the analyte and so we can look at this in terms of a two pixel system which is the sensory motor photoreceptor uh that can spatially resolve light stimuli and then the information is sent to the effector on each side and it drives behavior and then over time you get these different curves that are ipsilateral you know they cross the midline and then you have these uh more evolved eyes you know with lenses and eye spots and things like that and then so you build up from that simple model to these more complex models and you can get these more complex behaviors like mutual inhibition bilateral divergence signal delays and all this and so the reason i'm showing you this is to show you that uh spatial perception and spatial cognition can be relatively easy you don't need to have a representational system you don't even need to have that many cells and but this is also it's a hypothesized precursor to visual system so if this looks familiar this is because it is the sort of bio cybernetic approach and it's similar to breitenberg's approach to bio cybernetics with vehicles and building vehicles that uh approximate small connectomes nervous systems things like that you can also look at spatially distributed light sources which require sensory arrays internal pathways and viewpoints on the light sources so things now you get embodiment involved and body location and posture become important and we actually had an interesting discussion at one of our saturday morning neurosyn meetings on breitenberg's neuroanatomy and synthetic biology work breitenberg actually did a lot of neuroanatomy before he proposed the vehicles work and he built these he investigated fly connectomes quite extensively looking at their neuroanatomy looking at their uh their output behavior and this is kind of what led him to the vehicles model

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we're kind of working on as well and we want to fit this into this model of spatial intelligence so next is the path integration and taxis view and i say photo chemo thermal signal because those are the different types of taxes that we want to talk about here so photo taxes are light signals chemo taxes are chemical signals thermotaxis thermal signals and thigmotaxis are touch signals and taxes refers to when you have an environmental signal in that form and then it forms a gradient in the environment and the agent or the organism will follow that gradient from where they are to the highest concentration of that gradient so the toxic movements are organized not by some great regulatory mechanism but just kind of following the gradient at the next highest concentration so this kind of relates to path integration so i'm going to talk a little bit about path integration which is sort of wayfinding or homing if you are familiar with like ants or something like that where or insects where they use path integration quite extensively and it's kind of remembering space and remembering sort of the map of the space around you and then kind of going from a home location knowing how space is transformed and then moving back to the original home location so you explore space a while and then you can always find your way back home and so that's path integration it has an interesting relationship with toxic behavior we'll get into what that is so we have these two papers possible uses of path integration and animal navigation and then principles of insect path integration and so this is path integration and ants so you can see it's a very complex set of behaviors it almost looks like our second phase of behavior is the uh levy flights where they kind of move in a certain direction but they make these leaps and uh this is kind of you know what what happens in ants where they leave home they they foraged in the environment and then they have to return to the nest and so they're using their ability they're using pheromones are using their ability to navigate and remember space and remember how it transforms when they turn their bodies and then make it back home so path integration is the basis for cognitive mapping path integration will happen in other organisms as well including mammals so this isn't just the thing that you find in insects but insects make great use of this so this is the basis for cognitive mapping there's a limited use of landmarks and resetting processes so they use landmarks but not exclusively they also reset their uh their calibration tool so usually they measure footfall or something like that and they use that as a heuristic they also have reset their footfall information sometimes so this

there are all sorts of mechanisms involved in this so the information needed to compute a trajectory involves a limited approximation of distance when the target is close but when the target is farther away you need to have a little bit different strategy sometimes you can use the direction to goal which is fixing and resetting your position and then matching behavior or having a coordinate system on top of a map and having that coordinate system be the guide to getting back to home base and so there are these different ways in which information can compute a trajectory and perform this path integration now there's some very interesting connections with taxes because like with path integration taxes are cases where the organism goes out explore explores their environment maybe they're foraging or maybe they're doing something else trying to find some delicious uh meal of some chemical chemicals or something like that and then return home or return to some other place and so instead of sort of these other mechanisms that we think about we actually have these gradients that one can follow you follow the gradient to its source and you then find your reward so there's this very clear signal environmental signal of where to go but nevertheless you have to it's not a straight line it's not a straight shot to the source there's a lot of exploration and there are mechanisms you need to use to compare two different places where you visited so you know where the greater concentration of this thing is so we can actually build this interesting table which is to generate a sort of a contingency table for any one position as you're tracking or whatever it is towards the source so there are three different behaviors you can go towards the source which is positive you can partially go towards the source which is angular or you can be negative and move away from the source and so this is all based on local information comparing two different places revisited and moving in that direction and as you noticed with path integration the path is wavy it's curvy and and it doesn't it isn't a straight line and then we can use different types of in sensory information so we can use light photo we can use chemistry chemo we can use touch

figma temperature thermo hardness duro and electricity or electro so these are all known taxes and sometimes

you can use more than one taxes you could use photo taxes and chemo taxes or photo taxes and thermo taxes

and integrate them in a way that helps you to improve your performance so for each of these then you can say

if you're at a photo taxes you can move towards the photo taxes or towards the thermo taxes or you might move

towards the photo taxes and partially towards the thermo taxes and this is all based on local information so over time it will average out to this path towards the source but it does produce some interesting dynamics especially when you have multiple modalities and so that relates to path integration of course because the source might often be some sort of you know food source and then being able to return home from there is also a function of sort of following the outbound signal or maybe following sort of the decreasing gradient these are mechanisms that don't involve the type of hippocampal spatial cognition we talked about earlier but is nonetheless part of the spatial intelligence we can also look at agents and we did this interesting work with agents where we're doing this sort of dendritic

computing with agents so we had these breitenberg vehicles with antennas and the antennas have these different sensory uh sensors on them and they take in environmental information in the spatial array so you have sort of this beam form of spatial information going from left to right and it comes into this uh antenna or this array and then it encounters these transformation points before it goes in the nervous system so the way dendritic computing works is you have sort of the analogy of dendrites in the brain where you have these uh these tree-like structures where environmental information hits the tree-like structure and all of these sort of uh parts of the array interact with one another or you know maybe the ones to the left take in more information than the ones to the right given the sort of the direction of the source of the information and then that influences how it's processed by maybe like a logic gate or some sort of transformational process before it gets into the into the nervous system and so this is an interesting sort of paradigm because you can do a lot of interesting things with spatial processing you can have environmental information that is directional it could be far to your left or far to your right or it could be right in front of you and that information is processed differently even though it's the same source so if you think of taxes of course your taxes are going to be different your information is going to be maybe refined or maybe it'll be obscured by this kind of paradigm um but you know it's it's going to influence how things happen and of course this is sort of how you know if we have if we care about embodiment we're going to have this sort of array of sensors on the outside of the organism it could be a visual system it could be other sensors and so we need to consider how those are processed spatially and how that influences behavior this is another example of sort of linking phenotype to the sort of processing where the surface perimeter is very curved and you have these these dendritic trees are these arrays that sort of conform to the shape of the the body of the agent you have what we call sensory beam forming this is a an idea from engineering where you take the different orientations of sensory information and process it based on its spatial location or spatial orientation to the sensor and then to the organism as a whole so this is very different from the way we kind of usually think about spatial processing as sort of like this undifferentiated signal that then gets processed in the brain this is actually it's processed in the placement of the sensors on the body we also have sensory modality specific gradients this is from another paper i published so in this case each sensory modality forms an input factor and this doesn't necessarily need to be an embodied agent but this is like uh where we have different sensory modalities and the agent is sensing the modalities and the modalities form a gradient in the environment so each sensory modality forms an input vector to the agent a directed vector describes the direction of exploration along the sensory gradient for each vector so each of these vectors describes a certain modality it describes a direction in which it's it's located and the direction of exploration that's expected and so you can have these different vectors that are aligned or they're misaligned and so sometimes visual information is in a different place in the environment than auditory information or touch information those all have to get integrated uh either you know sometimes you'll have an angular displacement sometimes they're all coming from the same direction and so that's easy to determine the the sort of the direction of behavior expected

behavior and so this fits into this model of open loop intelligence augmentation where we can have this model of intelligence as an open loop amplifier so something like what W. Ross Ashby predicted in his cybernetics work

uh but instead of having uh this sort of model of the environment and sort of like an intelligence or an internal model we have an intelligent agent with a world model and then we have this uh formulation of intelligence which is sort of the uh sensory information that's incoming feed forward behavior the behavior that's generated and then the selection mechanism on that uh sensory processing is incoming sensory information with selection that determines your initial sort of intelligence and then there's some model of the world that you refer to to refine that and so thinking about spatial navigation in a mathematical framework we're interested in spatial navigation as this continuous set of dynamics so we can use a lagrangian to describe this we can draw this from the equations of motion so we're interested in spatial navigation not necessarily as an agent that's visiting a number of locations and looking at the pretty squiggly lines that it generates but we really want to know what the equations of motion are because the equations of motion can tell us something about the behavior of the agent

and in turn what they're doing to process information to produce this trajectory so this is a kinetic description and navigation where the segments of motion or output of place in grid cell representations for example where other types of spatial representations are put into the environment as behavior and this is an active process so as representation changes so to do the equations of motion this is a sketch of maybe what this looks like in terms of the taxes so we have an agent bradenberg vehicle it's integrating taxes from multiple sensory sources and produces a lagrangian description of this trajectory that goes through space you start near sort of a thermal gradient so you get these thermal taxes and then you leave the thermal gradient and you go to the chemical gradient and you venture through the chemical gradient you come back out to the uh ph gradient and then you venture through that you come back to the fignax gradient and then out so you explore these different gradients as you encounter them sometimes the gradients will overlap but it basically describes sort of the spatial behavior in terms of these guiding cues and then the internal mechanisms you know that's something in this model it's a black box but we can think back to some of the potential mechanisms that we might use to produce this behavior so i want to go back to sort of the internal aspect of intelligence and talk about the neuroethological view neuroethology being sort of the study of the brain in naturalistic contexts so again we revisit this idea of spatial congruence in this case we saw that back a few slides back that we had those uh sensory vectors that were maybe they were aligned and maybe there was displacement in this case you can see kind of what that looks like they train these cats to respond to signals that are both congruent meaning that a visual signal and an auditory signal are coming from the same place and signals that are incongruent so the auditory signal the visual system or this visual signal are coming from different places and then the food reward is where you get the

food right so you know it has if the signal is coming from the same place the cat goes to that food tray gets the reward if the signal is coming from two different places the cat has to sort of figure out where the food reward is is it the auditory signal or the visual signal and they have to disambiguate the two signals to get the food reward so incongruous sensory cues or feel spatial biases not all locations or orientations are equivalent so it has to sort of sort through even though it has all these orientations and rotation spatial rotations and spatial location information stored in its internal model of the environment it still has to disambiguate a lot of information to make decisions to make the right decisions without error we you know we have this model of coincidence detection spatial coincidence detection is rooted in the superior colliculus you can see a model of superior colliculus at the lower right and so these experiments are very famous for revealing yet another place where spatial processing happens related to this is the chemo affinity hypothesis this is another famous experiment where it's more about spatial framing so the idea is that you have well in speary's uh experiments on this he took a frog and he cut the eye out and he rotated 180 degrees so when the eye is in its normal position the frog can shoot out its tongue at like a fly and capture it pretty quickly and do this pretty accurately but when he moves the eye 180 degrees he rotates it in its socket the tongue goes in the opposite direction of the fly almost consistently and so the idea there is that you have this spatial frame that's based on the eye and in the way it develops that spatial frame is put into place and then when you rotate the eye you're basically shifting that spatial frame 180 degrees off the spatial representation of the optic tectum is established in development and it stays like that throughout adulthood when you shift the sensory array the corresponding reference for your shift so again we have these two cases of spatial incongruity one where you have two senses that are decoupled the other where you decouple the sensory array from the target in the environment and so that leads us to this neuro simulated morris water maze and so this is interesting because you have visual and auditory sources and they're decoupled from each other not only are they decoupled but you have multiple sources around the arena the idea is these vehicles will

travel and try to find the corresponding visual and auditory sources so there's kind of like this uh these are a great this spatial gradient of visual information and auditory information and in some places it's mixing in some places it's independent and so these vehicles are kind of exploring their space trying to fix on some source and exploring in space and so this is kind of a continuous maze if we think back to the mazes that we used for uh you know hippocampal learning this is a continuous maze and this is something that will you know maybe tell us something about how we can process spatial incongruous spatial information and very simple nervous systems and one more detail about this is that sometimes these sensors are all located sort of in the same direction you have on this vehicle here you have three sensors all pointed forward but on this vehicle but on this vehicle you have sensors that are pointed in all different directions you basically mutated them so that they're on different sides of the body and so that affects processing sometimes that leads to a periodic orbit sometimes it leads to something else and we don't know we haven't run these experiments but this is just kind of a thought experiment uh where you know we're kind of thinking about varying the sensor uh location varying the sensory

source location and then having this naturalistic behavior where we can observe gradient following and behavior output behaviors that are spatially interesting so if we have the spatially incoherent framing plus spatially incongruent multi-sensory information it can lead to release interesting dynamics so then the next two papers are the brainstem integrator for self-location and memory and positional homeostasis and zebrafish and topographic maps of visual space in the human cerebellum so this starts with a quote from this vaness paper from 2019 that's the second one topographic maps of visual space in the human cerebellum and this is uh as follows retinotopic mapping under fixation precludes a distinction between purely retinotopic or eye centered craniotopic or head centered spatiotopic or world center and other spatial encoding schemes this is in line with what we've been talking about we've been talking about different types of sensory centered uh processing head centered processing head direction cells etc and then this allocentric view of the world where things are spatiotopical we have this uh retinotopic mapping that in humans at least or in mammals is precludes these distinctions and so this actually points to the cerebellum is another site of spatial processing the cerebellum encodes multiple polar coordinate maps of visual information that provides information to the effectors so now we're not talking about the sensors we're talking about the effectors so now this is the outbound information and the cerebellum encodes these maps for that outbound information to be transformed okay so now we're dealing with embodiment we're dealing with the consequences of embodiment

the cerebellum the cerebellum is known for producing smooth movements it's involved in motor control it's involved in smoothing out motor control and having these predictive models is very unique architecture allows you to do that and o'keefe and nadal of course also had something to say about embodiment which is that the movement of an observer's body changes position but does not alter the position

of objects this is distinct from relative position the body changes position of course as you move it doesn't alter the position of objects though because you have sort of a framework for thinking about different transformational views on objects so if you're looking at an object from the front or from the back or from the side you can put that together and know that that's the same object this is a graphical abstract from this yang paper and cell this is a zebrafish it shows that the cerebellum and the inferior olive are involved in this location error signal and then it's also involved in determining velocity and also then you know the zebrafish is moving with a change in position so it's locating the error signal it's correcting the error signal and generating a certain velocity that changes the position in an adaptive manner and we can think of this as a control system where we have this input that leads to swimming behavior so we have the neural coding the behavior and these brain-wide circuits that control this centered on the cerebellum the inferior olive and so we can think about this in terms of a map so we have our cellular cerebellar map it's taking in information from different parts of this organism so we have the a lot of things in the head that are mapped to things in the hippocampus and then those are produced outputs for those things so we have output behaviors uh being uh mapped from the hippocampus to output behaviors um and this is very much like sort of a homunculus map that you see

in the neocortex um and so you know we can actually replicate this in a vehicle we can have this little area here where it maps sensors from different places to these maps and then an output to these effectors so you have inputs and they get mapped to different cell populations which have uh sort of a mapping to a specific effector output so in this case in vehicles or wheels and you have these uh sensors it could be light sensors or vibration sensors bringing in information and so you know we can kind of classify our different uh inputs and our different outputs and this is kind of what you see with the topographic distribution of output neurons in the cerebellar nuclei and cortex to somatotopic maps and primary motor cortex so this just shows for different uh effectors like hind limb or proximal forelimb or distal forelimb there's a different distribution of cells for each of those types of output in the cerebellum and so looking at this more closely you can build a start to build a network from these different uh cell types it adds self it adds a functional layer to these cell types networks and vehicles and it gives us this ability to map things outward as well as from the sensory system inward there's this core sensor effector mapping where you get spatial biases biases in these interneuronal networks you get information exchange among spatially sensitive interneurons and then peripheral processing networks which are shown in the different colors here are specialized sub networks for timing differentiating and integration of activity so what we're trying to do is build a very small nervous system that replicates some of these findings the next i'd like to talk about these two papers the first is omnidirectional sensory motor volumes and electric fish and then spatial planning with long visual range benefits escape from visual predators in complex naturalistic environments so this first example shows the weekly electric fish the weekly electric fish are interesting because they send out this electrical signal around their body and these descends prey in an environment where they can't really see anything it's a very murky aquatic environment and they have this is what they call active sensation so the electrical signal that's sent around their body from the electric organ near the head to the tail where there's a receptor is this electrical field this volume and so uh in the first paper that i talked about this they've uh reconstructed these sensory volumes around the fish at different orientations and then these dots involve sort of prey that are in the electrical field so that the fish can sense the location of the prey so it can move towards those prey and capture them so you can see that the prey are spatially located spatially uh segregated within this volume um and then the fish have to make a movement towards that and of course there are different different perspectives on this volume so if you look at the side view or the back view you can see that there's a different distribution of things and so one of the thing other things that they explored was this habit-based versus plan-based action selection so once the electric fish senses that they're prey within the volume they can go and capture the prey but they have these two strategies called habit-based and plane-based action action selection basically they've did these simulations uh where you could look at you know what was involved in this and and i'll just

tell you right now that habit-based is not predictive and plan-based is more predictive
and so habit is based on a model-free approach so habit is where behaviors generated by state action values
are not related to immediate outcomes they're not adaptive they're just state action values and there's
no real prediction um and then planning is a model-based approach where behaviors are best based on
action
outcome knowledge structure so a knowledge structure that helps prediction future states and expected
outcomes
are modeled and this is very adaptive so you know you know kind of what your you what future states to
expect
you know what outcomes to expect and it's much more predictive than the habit model which is kind of just
output of behavior and so these are interesting because we talked about in last year's talk by our group
about the role of habit and planning
in sort of agents
and and thinking about uh cybernetics it's what they call goal-directed behavior and so this is an important
distinction that
that is also important in the spatial intelligence paradigm finally we want to talk about the view
from embodiment in g slam g slam stands for generalized simultaneous localization and mapping so again
we've
talked about embodiment we haven't talked about g slam and so we'll talk about two paper g slam papers
one is by adam saffron and colleagues generalized simultaneous localization and mapping
g slam is unification framework for natural and artificial intelligences and then this other review on a visual
slam for
robotics evolution properties and future applications so in these papers basically they think about
um spatial processing as the sort of control system you have a control mechanism u sub t discrete time
steps t sub i and then sensor observations o sub t so you have these three components
elements then it leads to an agent state x sub t environmental map m sub t and that leads to spatial
representation and behavior so it kind of looks almost like a reinforcement learning model with input from
previous states the environment and then it gives you the spatio-temporal representation but of course there
is
this aspect of the classic sydney harris cartoon where he has he's very vague in step two and he
says then a miracle occurs and so we don't necessarily know what the filler is in this between agent state
the environmental map and then defining those and then what what leads to spatial representation and
be spatio-temporal representation and behavior it's a it's a hazy box maybe you can see someone inside of
it and i think the things i've talked about in this talk at least from the bio biological standpoint can help
but we can also think about saffron's idea of latent slam and this is a basis for map updating so we're
not totally lost here in slam we can use these kind of latent spaces to help us simulate spatial
behavior and then things beyond that but uh latent slam is an evaluation between currently represented and
novel experiences there's what they call view pose matching based on place cell and grid cells so they
use the place cell grid cell model so when current observations match information then loop closure

and the representation happens and that's confirmation that that observation is in the representation in the weight and space when no node can match existing information in the representation new nodes are inserted that's this graph relaxation so basically what happens is you have this latent space that grows with experience and so you know you either confirm that you've been you have this this experience of the weight and spaces can or this structure of the weight and space is either confirmed or disconfirmed with environmental information and then when it's disconfirmed you add nodes to represent that so next time when you encounter something similar you have that representation in place and so a couple words about the loop

loop closure algorithm this is from introduction to loop closure detection and slam this is where you have a form of error accumulation that occurs similar to path integration in animals and so in path integration as well as for this form of error integration you count the steps along a path and when a landmark recurs the map is

updated to reflect that information and so if we we state this kind of in a pithy way we can say keep wandering until you find a loop closure wait a minute i've been here before so you wander down the beach and you keep going and you find a loop closure until you kind of go all the way down the beach and back to another point that you've been to before so there are connections between this kind of weight in space approach and a cybernetic system so latent slam attacks attempts to recapitulate a neuroanatomically interesting weight in space model latent space allows us to compress a full state space where we don't need knowledge full knowledge of every state in control theory state space models serve as a path to dynamical systems like we talked about with the lagrangian and and all this about trajectories in a cybernetic formulation however we can build upon latent slam by using a state space a growing network and causal relations so we can actually formulate a solution to this in the form of a cybernetic model and this is something i think we also talked about last year we'll get to this in a minute but this is an example of graphic relaxation algorithms so this improves your eulerian path estimates for your network structure by checking if a shorter path can be found through alternative edges in this case adding edges so you explore this network you see what the path lengths are for each each traversal of the network and if um you know if it if it can be found great if not it adds a node and you have this shorter path but this is kind of the way you can do this in a cybernetic context these are these complex referential or quasi-causal relations for different control outcomes this is from last year's symposium where we propose this uh cybernetic model of feedback loops that produce not just the past present future but also alternative futures and complex outputs associated with all of them so you can model a feedback loop as sort of um going back to the future basically um as you have a first order closed loop feedback context and then it produces a complex output with a lot of different causal uh consequences and and with a lot of sort of uh causal implications but then you also have these alternative futures that can be generated from the same system another way to think about this in space is you have these teletubbies that are you know there's one in the present maybe they were here in the past here in the present and then you have these alternative futures maybe a future state over here and an alternative future state over here there's a spatial location associated with egypt the

cybernetic model recalls douglas hofstadter's i am a strange loop where in chapter four he talks about goals and loopholes and he talks about how self-applied operations can produce these circularities that are very interesting not only from a causal standpoint but from a spatial standpoint as well and so we can also think about ergodicity and how that can also contribute to the sort of spatial learning so ergodicity is this idea that if you visit enough locations in a space you can produce a mean path that describes every location in that space so the mean value is something that gives you full spatial knowledge of that domain and so this is a paper where they're using a robot to learn all the possible states of a movement path the mean path is described by robotic systems this mean path is optimal and this can actually help assist people with disabilities people who can't move optimally in space if you have a robot that can sort of explore all the spatial locations in maybe like a movement space like a workspace or in a room or something like that then you can assist people when they make mistakes and help them cover up their mistakes through knowing sort of what the mean value should be at any location so again our graphic here in our pithy statement if you want walk down the beach enough times you can predict a path that matches the lap of the waves from a mean value so you can actually track the path of the incoming tide and walk right along the water's edge if you've walked along this beach enough this brings us to this idea of an ergodic ensemble which is where if you sample a system enough times any given ensemble is equivalent to the time average that's sort of the formal definition learning increases information but an internal representation may provide ergodicity so an example of this is for native language speakers a language tends to be ergodic versus english as a second language speakers where the language tends to be non-ergodic and this is from a study where they looked at native speakers and english as a second language speakers you have to build this strong internal representation characterized by sort of native language ability when described in bayesian versus statistical mechanics terms so ergodicity is usually described in terms of statistical mechanics individual cases are not equivalent to generalized maps and so this is something interesting about this that we can use this kind of ergodic example we can use ergodicity to describe spatial navigation but we also have to take individual experience into account sometimes um spatial navigation for one person is not ergodic but for a population it is so what have we learned uh the first thing is that a representational system that is anatomically explicit like such as the hippocampus and evolutionarily conserved so the hippocampus invertebrates provides the hardware for quick efficient spatial learning so the longer you have that structure around evolutionarily the more refined it can be the more ergodic your you know if you acquire something in development then it's you know it's ergodic and you have the hardware to carry that out if you have to use sort of a an expedient neural architecture for this to learn it or it's not something that is typically learned from development then you're not going to have as efficient a spatial learning number two optic flow mediates spatial perception in terms of both the body egocentric and environment helio centric and so we have these different frameworks for the body and for the environment optic flow mediates spatial perception in that way viewpoint representations or frames of reference also depend on sensory specializations established in development so technically the sensory specializations are established in evolution but you

can

have these things play themselves out in development and they influence the viewpoint representations across species and how you know for different sensory systems for different uh neural architectures they actually mediate spatial learning number three path integration produces dynamical trajectory with complex parts so this is again path integration another form of spatial intelligence where you get this dynamical trajectory where you can decompose the parts of the trajectory it tells you something about in you know uh periodic spatial intelligence or at least the behavior of spatial cognition and so path integration results from sensory and internal information as well as processes of regulation and prediction number four a body can be characterized by a sensory manifold while sensory motor integration acts as an immediate

source of spatial bias so again we have this the body plays a role and then sensory integration also plays a role

and so sometimes our body is getting visual information from one location and auditory information from another

sometimes our bodies can act to bring those together sometimes not and so this is an immediate source of spatial bias

action selection operates on the integration of immediate or longer term information so we can have a simple model

of immediate uh sensory information like in the habit case in the electric fish simulations or a longer term model of planning

uh where sometimes we can use a world model like we talked about in the uh intelligence augmentation example

where you have this model of the world that you draw from so it's a longer term context number five unified models such as g slam

serve as a framework for embodied information processing this unifies the role of hippocampal path integration

circuit optic flow and the spatial bias of sensory reception so g slam really unifies a lot of these mechanisms into one framework and then six which is an additional thing and this is kind of more of an insight

enough free or random navigation can produce a good enough cognitive map by relying on an ergodic ensemble

of spatial information so if we just let our simulation run forever our agent explore forever it can produce a good

enough cognitive map it doesn't necessarily need a lot of architecture but the architecture deals with a lot of these other aspects of structure

that random or free navigation may not solve and so again with the neuro we're dealing with potential brain centers connectivity mechanisms for spatial representations and cognition across animals

the simulate we're using boxes and arrows or cybernetic type systems spatial distributions of sensory information

mathematical functions that describe behavioral output simple bio cybernetic agents and models of sensory processing from the literature

so in future directions we need more complex models more exploration of causality and systems context
and systems context expansive and reconfigurable networks and of course more synthesis of all these ideas
thank you for your attention
thank you for your attention

thank you for your attention

thank you for your attention

thank you for your attention

Thank you.

Thank you.

Thank you.

Thank you.

All right.

We're back with our next session.

So I will let Stephen introduce the speakers.

And thank you all for joining.

Go for it.

Okay.

Thank you.

Daniel, thank you very much for having us here.

This is a wonderful opportunity because we all come from different backgrounds, but we're very much united in that active inference has a tremendous potential to help more people actionably in the relatively near term than we think is realized.

And we've been coming at it from different ways and learning from one another.

So we appreciate this opportunity to share this.

Giacomo, I have to tell you, before we started talking with this project, your work on light touch and contextual work is something that I have quoted and that I believe has a tremendous relevance for my profession.

I'm a chiropractor.

You're a PT.

Jim, you're a body worker from a somatic point of view, massage, but you're taking Marco Blanket in a whole new direction.

And I'm looking forward to seeing your presentation on Friday.

And Dave is our academic that's going to be leading us.

Dave Newell is at UECC.

He's, which is one of the premier chiropractic institutions globally.

And he's working with a new group of pain inference, of active inference focused on pain.

And that's basically the name of the group.

And we look forward to Dave's work pulling us together.

So with that, Dave, take it away.

Thanks, Stephen.

So I'm going to attempt as far as I can to share this.

We're going to try and make this relatively informal conversation.

We have a few sort of kickoff questions that we're going to try to use to unpack each of our guests here that Stephen's outline in terms of their background, their interest, how they got to this particular point.

I think Stephen and I, but I think the two of us did watch Carl's plenary introduction.

I must admit, you know, listening to that, I firmly put myself in the developing category

when it came to active inference.

And I've had some interest in this for a couple of years and from a particular background before that, which we'll probably get into and all of us will.

So having said that, you know, we've each come to this in our own particular way.

And some of the questions that I ask and I suspect perhaps some of our participants here will ask as well will be somewhat rhetorical.

And so we'll do our best to make it entertaining, perhaps informative and see how it goes.

So I'm going to really sort of kick off with each of you, really.

And I guess I'm going to go I'm going to go with Jim first and go back that way through Giacomo and then Stephen.

And, you know, what was your journey?

Tell us something about your journey in terms of being interested in active inference.

Why it struck you as a framework that excited you?

You know, what's unique about this?

Because it's pretty esoteric.

I mean, like it's not all over Facebook and the national newspapers.

So it's still, you know, relatively outside of those people that are really into this, you know, a rather niche idea.

So so how did you get into it, Jim?

And let me let me try to answer that.

So first of all, Stephen, I'm not sure.

I just spoke to Daniel online by email.

I think that my presentation, which is a recorded video that I finally updated, uploaded last night in a fit, in a rush to make the deadline.

I think it's going to be available at 2230 UTC today.

But so I don't I don't know if I'm presenting on Friday, but it would be a video that's already recorded.

And I go into everything in depth there, of course, it's about a 50 minute video.

And so my background is I'm a my academic training is in his historic history and social theory.

And and then my my profession or my love really is touch.

I just love touch.

And I'm one of those people.

And and and I'm a deep tissue structural body worker.

And I've worked with the fascia with the my focus was in the deep core anatomy.

As I worked on people, I came to understand that everyone is really dealing with what I call postural collapse syndrome, which is a failure of the postural, the core anatomy, the postural system.

And so I've spent the last whatever 10 years really analyzing that.

And when I read Sherrington, so and I'm a geek, so I'm reading all over the place that I'm trying to understand interoception at first, which I got from somatic psychology.

And I thought, oh, everything is interoception.

Then when I read Sherrington, I understood that that that paradigm, that that whole notion is completely wrong.

And it's proprioception.

And after that, I connected with with Carl Fristin's work on the topic and the work of the active inference community and the whole idea of the Markov blanket and a model of life.

Now, Sherrington was the one that pioneered that neural architecture.

That model of life in 1906 was never developed until Carl Fristin really took the bull by the horns.

And and I think that having a model of life is such a powerful thing, a perceptual model.

And it's a spatial model.

So for those of us into somatics, the spatial reasoning is really kind of the way we work.

Rather than a top heavy kind of intellect led reasoning, it's more of a ecological kind of openness to the ecological affordances that are provided.

And and those are deep within us.

So so I got really excited.

I just love the I mean, you know, complexity theory, the cybernetics and and and as a historian, my focus has always been on what I call the somatic tradition.

Two books were published in 1543.

Copernicus was a mathematical one.

And that defined the mathematical tradition, which has become ascendant today, of course, quickly became ascendant, starting with Newton.

But the somatic tradition throughout history has been a vital one.

And that is asking a different set of questions than the mathematical tradition.

It's about life and complexity.

And Vesalius wrote his Fabrica, the fabric of the human body, also published in 1543, the same year as Copernicus.

And that for me began that somatic tradition.

The only way to understand complex living structures is through complexity theory.

And I think that active inferences.

It's just definitely the most exciting and comprehensive and really profound expression of complexity theory that I've come across.

I'm a big fan.

That's kind of my background there.

Brilliant.

Now, that's really interesting.

I've got a couple of things.

We'll come back to that in a moment.

But we'll move on to Giacomo, I think.

Yes, thank you.

Yes, I'm Giacomo Rossetti.

And my background is from physiotherapy.

I'm both a clinician and also a researcher.

And during my journey in research, I have had the possibility to investigate and learn some points from the contextual factors, placebo and nocebo effects.

And through this journey, I have had the possibility to read some points regarding the active inference.

And I find this proposing theory really interesting because it fits well with the real-life clinical practice because it can help us to explain the unexplainable things that we encounter in the clinical scenario every day.

So, starting from placebo, nocebo and contextual factors, we can conceive these elements as strategies that can have an impact in terms of active inference and can modulate the perception of patients and also have an impact in terms of therapeutic outcomes.

I think this is a really interesting theory that we should develop and investigate well using networking and different perspectives from different fields.

Brilliant.

Thank you, Giacomo.

Thanks, Giacomo.

Stephen.

Stephen.

I'm a doctor of chiropractic who practiced for 26 years.

And in my somewhat schizophrenic profession, I've played a number of roles.

But as an undergraduate in college, I was between psychology and biology practicing.

I spent time in Prague with Vladimir Yanda, if anyone's in that world, upper cross, lower cross syndromes with words that he created, concepts that he created.

And also, I was into yoga.

And in my practice, I realized that simply manipulation by self was not fixing the problems once there was a motor pattern that occurred.

And so, we came at it from a biomechanic and yoga point of view with a little bit of mindfulness.

I began lecturing in 2003.

And then in 2011, the course that I've been teaching was posture and motor control exercise integrated with chiropractic, mobilization and manipulation.

The book that caught me was a book by Bud Craig, How Do You Feel?

A Neurobiological Moment with Your Interoceptive Self.

Bud Craig opened the door to interoception as a sense that was separate and distinct from proprioception.

And within a lot of the psychological research now, the mind-body research, interoception has become a big topic.

And when I started looking at interoception, then I ran into active inference.

And it's the same grounding because interoception is one of the inputs for active inference.

And it's something that is not being realized in my mind in just about any clinical way.

Giacomo, I believe some of your work is one of the few that are touching on that linkage from a somatic point of view.

And I believe that there's a tremendous amount of good that can come from creating agency on the part of people, giving them the ability to have self-efficacy and to harness active inference as a framework to be able to systematically import that to people in different clinical situations.

Back pain to psychological issues.

Thank you, Steven.

So me, I guess, I sort of have been teaching chiropractors for many years.

I'm now part of a wonderful university called the Health Sciences University in Bournemouth in the UK that teaches many healthcare professions now,

including all of the manual professions, physiotherapy, osteopathy, chiropractic.

We don't do massage as such, but I have a PhD student who runs a school actually in the UK who does massage.

And I think over the years, I've become interested in the mechanisms behind how therapy encounters with manual therapists,

you know, are just doing the thing that they're doing.

I mean, I think there is there is there's now, you know, plenty of evidence suggests that visiting a manual therapist has really positive outcomes.

But I think one thing has been missing over many years, very frustratingly for scientists like me, is exactly what is going on.

You know, people do all sorts of different things to patients, whether it be manipulation,

whether it be some form of touch, whether it be exercises, joint mobilization, all sorts of different things.

And we know that that things like, you know, exercise from exercise to talking therapies, to manual therapies, to a whole bunch of other things seem to work relatively similarly.

They work as well as each other in many ways.

And so so I got really interested into why that was that was happening.

You know, what are the mechanisms going on?

And through that, I obviously came into the world of placebo and nocebo like Giacomo and found that somewhat frustrating

because there's so much historical ladenness that goes with the language of placebo and nocebo.

It's a sort of uphill battle to get clinicians to listen to you when you're talking about placebo and nocebo in many ways.

And and out of that then segued into contextual factors, which are these these these things in therapeutic encounters

that can essentially change the meaning of what patients are expecting to happen.

And and really then stumbled across.

I think I I don't think this is right.

Remember, there was a paper by Angara on Kapchok at one point who talked about predictive processing as being a.

A particular, you know, potential mechanism behind placebo.

And I think from that sort of just fell into the active inference framework community and, you know,

was so enamored by the breadth of that framework.

It's parsimoniousness, the potential width of things that it could start to explain, you know, notwithstanding that it actually had a you know, it told a story of a mechanism of life, let alone clinical interactions.

And and I was so enamored. We just set up a center of pain and active inference research at the university.

So that's how I am now. That's how committed I am.

And I think this framework. So. So, yeah, very similar stuff.

And so one thing I would say would be and I think this again is one of these rhetorical questions.

You know, why do clinicians in your your viewpoint need this rather esoteric new?

45000 foot view of life framework, you know, because I'm sure in all of your professions, you know, they will say, well, you know,

this is all very well and good, very interesting, but, you know, you know, my patients are getting better.

What would you say to that? Anybody?

There's a book by Simmons and that begin with why.

I believe that active inference is an underlying why for a lot of different things that has not been effectively articulated nor communicated.

I would say very much that the effort to talk about it from Carl Frescan's point of view from the prior presentation today, that are things that I don't know a lot about personally, but there's congruences.

And I think the more that people in different professions can look at those conference, look at those and pull those truths together addresses some of the issues that I believe are coming for the active inference community, which is trying to spread wider.

And so having a common why for different schools of thought like we have here within one trench manual therapies and making those congruent with other schools of thought to become more of an accepted truth is this is the way life is.

I believe that's because that that's the underlying to me fascination and promise of of active inference ultimately.

Yeah, so unifying mechanism.

Yeah, so unifying mechanism. Really, that's I would agree with you. That's tend to be what I found.

Actually, segueing out from that or going on to, you know, using that as a platform, then can each of you and I'll start with Giacomo.

Can can can can each of you sort of then tell us a little bit about how it's informing your own work, particularly because I think you all practice still how it's informing that on the ground.

Practice focus people in front of you, real world stuff.

And and and, you know, because I think that's really important.

It's so easy to talk about the sort of high level stuff that we're talking about and listen to people talk about that, you know, even not not that I can do this.

But Stephen and I were talking about this, even the sort of all the maths behind it, which completely discombobulates me.

But the you know what what is going on on the ground?

So Giacomo, give us give us an idea how it's how it's you know, what's happening in your practice.

Yeah, I think that active inference helped me, but also all the other clinicians for many reasons.

First of all, it can symbolize an overarching framework that help us to explain why different treatment strategies can help and have an impact in terms of therapeutic outcome.

In our patients independently from which kind of treatment we decide to apply.

But it works as a general framework that helps us to understand what happens in the clinical encounter.

And as a second reason is that help us to explain why sometimes the different treatment solution that we apply with our patient doesn't work because there is a great complexity in the real life that is somebody unpredictable.

For this reason, we need a framework that helps us to explain what happens in the clinical practice, because, you know, when we treat the real patients that sometimes we are challenged because the patient has a very strong prior, strong belief that are really difficult to be changed.

So in this, using this model, we can, for example, use the different element presented in the therapeutic arena, named as contextual factors, in order to modulate the perception of patient and help them to challenge their prior, using our words, their rituality, our techniques, our verbal and nonverbal communication.

And all the other cues presented in the therapeutic encounters that can be combined with the specificity of our techniques.

They can act as a boost that have an impact for our patients' perception.

So in this reason, and for this reason, I think that the active inference can really help the clinician to understand what they are doing and also understand the patient response in the clinical practice.

And for that reason, there is a strong need of more research and more awareness among clinicians regarding this topic.

I think that's a good thing about this topic.

Yeah, that's great.

I'm going to go to Jim next.

Hi, yeah, well, you know, the title of my video, my presentation is Exteroceptive Bias and Somatic Belief Updating in the Global Model.

And so I'm going to.

Unpack a little bit of that for us.

Oh, I'm going to.

Exteroceptive bias, then, is an innate natural setting in the system.

If we think about the receptors on the outside of the of the of the organism.

Right.

We have distance receptors, touch receptors, taste, smell.

These are distance receptors.

And we the part of the brain that's associated with exteroception is also the surface, the cortex of the brain.

And interoception then would be the viscera.

And then that would be the insula of Bud Craig's famous inside there.

It's the insula.

And but but for proprioception, it's going to be at the at the root here in the cerebellum and the spinal cord.

Now, so exteroceptive bias is just the situation in which the organism is running its outer sensors.

And as soon as something is born into the world, it has to immediately look for warmth, for shelter, for kin, for food, for safety.

And that never stops.

So there's a certain innate bias in the system.

There are an awful lot more inputs coming from the outside than from deep within, for sure.

But exteroceptive bias then can become a control bias or sort of a vicious cycle, a positive feedback loop.

If we're repressing the inner sensibilities, which it's the postural system.

So it's all balance.

It's all integration.

It's all stabilization, centering and grounding.

These are the kinesiological functions of the system, but they're also the cognitive functions.

And without those, then we can jump into, you know, basically, we're just thinking too much and we're stuck in our heads.

So when I sit down, when you sit about my work, right?

Like for me, a classic example of my work is like, I'm over at the cafe and someone sits down.

And, and, you know, they're talking about their pain or their issues and everything.

And it's like, let's get out of our heads here.

And I say, come on over here and sit down and let me work your arm or your hand, or let me rub your shoulders or do this in their coffee shop.

And, of course, you know, with consent, but, you know, getting out of our heads is sort of a huge, important theme.

And as manual therapists, that's our specialty is coming into the body.

And as a manual therapist, I see my work both theoretically and also in a tactile sense is as a somatic educator, helping people to understand that interior perceptual system and come into their bodies and have that grounding and centering that's available to them when they need it or when they want it.

You may not want to be doing your grounding and centering if you're in heavy traffic, but you definitely don't want to be dysregulated internally either.

So there's, it's that, that underlying system is always running.

But what I think we're seeing in the world, the surprisal that we're seeing in the world is overreach.

We can call it technological overreach or they get this anthropocene and that the society that we built is really very, too top heavy, too intellectual.

As organisms in the world, we need more balance.

And so that's kind of my focus as a somatic educator, as a person.

Yeah.

Can I say something?

Yeah, go on, Stephen.

Yeah.

Anyway, a point that I believe to making active inference possible is that we're seeing a lot of clinically actionable in the field on the ground with clinicians is making it something that is what you said before, Giacomo, a boost to whatever else we're doing.

And I would even amplify that by making it a synergy because it's not just, I think, 20 percent better. I think it's more like potentially 200 percent better.

It can make a dramatic, dramatic difference because it can not just be in the moment of the therapeutic encounter, whether you're doing a manipulation or I am, however we both do it.

It's connecting with them as a person.

And Jim, I think this comes back to what you were talking about.

Yeah.

But a point that I think needs to be made that ironically enough, Carl Friston said earlier, what he said was how we see the world and how we react to the world is what he was talking about active inference being.

And when he was talking about seeing, I don't think he was talking about visual, he was talking about the gestalt of the self of the person.

And my add to that, my riff on that would be how we somatically and perceptually feel the world and how we react to the world.

And to that, in terms of making it actionable, something that is a very much an interoceptive insight is that it needs to be moment by moment.

It can't be a cognitive thing where now you know that.

So you do it.

The interaction needs to be now and now and now very much correlating the contextual communication, which David, I like.

I teach things using something that that would describe, but I know I've not used that word in how I lecture, but that's exactly what makes sense to do, because contextual means in this context, this is what I'm doing with you.

And the synergy between feeling that on a somatic basis, both proprioceptively and interceptively and they're connected, but they're not the same.

It has a much greater salience from a sensory point of view and less relevance in terms of shifting behaviors.

Yeah, that's fascinating because in a way, I think we've all come to this in different in different angles.

You know, I think one of the things that has frustrated me with with with the profession that I've been most associated with is chiropractic.

But I know it's also something in osteopathy.

I don't know much about body work and massage, but I know that there are similar conversations going on.

And and I think there is something similar, perhaps in terms of physiotherapy as well.

And what I guess what I've seen through the lens or the telescope, even looking at the wrong end of the telescope that I've seen is.

Too much body centric explanatory frameworks.

Now, what's really interesting about this discussion we just had here is and I would say that the Giacomo and I probably, you know, if you're going to look at a spectrum, sort of a Cartesian spectrum between mind and body with mind at one end and body at the other.

I guess the Giacomo and I have become very interested in the mind stuff.

And I know that, you know, we've got some colleagues, George Estevez and Mike Thacker, Mick Thacker, that will that will very much be Steven and Jim's end of the spectrum, which is which is very body based.

But there's something about the body based now, certainly the manipulative professions, perhaps where, you know, there's almost a complete not a complete but, you know, a tendency to sort of not worry so much about what people are thinking.

I think that's that's why we've headed in was certainly why I've headed in that direction.

On the other hand, there's something about a body focus in a different way to find the illusion.

I think the old traditional explanatory frameworks are around, but you find a lesion.

It's very biomedical. Find the lesion. You fix it. You get rid of it.

You know, if you're if you're a surgeon, you take it out. If you're a manual therapist, you fix the lesion in some sort of way.

But that's not the body nest that you and Steven are talking about.

You're actually talking about a very different body.

But would you say that all of us are actually trying to find some position in the middle where we feel that we can get these professions that get these clinicians to just be more balanced between these two things to get real holism in a way?

I used to think that back in the 70s and 80s, the word holistic came into vogue and I despised it because it quickly became a mantra for selling little slacks of snake oils and gadgets and gizmos.

I disagree with the question. It's not mind or body.

It's not it's not even mind, body or body, mind. It's one gestalt. It's one thing.

I don't care how much you're focused on your mind. If I drop a brick on your toe, your focus is going to change quickly.

And that's part of being human in a moment by moment basis.

So from clinicians using that as leverage, using that knowledge as leverage and making what we're doing with them something that is giving them agency that they can understand in their in their language that correlates moment by moment.

And leaving them with a psychological feeling of success and positive response.

So, yeah, I would agree with that.

Yeah. Jim, you had something to say. And then Jacob.

Thanks. Thank you.

I wanted to say that I think that probably for me at least the singular contribution of active inference, which is also I point out in my talk and I go into it very extensively,

uh, Sherrington's original, um, his probably his greatest achievement is his, the model of life, the model, a neurological,

he calls it the neural architecture of the animal as a whole, the perceptual model of life that is scalable.

And it's scalable from, from abstract systems like the Markov blanket to the very intuitive example of the cell to the vertebrae body.

And, and, and, and other kinds of systems, but, but it's certainly makes.

Hi, Daniel.

Daniel mute your mic.

Um, so, um, the model of life, I think is, is, is a way to look at things.

Uh, so it's three perceptual systems with three different fields in the body, the outer surface, the inner surface, interoception, and then the core proprioception, all of which are minded and embodied at the same time.

the same time and this is just a different way of seeing things i think it's probably the single

um most exciting and powerful insight um in addition to the notion of active inference

itself a kind of a active inference then takes the mind and the body and brings them together

it's active and inferential at the same time you can't break apart the sensory motor loop the

perception action cycle and this is a real shift in thinking so that that when we touch we're

touching a person we're we're touching a person in different ways at different levels with with

boundaries set up within those levels so that i can be touching you more superficially or touching

you more deeply i can be touched more superficially and be touched more deeply i think it's an exciting

massive i think i think historically speaking it's a civilization level paradigm shift yeah i agree

i do agree yeah yeah no thanks to you jackama yeah david i totally agree with you regarding the

challenge between body and mind theory because we should also and always consider that the things

are different if you you are living in the lab or if you are living in the clinic so if you are living

in the lab but you can ask about something related to the body or to the mind but if you are living in

the real life clinical practice and you interact every day with a patient you know well that your life

is composed by moment my moment interaction where you apply some tools that are related to your clinical

background and expertise such as manual therapy exercises physical therapy agents and concomitantly

you modify the context using different types of touch for example modulating different types of

verbal communication and non-verbal communication in order to synchronize your body and your person with

your patients in front of you so in this if you consider this real life situation we should stop to

oscillate like a pendulum from one extremity to another but we should try to be balanced

in order to strength both the positive elements of the different theories and in this sense i think that

active inference help us to have a general framework that can help us to explain what really happens in

the clinical practice so i think that if you are living in the clinical practice you encounter different

challenges every day and the models the framework of active inference can help you to

give an interpretation of meaning of what the patient present in terms of pain and dysfunctions beliefs prior

and try to challenge them no that's that's great i um one of the things that i thought about is you

know that there is a sort of rhetoric that goes with um you know an integrative approach to whether we call it

holism whether we call it integrated medicine or integrated approaches and it feels to me that you

know um like you all of you have said in different ways that activate and i agree with jim i think it's

a paradigm shifting idea um and of course it's got a long history it's not you know as much as carl

has been the sort of contemporary uh horseman that's take this forward it's obviously come come all the way

you could say it goes all the way back to plato's cave and and and you know for sure yeah exactly but

but the um so it's interesting what you were saying jim and i think the because you said everything is

minded i like that phrase and um that it doesn't speak to this sort of hierarchical um models that all

hierarchies i know that um make our colleague um and uh i don't know what carl and the guys would think about this

but i know that there are other authors that have suggested that you know that everything all from cells upwards have a markov blanket and they also have a model of themselves and they act you know within their own

environment to confirm their own existence by by by generating that model and acting in such a way as to keep that generative model going so you know it sort of gives the idea that there is a sort of mind or a model at every level and i guess what we're doing we i guess we've got somewhat interested in different levels of the hierarchy of those models perhaps you know we could say you know um we're interested in prize because we're interested in and sort of high hierarchical models of meaning and culture and what this really means and but yet there are all these models all the way down i think you actually talked about the core being a being a sort of model of its own that had its own markov blanket and so on and maybe maybe yeah maybe talk to that and i know stephen has got some similar ideas as well so you know

pitch in so i um i love that uh thank you for that introduction um the thing that i played around with um in my talk is the idea of of a double blanket theory now we know that there are two blankets a model has to have an internal model that you know there are two blankets in the model but the inner blanket tends to be um subsumed by by just the calculations of of of its parents and and and um and siblings and stuff and then i don't know much about statistics i think it's fascinating but i'm a historian um so what i the way i play around with it is i i call it blanket politics and the inner self self has now interoception is a little different it's a little more amorphous but the proprioceptive it's a very concrete sense self-identity it has very concrete intentions and goals in a concrete view of life that's different from the exteroceptive the outer blanket the inner blanket and the outer blanket of course are integrated profoundly integrated in terms of of proprioceptive priors and exteroceptive you know sort of um maybe posteriors or actions there's no distinction but at a certain level um it's possible for there to be blanket politics and i think we can talk about that in terms of uh fristen's uh disconnection hypothesis but also i like to think about it as a cultural theorist in terms of um the double bind gregory bateson's double buying and from there starting to really think about things like like psychopathology and schizophrenia but the point being that that we really have different selves and there are different agendas involved and if we're really too and maybe too internal but that's rarely the case far too external then that can set up a kind of a um situation uh like the double bind and as i call blanket policies i want to say one thing about the relationship of the mind and the body and just throw this out there for me the relationship of the mind and the body is like a rider and the horse except that you're both the rider and the horse and the horse knows the rider the horse knows where to go two o'clock in the afternoon we're going up the mountain trail to the spring 11 in the morning we're going downtown to go shopping and if that rider has a heart attack or falls unconscious the horse is going to turn around and head right back to the to the paddocks um to get fed and and and the horse so the mind knows the body the body knows the mind

as manual therapists we're really not fixing people of course we're fixing people i fix people but if we want to really get um get into the semantics the body heals itself and the collective body the collective society also heals itself the point is how to get out of top-down control and start to really post you know when you post on a horse you get into the rhythm of that canter and the whole thing works smoothly how do we really find that um that connection with our patients and enable and help and assist the body to heal sometimes there has to be an excision a surgical excision but in general what we're trying to do is really facilitate the self-healing processes of the organism yeah that i what comes to mind is before i go on to steven and then jackham as well yeah um that really what we're saying in is in dysfunction of whatever that may be whether it results in pain whether it results in um anger whether it results in um um anxiety you know whatever that there are lots of situations where the body gets in the way of the brain and the brain gets in the way of the body and and in a sense you know that what again it goes back to is there some sort of way where you know using your analogy where the horse and the rider um notwithstanding that they are one but they actually start to function as one and then there is a and the dysfunction is when they're not functioning as one and they're sort of getting the way each other anyway that that's a great idea so steven yeah humans are unique in terms of the hierarchies of complexity that we've been talking about which one of the things that attracted me to complexity theory because in the lab you learn about lower animals and earlier today and the truths of those things up to the mammals are truths for life but humans have a number of uniquenesses both on biomechanical as well as a cognitive uh framework that both interact to connect mind and body where the first creature on earth to ask what is it mind or body no other creature has posed the question with the first creature on on earth to have to be an obligate biped that is a mammal birds of obligate bipeds but they fly our superpower is cognition and thinking about things and being able to have choice so in the rider or the horse the rider is the only one that can say you know what um i just remembered something someone asked me to do something i forgot about that let me go back and and and change directions with a horse we'll keep on following that pattern and for most of our lives we keep on following the path that we followed before and that's one of the challenges with any kind of rehab giacommo with with patient compliance that we all know about and that's why from an active inference point to do getting to people in the moment is where there's so much leverage what everything that that you said is make is being present with them and relevant to them which means addressing what they're perceiving with what you're doing and how can that affect what they're perceiving and building a virtuous spiral to take this to a whole different level and when you look at it from even another point from the overall wellness besides the pain point of view all the quadrupeds their their locomotion and the diaphragmatic function are linked and when they're breathing and running they are breathing in the same sink humans are different we've disconnected our breathing moreover we're the only creature that can intentionally i'm going to stop breathing and hold my breath other animals don't have that as a volitional and we've used that ability to control and then modulate to develop language and language has become everything else that we've done from a cognitive point of view so i think that beginning with the fact that we're getting vertical vertical and the first leg that we talk about

is postural balance it's not just posture if i'm standing here like this what looks crooked may appear crooked but an overall biomechanical level it is physically balancing from a biomechanics point of view and from a headspace point of view your posture is how you balance both your body and your mind yeah no that's great giacommo did you have anything to add to that segue outwards which is entirely my fault i do apologize no no problem i was thinking about the previous uh conversation regarding the rule of touch and as a clinician i'm thinking that for example when you decide to apply a therapeutic amount of therapy techniques for example and you are informed by evidence that you can modulate that shorman medium terms the therapeutic outcomes you have tools but the active inference help you to explain how to apply it if you want to have an impact in terms of patient interceptions for example when you decide to apply a manual therapy techniques and you use for example a very hard pressure if you are not in a moment synchronized with your patient you and you are not aware you are not in a listening phase you can adapt you cannot adapt the touch so it's a very it's a fundamental step because you have a powerful tool but if you attack if you apply it in a painful way to your patient there is a risk to have a worse outcomes so you need to modulate the touch in order to get the positive outcomes and you can do it only if you are synchronized with your patients so this is a dance yeah it's a dance it's fundamental if you want to change and try to challenge the prior of the patients you be at first to be synchronized with your patient in connections

the purpose would you say is yeah gone gone gone yeah that's great so would you say i mean the question like a rhetorical one um you know how do we stop some hard-edged um scientific uh manual therapists have been doing this for years how do we stop what we're talking about just sounding like new age woo that that is one of the problems here obviously we don't think like that but i mean in a way i'll talk about dancing about holism about you know horses and riders and whatever do you think that it is that is the role in a way of this framework that the active inference framework is is something that is entirely scientific it's uh we'll talk about empirical evidence in a moment when we finish but but that that stops it you know um feeling as if we're sort of we're being very woolly because actually isn't woolly whatsoever jim yeah i want to say that it is woo woo it is yes and that's very important and so there's two fields in the model there is the surface field and then there's the deep field now the surface field has language volition intention communication and it has a kind of consciousness that we associate this is who we are um we identify that as reality and it's the evidence that it produces is other evidencing and but there's another field and that field is haptic and that awareness is a haptic awareness and haptic awareness in the deep core doesn't have access to all of these different um expressive mechanisms that exteroception does it doesn't have distance receptors it doesn't have language it's a pure haptic or um um uh a touch sense that's generated so that's very important because the body can go over here a little bit too far but that system is going to recalibrate and reorient and establish and reestablish stability and balance it's a very important function but it also has a form of consciousness and awareness we look down on it because it's not verbal visual and symbolic but it is a form of consciousness and so when you walk

into a yoga studio you know the first thing you establish is epistemic trust
uh a meditation we would call that therapeutic alliance wouldn't we if you were in some other clinical setting
perhaps but going just a little bit deeper into that system and working in that quiet field that deep
field and um there really is a kind of a um the semantics are different in that language and i think one of the
things that active inference is doing is trying to create a semantics or a sort of a grammar and a vocabulary
which you can hear in carl fristem and others talking that we ourselves are starting to practice using
that really expresses that that innate active inferential um potential that comes from the deep
field which is woo woo and we need to be able to affirm that for what it is it's the deep field it isn't
the exteroceptive field but but there needs to be kind of um balance and integration between them i think
that is a lot of evidence in work that needs to be done for that yeah no that's great any before we
move on to something because i think we will wrap up and i just wanted to wrap up with um i think we got
an hour so and i know that uh giacomo's got to go so i just wanted to to wrap up around that and
what do you think are the barriers that that get in the way of active potentially might get in the way of
active inference in our fields be you know gaining traction or or perhaps not just our fields but in
in the scientific community generally particularly around anxieties around limited empirical support
for example and um you know accusations of non-falsify ability i know jim has got something
to say about this uh because we talked earlier on and i brought this up but i mean just let's just go can
we start with stephen what what do you think do you think there are there are a few hills to climb
here in terms of that that scientific um support i think that we have more than the seven hills of of
of rome here uh there's many different ways of climbing it um one of one of that one of my efforts
to correct you i i no longer practice i've not practiced since 2000 and 2008 pretty much uh i've lectured and
taught and that's given me a a viewpoint of the world and the professions um i i spoke this morning
about with uh tampa general hospital where we've spoken to pts and ots over there and i think giacomo
a lot of the things you talk you've mentioned before pete ots are doing as much in the moment by moment
more than pts more than chiropractors now in a lot of the daily practice of what they're doing uh to answer
how it would look differently i'm speaking in two and a half hours um about the paper that we did in
that was published in frontiers of neuroscience intercepted posture pictures and taking a picture
of someone's body because posture is important not pathologizing it not making it a disease that i'm
going to fix cure but observing their perception internally of what it feels like to stand tall
and capturing that in a systematic way that you can do it with an objective
to be able to preserve what the body's doing because i think that there's going to be different
techniques and i i put on a personal point of view i may prefer a more active osseous technique and or
light massage or with giacomo you might prefer a lightness of the other way around and there's
ultimate variations within all of these somatics uh i i don't like wearing wearing we're wearing
turtlenecks it doesn't feel good to me you don't like it's not a matter of what's right or wrong it's a
matter of what works for that individual and your point i totally agree with we need to connect with
that person for their body for their mind for their priors and moment by moment help to reshape their
perceptions towards their be actions and behaviors because otherwise it's academic which is fun but it's

not real and that's the challenge yeah so so there are many there are many doors into the system at many levels and and you have to figure out the particular door that you're being faced with so um yeah uh jack and then we'll let me end with jim that's okay anything to say about your worries about the uh empirical support around this framework you know will it will it just fade away and and become like you know the the ether hypothesis you know back when and then discover that it doesn't exist i'm sure it well what do you think i think there are different uh issues and challenges up to now for example that we don't have too much paper produce convoy involving patients with musculoskeletal pain or chronic dysfunctions and this is an issues that we hope that evidence will cover in the futures another issues is related to the lack of knowledge of clinicians and researchers if you don't know it how can you use it how can you investigate it and the third possible issues is the bias of patients they try to defend themselves using the previous frameworks and asking themselves why should i update my way of thinking way of way of acting but we know that science is growing and networking share different perspective and try to see the things from a different point of view so i think that the model is really interesting but now what we should do is to move a step forward inform evidence producing literature and also inform clinicians with some tools and also some suggestions in order to help them to apply this and integrate the model in their clinical practice brilliant we have opportunity and responsibility but we should move it forward yeah yeah yeah no agree and i think i think all of us in our own way are trying to do that um jim i'm gonna give the sort of semi last word for you uh and um and then we'll we'll finish up i i mean i think um i think in terms of active the active influence community so long as it remains heavily exteroceptively biased it's going to continue to to be kind of insular and and and um there there are going to be major kind of gaps in in in the theorization i think that um the impact that active inference hat can have on the clinical environment the clinical experience is not different than the impact that complexity theory is having the biopsychosocial model and also the structural model of the body this is a major revolution this together with active inference the a civilization level paradigm shift is very multifaceted and there is a cultural expression of it as well in movement movement but the um understanding of the body as an integrated whole i think the most important single discovery in anatomy is the rediscovery of the fascial matrix and understanding how isolated musculoskeletal disorders are part of a of a global bottle a a a a structural uh system and this is something basic to what structural body workers do and the book behind me is a really great source of this kind of new synergistic or i would even say cybernetic anatomy that's just a new revolution that hasn't been developed just like active inference itself much of the terrain has not been mapped experientially and so i think that we're really um we really are um at the very beginning of something that's quite exciting yeah no i i would agree um daniel has been very kind and uh just ask us one question i i know giacomo has to go um well maybe i could just ask this question to you very quickly giacomo um and i think we've sort of answered it to some extent if you can think of anything else but he says uh you know how would the encounter setting look different um in what ways would this be technical algorithmic educational certainly education interests both of us giacomo i think that would be something we would say would be would be important but um any any insights just

very quickly into that question i think that the active inference help us to apply what we just know
for example in terms of contextual factors we know that we should use a verbal and non-verbal communication
and the use for example of therapeutic touch but using this new framework we can decide how to apply it
but this strategy arises from a step-to-step evaluation one moment one moment evaluation so i'm interacting
with the
patients if i notice a particular anxiety i try to modulate it using a verbal communication that is not
a rigid way of interacting but it can change and can be modified and modulated on the basis of patient's
response this is an example okay so perhaps one way to to sum that up is that although clinicians are
already dancing with patients to some extent that they they could dance better um and uh and you know
perhaps be as responsive to the person standing in front of them as as as how they feel they should be
moving
their uh their feet as well in this thing maybe maybe you know if you can think of the most complicated
dance to do maybe that's what clinicians could do better jim one last thing jim and then we'll have to go
a terribly quick little point i'm just looking at the little chat um in what ways would this be
this change of the clinical environment be technical algorithmic educational etc i want to add the word
somatic
right and i think the somatics is really the um the the the area for pioneering right now um and in all
that that means on a very profound level brilliant fantastic daniel are you there yep so i think i think
we're done um that that was real fun thank you so much for that question i think we've been uh we've
enjoyed it i think i think uh this sort of stays up on youtube anyway doesn't it so maybe people will
still have a chance to watch this um beyond this this live session is that correct absolutely same url
thank you so much yeah yeah that was for the opportunity right yeah really fantastic opportunity
and and thank you for our um uh our somewhat holistic way of approaching this and uh uh and uh yeah i
think
we all really enjoyed that so hopefully people will enjoy that in the future thanks so much daniel
thank you bye thank you all right bye
bye

All right. Next up is going to be a pre-recorded video by Sonja Didieger. And the title is going to be described in the video. So a 20-minute video by Sonja. Here we go.

Hello. The title of this presentation is Active Self-Referencing. The concept of self-reference might be that which joins complex self-modeling with reference problems in formal languages across a spectrum. And before beginning, I want to thank Shanna Dobson for the conversations that we had leading up to this. Unfortunately, she cannot join this presentation. That was the original plan. So what I will be showing are a series of speculative proposals slightly all over the place, but hopefully interesting. And in keeping in line with the theme of the presentation, there will be no referencing to other authors and sources. These are tacit in the background. And hopefully, if you are interested enough and you would like to know more about this, you can reach out to me. And perhaps later on to myself and Shanna as well. We will try to write this paper for which she was going to cover some of the mathematical aspects that we were interested in exploring. Having said that, let's move on to the presentation.

So in the context of thinking ourselves as predictive, actively inferential organic creatures, we can

understand reference as shared models encoding possibilities for action. Anything that gives us any kind of information is capable of orienting ourselves in a specific direction.

All linguistic, logical, etc. reference are inherently ambiguous. We need to actively infer.

Reference require context for concretization or else they remain unactualized plans, imaginations, memories, memories, memories, whatever we want to call them depending on the context.

This becomes very important when we're going to think about self-reference, because in a lot of the situations during which we can contemplate self-reference, there is a sort of suspension of action, as we will see.

So simple reference with relatively easily concretizable possibility spaces, such as, let's say, the referent cat, are, I dare to say obviously, less computationally challenging than complex ones, such as negation, for example, or the concept of reference itself.

Here we can think about situations that we have decided to nominate as paradoxes, contradictions, and in some contexts I have really stretched the boundaries of this sort of conceptual space and included all sorts of problematic tautological statements and situations during which things have to somehow return to a set point for now.

All of these phenomena are cognitively challenging reference moments during which bifurcations, during which information can be created, during which something can be contemplated and possibilities for action become possible.

These are indeed moments during which we are challenged by the problem of reference, because as things need to actualize themselves when there is no immediately apparent possibility for action, things get rather tricky.

Of course, we can think about this rather well-known situation where we have a statement which refers to itself and in many different ways, but also in one simple way contradicts itself in what it is stating.

It's actualization is rather awkward or its actualization isn't, we could say.

So here we have this pointing to the sentence.

This sentence, what we are pointing at, what we are referring to, is what follows after this.

So this, when we are looking at this, this right now, this is what is true now.

As we move forward, because we are inclined to do so, reading is something that kind of happens before our very eyes once we are able to read.

We cannot but not read, almost.

This points to the sentence, this is true now.

This sentence is true now, or sentence is true now.

And now, this is true.

The fact that it is a lie.

So what just happened, actually in many ways, did not actually happen.

It did not get actualized.

But it is actualized before our very eyes.

So what is happening here?

We can also just simplify this and we could go down many routes.

But we could just simply remove the sentence bit, of course, and just understand that this is something that says that it is not.

This is a lie.

And that is a problem.

One of the problems of ambiguity and sort of the actualization of possibilities and statements is that this goes all over the place, of course.

Different types of contexts will lead us in different directions depending on what it is that we are, depending on the context in which we find ourselves.

So obviously all references, as mentioned earlier, are ambiguous.

Everything requires context for disambiguation.

And in the space of, sort of in the metacognitive space of thinking rather abstract future possibilities, these things of actualization versus non-actualization become confusing for the actively inferential agent.

Again, so this situation is contradictory by being what it is, because the future, as we know, affords possibilities, which we can sometimes calculate as probabilities.

Sometimes.

So, what is remarkable in an active inference context is this predictive challenge, right, that we encounter when we enter these self-referential loops,

when reference kind of models reference itself, when we have to say something about the very sort of fulcrum, the very substrate, which is enabling the capacity to witness possibility.

So, reference, and this kind of sort of metacognitive capacity to self-reference or of self-reference, is an incredibly powerful device for witnessing and or creating new information.

This is the discovery versus invention challenge, which, if we are given to entropy as we know it, is or can be conceptually collapsed as one phenomenon, which is the inevitable recombination that we are given to of systems stacking upon systems,

which we can only understand in terms of how they orient themselves towards something, that is how they, you know, minimize uncertainty for themselves and remain being what they are through these aforementioned reference processes as a minimal condition.

So, what predictive function does self-reference serve?

What are we, what are we arriving at when we are able to witness this?

What is it that we are doing when we engage in the witnessing of reference?

What is it that which enables the very witnessing, right?

Is it a foundational feature of self-modelling or is it a side effect that we kind of inevitably developed because of the capacity to actively infer, to sort of be in a situation and need to experience moving on to a new situation?

So, again, well, as we were thinking about it, this would amount to the proposal that kind of basic self-reference is a prerequisite for selfhood,

where selfhood emerges gradually, evolutionarily, developmentally, through the self-surveilling and synthesizing of one's own information processing.

And this is what becomes confusing or contradictory in formal contexts, again, in sort of purely witnessing contexts that do not yet actualize.

For example, when arithmetic says something about itself, because being the creatures that we are, we did not always have reference.

It could be speculated because, well, in general, we are used to self-reference being an information-creating endeavor.

We inevitably, actively infer ourselves towards something always.

There is always, there is always orientation, whether we are highly aware of it and have very concrete mapping of what it is that we wish to follow,

or whether we are simply, let's say, inevitably given to whatever actualizes itself before our very eyes for us because we continue to exist.

So, when something is incapable of stabilizing the informatic result that we usually expect from referential exchanges, no inferring is possible, no action is possible.

So, what can also be said under speculative investigation here is therefore the concept of self-reference as that which enables exility, uniqueness, because it creates information.

And in general, given to language as we are, most linguistic or communicative phenomena in general are representable as reference within shared generative models.

So, there is always the expectation whether, I mean, there are many ways in which this could be said, but because language exists between creatures and we expect that meanings are, you know, relatively shared, reference often point to more or less the same thing.

The actualization of things, even if we cannot parse it ourselves, is happening elsewhere.

That is always the expectation, that is what language is sort of doing for us.

But reference itself, so the substrate that enables this linguistic modelling, cannot be modelled as a referent within its own system, of course.

And this is much like the self, which remains computationally intransparent to itself.

A thing cannot fully account for its entire structure.

That is quite a simple and at the same time incredibly complex thing to realise.

The loop that is then created, what is this loop of self-reference and sort of something being able to maintain itself as a thing that has to check itself for existence permanently, pretty much.

This is sustained by agents, first modelling themselves as agents, of course, to effectively act, but this is always incomplete and complex in many ways, as has been mentioned.

So framing it as such, self-reference can therefore be interpreted as the process of representation of thisness or individuation, hexiety, a thing becoming aware of its capacity to act.

So act uniquely creating information each time.

It's sort of the, what we can hang on to, what we can understand as remaining the same in the face of eternal unpredictability unfolding before our very eyes.

So we need to hang on to something, and that something is the self, of course, and that is sustained by the self being able to check itself for existence.

That's sort of self-referential thing that anything that is, that we can call conscious, and in many cases incapable also of metagognition.

So that is, that is what we are doing, and when that function is not performed, because we enter other hierarchies of stacking reference upon reference upon reference upon reference,

we enter loops where it doesn't seem as if we can create or we can witness novel information.

So this sentence is a lie, for example, that is either, we can contemplate that as something that is creating a lot of new information,

because it's creating this sort of back and forth paradox where we need to go back to the beginning, to the end, back to the beginning, to the end,

in order to understand what it is, or try to understand what it is that that thing is doing.

But there is a kind of symmetry that doesn't work for us for predictive purposes there.

So we can witness these formal tensions, and there is no active information giving resolution in this sort of self-referential, self-proposal,

that we usually witness as being capable of giving novel information because we are given to uncertainty.

So maintaining this recursive process sustains the experience of continuity through memory and planning, which cannot be symmetrical, but can be intertwined through reference points.

These are the things that we hang onto, again, for sanity.

And for all cognitive systems, lived experience necessarily involves the continuous rehearsal of self-predictions,

as already mentioned, and the permanent updating hereof.

And again, as already mentioned, this becomes extremely complex, uncertain, if not unmappable,

literally unmappable, in a dynamic communicative landscape where we rely on others' self-referential processes,

not only their selfhood, let's say, but also processes which depend on specific set points and sort of formalizations.

So, a self-model which is capable of self-reference, as linguistic communicative capacities, updates all the time its predictive self-referential tension with itself,

which creates new information.

And this effect is mirrored in the larger communicative landscape through other processes of self-reference.

So, because we experience this, we also witness it as happening in examples such as,

this sentence is a lie.

Or, as I like to formulate it often, this lie is a sentence, which I would be happy to talk about, but there is not time for doing that here today.

And, of course, this has implications for understanding metacognitive prediction limits and the emergence of perspective and selfhood in all actively inferential agents.

Because what I think I would like to say here is that the reliance on the idea that things are actualized elsewhere, we might not be, let's say, capable of witnessing what our neighbor says about an event that happened when we were not there.

But we are able to understand that as having happened or something which might happen.

So, the mere idea that actualization is possible and that this is actually what language gives us, the sort of representation of possibility through a shared linguistic niche.

So, the more interesting thing is that we can do is to create a very strange new domain when we create new language systems, new formal systems, which actually take reference elsewhere and do something entirely different with it.

So, I hope that this is a very interesting thing.

I hope that was not too convoluted.

These are all work in progress thoughts and things very much at the edge of what a brain can afford.

And again, as mentioned, thank you very much to Shana for the conversations that we had leading up to this.

And it would be really cool if in the future we can finish this paper with my half, which is what I just presented here, and then her half, which would be some mathematical aspects of considering this.

Thank you very much, especially to Daniel, who holds all of this together, and to the ActiveInference community.

That's it. Thank you.

Thank you very much, everyone.

Alright, next up is going to be a quick five-minute lightning talk by John Clippinger.

Here we go.

Hello, my name is John Clippinger.

In this lightning talk, I would like to talk about the end of extractive economics and arbitrage towards a holistic economics through adaptive inference, value creation, and verification.

So, some big ideas that I'll be throwing out here.

And it'd be difficult to get them all in in a really coherent way in five minutes.

But I am building on an earlier presentation called Agentic Finance Active Inference and Dynamic Portfolio Management that you can look at that provides the fundamentals for this.

But what I'd like to do is really sort of get these ideas quickly out there towards a new kind of economic system and a whole new kind of economic framework.

It's one that really goes from looking at different opinions of arbitrage and establishing markets and bid and ask towards a measurable uncertainty reduction rather than speculative opinion.

And combining Bayesian inference and thermodynamic accounting and multi-agent coordination.

It proposes a verifiable and adaptive energy efficient architecture financial systems capable of aligning real capital with real world outcomes.

So, what's the problem that we're trying to solve here?

And it's sort of fundamental to everyday economics and finance.

And that is that you establish value through bid and ask, through the spread.

So, it's what people are willing to pay versus what they're willing to buy for.

What's the spread?

What's the difference in opinion?

So, prices are really tied to differences of opinions, not some kind of underlying objective measure.

And these can be manipulated because there's real incentive to create spreads in order to make the difference in the margins of value.

And so, there's an intentional informational asymmetry that's always been exploited.

So, liquidity is not necessarily equal to value.

I mean, you manage liquidity.

You've got to create liquidity in order to create markets.

And you can change and manipulate liquidity by through interest rates.

But these are just ways of adjusting it within the context of different kinds of opinions, not saying there's any underlying principle of assigning value there.

And what it does is it creates an incentive to manipulate value.

And hence, you get bubbles and you get rug pulls.

And you get a lot of the problems that we have right now with current markets.

So, what we're arguing for is a core replacement.

Price is the capacity to reduce uncertainty for a unit of energy.

So, you use active inference.

It treats every agent as a system that spends energy reduced predictive error around targeted outcomes.

We then price the asset by the expected reduction and uncertainty normalized by energy and capital use.

Now, these ideas are pretty radical in the context of current economics, but they're not as radical as you might think.

Because when the interest rate swaps, when you have fixed rates versus variable rates or floating rates, there's something called the basically SOFAR, which is secured overnight financing rate, which is a broad basic measure of the cost of borrowing cash overnight backed by collateralized by US Treasury.

So, there is this mechanism that allows you to exchange things of different kinds of value proposition in order by virtue of the standard.

And now this is set by regulation.

But we're arguing that this could be set by basically a scientific method of assigning value around energy prediction and information reduction.

So, it creates a way of incentivizing markets.

So, if you look at markets as Markov blankets, basically Markov blankets of nested agents, they're trying to synchronize around a common outcome that minimizes uncertainty and aligns them all around a common outcome and stabilizes it.

So, you can create a welfare index, you can have some kind of outcome or some KPIs around a project that allow different players in a marketplace to get the signals, the signals they want in order to achieve a shared outcome that is beneficial to all.

So, this is the holistic notion of the markets.

We're not, it's not adversarial, it's cooperative, it's alignment of interest, a multi-scale alignment.

And you really have posteriors as prices.

Each agent published a verifiable posterior.

Each dollar is jewel, and they reduce X units of uncertainty around a desired outcome.

Settlement by evidence, basically, self-synchronization.

Now, what this does is obviate the need for the fundamental on-market mechanisms like bid and ask, liquidity pools.

So, you have decentralized exchanges, you have monetary policy, which is adverse to fiscal policy.

This combines them all in a new kind of metric that is very powerful.

So, if you're interested in this, please give me a note.

I'm at John at firstprinciplesfirst.ai, that's fp1.ai.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

We're back with Entropic Motivation, a panel discussion.

Alex Kieffer will facilitate and share some slides.

Thank you all for joining.

Go for it, Alex.

Thanks a lot for joining me.

I'm looking forward to this discussion a lot.

Since we have an hour and this is the deepest theme I can think of to discuss, I didn't want to spend a lot of time on introductions.

But just really briefly for people who are following this.

I'm Alex Kieffer.

I'm really thrilled that everyone has decided to join today's panel.

We've got Daniel Polanyi, who, again, I'm not going to give lengthy introductions here, but Daniel, of course, has developed the empowerment framework for understanding motivation and agents.

So Jorge Ramirez Ruiz has worked on the maximum occupancy principle, which we'll discuss today as well.

Adam Saffron is, I think of Adam as like an active inference in physics-inspired machine learning kind of generalist.

But I also just love discussing philosophical topics with Adam.

Similarly for Dalton, Dalton has also done some work on, well, you all know Dalton here, but on the free energy principle, developing some of the mathematical foundations and also with a particular emphasis on its relation to things like maximum entropy and maximum caliber.

So obviously relevant here.

And Sander van der Kruijs, some of you have only met over text.

So I apologize if I didn't pronounce things correctly.

But Sander is a psychologist who works in the predictive processing kind of active inference space as well.

And I recently came across some of his writings.

There's a great blog called There is No Reward.

So I thought that each person here would make a great contribution to this discussion.

So I wanted to just say a couple of introductory things and then we'll move away from slides and just get into a discussion.

So the overall theme here, in my mind anyway, is what role, the question is what role does entropy, in particular like the maximization of entropy under constraints, as we'll talk about, but like what role does this play in intrinsic motivation?

So, you know, entropy is one of the probably most uttered words in cognitive science in this, you know, 20, certainly 21st century.

So you can define in many ways, but basically I just mean the, you can think of it as the average surprise or the spread of a probability distribution or the amount of uncertainty associated with the distribution.

And of course it has a thermodynamic meaning as well.

And we get to that.

And by intrinsic motivation, I mean, really, this is just about motivation, but I'm just trying to focus ideas.

There are many ways of thinking about how creatures, you know, artificial and biological or natural naturally arising might be motivated and intrinsic motivation.

Again, many ways to define that, but I think of it as just kind of separating, trying to isolate or distill or understand aspects of motivation that don't boil down to like the kind of thing that's modeled in very vanilla reinforcement learning.

So that might be a straw man, right? But I mean, they're just like learning sort of to pursue some reward or like learning associations between certain actions and reward.

So the question here to, for me at bottom, the question here is like, is there a, an aspect of motivation in intelligent agents and living agents that, that doesn't just boil down to, to reward seeking behavior?

And that also might, might encompass things like fitness in the, in the context of natural selection.

So we'll talk more about that, but I wanted to get right into some questions here.

So, and please feel free to interrupt by the way.

Yeah.

Yeah.

So, so as I mentioned, I think one useful lens on this, which has featured in my recent work is to think of, to look at the connections between particular

particular accounts of what intrinsic motivation amounts to and how it might be related to the idea of constrained entropy maximization, which is important in a number of ways in cognitive science and physics.

But so I, I recently have been looking at these three frameworks, empowerment, active inference, and the maximum occupancy principle.

So Daniel, since you've joined us here, I won't ask for a general introduction to empowerment, but I wonder if you want to say just a few words about

what, what empowerment is like, what it means formally and how it, how it relates to how you think of it relating to motivation in agents.

Okay.

So the basic definition of empowerment is essentially it's a measure of the potential information that an agent can inject into the environment by its actions and pick up later by its sensors.

And it has multiple interpretations.

There are many converging ways of looking at that.

One way is to look at it is to maximize the possible choices that the agent could have in choosing its future that it can distinguish itself.

So it's not just able to choose different futures, but it must be also able to distinguish itself that these different futures have been chosen.

So the idea is basically that if you're in a situation where you have more choices, you are more likely to be, let's call it vitality.

It's a, it's a way of maximizing your vitality, maximizing your options to run away from death, which is zero empowerment.

Um, there's an interpretation in terms of evolutionary niche that your sensor remote niche mean is if you have high empowerment, it's well, well matched where the actuators and sensors are well matched.

They're not well matched.

They're not well matched.

You will try to match them better.

Um, there are of course other criteria like energy conservation, so on, but assuming you have given sensor and motor equipment, you will try to go somewhere where they are best matched, where what you do and what you can distinguish.

Um, are maximally compatible.

Uh, there's another control, the regular interpretation, basically it's combined control ability, observability measure.

And there are many others.

Uh, you can think of it as mobility in games.

If you look at deterministic games, you could look at it.

Money is a form of empowerment.

It's a potentiality.

It's not what you actually do.

That's in a nutshell.

Um, of course, that's much more what can say, but just to give you a quick upshot.

What empowerment tries to capture.

Oh, thank you so much.

So, right.

So you can already see how a lot of those things that you mentioned don't just boil down to simply seeking reward in a, in a more basic sense.

Um, but so obviously I want to open up more for discussion, but, um, I thought it would be good to maybe since we have people who are sort of representative of these, of these theories to sort of lay them out first as, as background.

So, um, so, so, uh, for active inference, um, anyone who'd like to jump in, but, but Dalton, I think you, you particularly in particular have done work on again on sort of anticipating a bit.

Um, pointing in the direction of, of the sort of duality between active inference and, um, entropy maximization under constraints.

So would you want to, we're also at this implied active inference symposium.

So it's not as if you really have to introduce active inference, but would you like to say anything to frame how you see these, um, these related being related at this, at this point.

Uh, sure.

Well, I can say a couple of things about, um, the particular view of active inference that I think would be most productive to take in these discussions.

Um, one thing I want to, to point out immediately is, uh, when you have, uh, an active inference agent, uh, you are looking at something that, uh, looks as if it has the policy of minimizing expected free energy.

So the actions that it takes in order to maintain its set points are, uh, uh, appearing to minimize expected free energy.

And, and indeed, if you build an agent and coded up such that it is an active inference agent, then you can do this explicitly by hand.

Um, the interesting thing about the expected free energy, uh, is that agents that minimize expected free energy, they both pursue.

And, and also goal satisfaction.

So there is a sort of, uh, balance between some kind of intrinsic drive or epistemic value, uh, and also a reward seeking extrinsic behavior.

Uh, and these are bridged in, in one quantity.

Uh, this kind of, um, on the face of it, it has a lot to do with empowerment in the sense that it's, it's not simply exploitation.

Uh, it has an exploration element.

It's, it's saying also, I want to learn as much as possible about my environment, uh, so that I can improve my future, uh, suite of possible actions or possible modifications of the environments.

I want to explore in such a way that I can eventually bring about an environment that might be even better for the goals that I want.

Um, as opposed to simply preparing a state of the environment in the goal that I currently want.

Um, more broadly, if you look at the free energy principle itself, uh, this same kind of dichotomy is there.

When you have a complex system, you have some kind of typical behaviors, some kind of, uh, states for which the, uh, system is characterized, uh, by which the system is characterized.

Say these, these characteristic states are things that we think of as in humans, lifelike states in control systems.

They are the set points that make the control system, uh, be the sort of thing that it is.

Uh, but within that, if you have a, uh, system that remains organized for long periods of time, it needs to have some kind of flexibility to respond to the perturbations that it's receiving from the environment.

So a system that very rigidly stays within a very small set of characteristic states is actually not very good at organizing, uh, because it is very easily dispersed by changes in the environment.

So again, there's a kind of trade off between maintaining a very rigid, uh, set of, of characteristic states.

Uh, those are your sort of constraints, uh, but also exploring within those constraints, what is possible and reconfiguring as the system goes along.

Um, so I, I think empowerment, uh, and constraint, maximum entropy have a lot to do with each other in the way they treat this balance between exploitation or reward seeking kinds of behaviors and exploration or, uh, dispersion kinds of behaviors.

Uh, they both allow a certain, uh, flexibility within some kind of confidence bar.

Uh, and I, I've thought a lot about this, um, and I know Alex, uh, and Daniel and others have thought a lot about it.

So I'm interested to see what insights come out of this panel.

Thank you.

Thank you, Delton.

I've been, uh, throughout both of these, these, um, sort of discussion, uh, introductions, I've been nodding vigorously.

I don't know if people can see, because we're looking at the slides, but we'll go to the cameras soon.

Um, thanks.

Uh, so, uh, um, Jorge, would you like to, um, introduce anything about maximum occupancy in terms of the, um, sort of the, yeah.

Well, just what, what, what the framework says, but also like, um, uh, what, what its motivation is in this context?

Sure.

Um, um, yeah, so I'd like to, I think the maximum occupancy, like, uh, fits very well with what people have said in, in the theme of the

of this, um, discussion, uh, because like the main motivation for this, uh, work was to think of how people usually think about like curiosity, you know, um, and exploration as.

You know, means to an end, you know, like, uh, version is very good because then you can, uh, reach higher rewards in the long term by just exploring the state space and access space.

Um, so we wanted to flip that around, you know, because like, uh, reward functions are, uh, in reinforcement learning, for example, are like, uh, when you're engineering, you know, you're trying to, uh, have a robot do something.

Um, you have to really try to engineer that reward function.

Um, and there's no real principle way to think about where reward functions come from other than what other people have said before in the past, like how reward functions can be evolved and so on and so forth.

But, uh, so we wanted to turn that around and say, well, rewards, but, uh, organisms think about as rewards are actually means to an end, right?

And, and, or should be more like means to an end, which you can think of many ways, survival or whatever.

But so we wanted to develop some kind of objective, um, that has the like minimal assumptions, you know?

Um, so we, we came up with, uh, this objective of maximum occupancy, like just from inspiration of the second law of thermodynamics and Carl Fristen has said it before of how there's a duality of, uh, like max path, uh, maximize path entropy.

And then minimize entropy, uh, for the organism.

Um, so yeah, so it, it, it comes with very well with this theme because, you know, like just maximizing entropy, for example, in this case, maximizing the future entropy of your actions and states.

It doesn't really make sense without constraints, right?

So it, it just maximizing entropy of your actions and states.

It's not an interesting thing to do.

It's just random walk, um, all the time.

Um, but when you actually think about constraints of organisms that have evolved, then you have very interesting things where they go away from those states that are, uh, as Daniel said, like low empowered, or in this case, you cannot access future action entropy and state entropy in the future if you die, for example.

So it has this survival kind of instinct, um, trying to work with its constraints to still maximize, um, path entropy.

Um, that's basically kind of the, the inspiration in what the maximum occupancy is about.

Um, yeah, if you have questions, uh, yeah.

Great.

No, thank you.

That's, that's wonderful again.

Um, so yeah, and I think that, that, that kind of perspective that maybe things can be flipped on their head so that, um, these sort of curiosity or novelty seeking, um, behaviors are in a sense, but perhaps more fundamental than, um, reward seeking, uh, is a, is a really interesting idea.

That's, that's, that's, I think been a major inspiration for me in wanting to do this.

Um, so before we, we move to sort of phase two of this, um, which would be, excuse me, to consider some specific questions, although I don't want to over constrain this cause others, there's, um, a lot to say.

Um, um, I guess, Adam or standard, do you want to say anything on the, on sort of the over overarching theme of intrinsic motivation or any of this, uh, any comments you'd like to make at this point?

Uh, Adam first, maybe just alphabetically.

Uh, no comments at the moment.

I'm just enjoying the discussion.

Awesome.

Um, yeah.

Sandra.

Yeah.

A few comments.

Um, I, I, I, I see some of the discussions that have been like historically been taking place in psychology and, and, and, you know, I think that's a lot of the discussion.

Um, and, and, and whether like this need for cognition or stimulation or exploration, what, what place it has within the other so-called basic, more basic needs.

Um, this is, this is, this is pretty interesting.

Um, that, I mean, that, um, there's now with this maximum entropy or with the empowerment or maximum occupancy, this possibility to invert indeed that picture and to, to, to not see the kind of, um, knowledge drive or stimulation drive or need for cognition as like the odd one.

Now it's something that is alien to the rest, but that is really front and center of, of, of this picture.

And that's, I think there's evidence in psychology, but it's never, I mean, the, the kind of evolutionary basic drives have, have, have always been the dominant thing.

So, and to see that there's a framework that, that doesn't deny that picture, but, but that can make it more rich is, is for me very stimulating.

Yeah.

Awesome.

Thank you.

Uh, I'm going to think, stop sharing slides now.

Um, so, um, is this, is this good?

Can we all see the cameras now or are we?

Yep.

Okay.

Cool.

Um, so yeah, so I did, you know, there were some other just slides with, with little prompts and notes, but I, I think I'd rather have more of a personal interaction as we do this, if possible.

Um, so I would, I just wanted to go through some, some, so there's a lot I wanted to get to today.

Um, but thanks for, for laying out the sort of, um, basic, um, ideas and frameworks here.

Um, so, uh, in, in compelling terms.

Um, so I, I had, I had three themes in mind to discuss and again, we can go as far off the rails as we want.

Um, that's the entropy part of the, of the, of the equation.

Um, but, but the first theme, uh, that I had written down here was on sort of the degree of the machine.

on sort of the degree to which these two aspects of motivation, okay? So intrinsic motivation

usually seeks something that's not just reward in a conventional sense. So we have reward-seeking behavior or risk minimization, and then everything else. And the everything else, I tend to want to

place under the heading of maximizing entropy, but I also don't want to front load that assumption too much. But I guess the question is, to the extent that we have at least two dimensions of motivation in general that follow out in many different frameworks for understanding motivation, like, is one of them more fundamental than the other? And we've already heard from a few people who want to flip the conventional wisdom on its head in a way and say that curiosity or entropy-seeking behavior is more fundamental. So maybe we all agree on that, who knows? But I'm curious whether anyone has a view on that.

I could call in particular people, I feel like a... Oh yeah, Daniel, go ahead, please.

I would say, from my perspective, I think the arbiter is evolution. So in evolution, there is no real thing like reward. Reward is a fiction that we employ to compute behavior to compute desirable behavior. And it requires the concept of utility, which has to fulfill number of axioms, which we know is not that easy to get if you actually operate with real beings, real organic beings, human beings. We know that naive utility theory doesn't perfectly work in the same way. So I would say, ultimately, what you want to understand is how does the concept of seemingly reward or seemingly reward-driven behavior, emergent evolution, concept of goals, for example, is also an emergent concept, in my opinion. It is coming through the fact that certain behaviors have to go through bottlenecks. Like I'm not talking about the information bottlenecks, here, just behavioral bottlenecks. For example, when you catch food, the food is localized, so it is a bottleneck to catch the food. It looks like a goal, but the reality is, if the food would be all around you, you wouldn't care. And it would not be a goal to catch food.

I always give the example, you're a tapeworm, sits nice and warm, and it's constantly flooded by food around it. It doesn't, the food acquisition behavior is not a specific goal, because the environment is nutritious as it is. And I do think that's what misleads us as humans. We like to categorize things, but this is an emergent phenomenon. I don't say it's not a real phenomenon, but it's emergent. It's not primary in my opinion.

Thanks, Daniel. So that's, I'm glad you brought up evolution. I think, so I take your point that, yeah, that reward is perhaps an emergent concept or emergent notion. But I think one question that I have here, and I will not take too much time, I promise, but one question I have here, which formed this idea of having this panel, is that I think in a broader sense, like, if we take evolution to be sort of producing variability and then selecting according to some fitness function, that the fitness function has received a lot of attention, but the producing variability is maybe the other dimension or aspect of this process. And that might have a lot of structure, or it might not, you know, I mean, there's, anyway. So, does anyone else want to comment? I was, I'm curious. Yeah, I don't want to overprompt people, but.

Yeah, it reminds me of, like, the studies on single-celled organisms that, when stressed, that seem to upregulate their mutation rates. So that there's just more bets made, and that seems like a more random walk that might end in a lot of attrition, but will also give, like, the necessary

structure for the future for the new environment, apparently. So, is that what you mean with the exploration and evolution?

I don't know. Oh, sorry. No, please.

I mean, I mean, this, the increased stress, right, what it means, in my opinion, evolutionarily, it means that you have a higher chance of dying out. And so, there is basically a change environment that actually reduces your vitality. And to counteract it, you create just more options, so that at least one will survive. Right? So, you don't do that when you're in a good shape. Right? You don't take risks. When, when you're doing well, you keep the risks under control. You only take major risks and the vital risks, when, when there are many chances that you might not succeed. So, I would say, you can understand it from that. Yeah, sorry. Yes.

Yeah, no, sorry. Just that, in humans, it seems to be the reverse, right? When we play, or when children play, it's when they're well-fed, when they're, they don't have stresses, when, so...

But they don't. It's reversible. Sorry?

It's reversible. This is a reversible risk. That's not really a risk. It's just free learning.

Taking risks, when I, when you say you increase mutation rate, you take irreversible risks. Basically, it's like having a company, and the company is in a bad shape, and then you start taking bigger risks to make it survive, then you would do if it's doing well. But play, plays the different play. Is a safe, it's actually an excellent point. So reversible is important. Yeah.

Yeah, I see. Cool.

I'm, I'm curious whether, so, I don't know, and it, perhaps, I'm reading into it, but, Jorge, in your work on maximum occupancy, I tend to think of, of this as somehow modeling, like, the desire is to stay alive, in a way, sort of for its own sake, right? It's, it's, you want, what you want is to be alive, and that is to occupy state actions space, path space. So, I don't know if you have any further comments that I might make.

Well, yeah, I mean, I think that, in this case, surviving is kind of a consequence of trying to occupy your future action and state path. Because when, when you die, you can't, right? So, but it's not the primary drive, let's say, of just surviving. Like, surviving is a very degenerate kind of a reward function, right? Like, if you survive, there's just so many ways to survive. But, but when you are actually doing some kind of maximum entropy on actions and states in the future, then you're, you're encountering some, some, you're, you're, you have very minimal assumptions of what you can do, but you're constrained about those things you can do. So, yeah, so that's kind of what I would say about, like, survival.

But I'm coming back a bit to, towards, like, what is kind of more fundamental. I think it would be more fundamental with rewards to what, what you want to achieve, right? So, like, if you're, if you're an engineer that wants to just make robots that do stuff, you know, like, if you can engineer a good reward function, then you could say that, you know, intrinsic, those intrinsic motivations that you give, you could give to robots, they serve the function to maximize work. In that case, kind of intrinsic motivations is not as, it's not the driving thing. And, and people, you know, have argued that reward is enough. But kind of the idea of maximum occupancy was like, well, reward is not necessary. And,

and I think that the, the, what we wanted to argue with this work was that, as Daniel has said, like reward is kind of like an emerging thing. It's something that makes you want to achieve other, something else. And, and in that sense, that kind of extrinsic reward, I mean, it's funny, because I like, I, we organized a workshop about like intrinsic motivations in it, in this RLDM conference, and, and kind of like, what came up with it was like, well, all rewards are intrinsic rewards. And in that way, I agree. But like, this rewards are kind of something that has evolved over time, to make you achieve something else, or to help you achieve something else. And so in that way, I would, in that sense, I would argue how that there's something, there should be something like a principle that makes you, over time, over perhaps evolutionary time, develop this reward functions, this extrinsic rewards that people call. Yeah.

Yeah.

Yeah.

So, so maybe a more, a slightly more focused way of asking this question is, so I mean, in a sense, it's, it's all down to evolution, but like curiosity, for example, like curiosity and like things like epistemic drives, and information seeking behavior.

With respect to that question, like, one question I had as well, do we must we view that as in the service of obtaining something else now? Okay, I guess we've problematized the notion of rewards somewhat here.

But like, is that sort of information seeking always in the service of obtaining some goals, that always a means to an end? Or might it be like a more fundamental aspect of what drives life and agents? And that's, that's, I think under some guys, I'm tempted to suppose that there's something fundamental about that, this other component of motivation, right? So there's the constraint, right? Then there's the entropy maximization. But so, so I think, Dalton, maybe I'll call on you here. From an active inference perspective, it seems like in a sense, these two things are on an equal footing, at least, rather than everything being in the service of risk minimization.

Right. And I suspected I might be called upon soon. So I intended on volunteering myself, just in case. Before I answer that, I just want to echo something that Daniel said, which was that I'm very partial to the view that reward is a useful explanatory fiction. And in particular, I'm quite partial to the view from the free energy principle, where we say, if a thing is existing for long enough to be observable, to be modelable at all as that thing, then it must look as if it's seeking environmental states conducive to its continued existence.

And that appearance gives the impression of desire, or it gives the impression of some reward, some goal-seeking behavior. But if you take something simple like chemotaxis, something like a bacterium or an amoeba, one could argue it would be a mistake to anthropomorphize that as seeking reward, as opposed to merely following a gradient, because if it weren't following that gradient, it wouldn't exist to be described at all. What is fundamental, what is physical, is the maximization of entropy. So in the end, even the imposition of constraints

is something that we do as a modeler. We say, here is a constraint that phenomenologically describes our observations, it gives us the right answer. But does that constraint exist in reality? That really strongly depends on what your sort of metaphysical or ontological position is, right?

So it becomes a little bit more subtle than saying actually that constraints exist as opposed to they're something that matter to us as the modeler, right? So that's sort of one component of the answer.

The other component is, it is often useful to describe agents broadly construed as balancing both of these things. Because oftentimes, there is that impression of reward seeking behavior. And when engineering agents,

when building things, we may want to give them reward seeking behavior. So active inference is telling a story

where both of those things are given equal weighting, that they're regarded as equal contenders in a competition.

And this proves to give very succinct, very explanatory descriptions of a lot of the dynamics that we observe.

So if you're in the business of flipping the story on its head and saying, if there is something to be observed for a long enough period of time, then it must be somehow balancing, preparing

a preferable environmental state with exploring its options to decide what could possibly be preferable.

If you're in the business of telling that story, then active inference is a very useful way to tell that story, because of the way the two things are balanced. But I think it's a somewhat deeper philosophical question, whether all systems or some systems, or no systems actually do this, do they actually have the policy of minimizing expected free energy? Or is it a useful model of what we observe as the dynamics of such systems?

Yeah, thanks, Dalton. I guess I have no problem sort of unconsciously taking a realist stance on that question,

which is often a blind spot I have. But I think, okay, I mean, but granted, it's a good point, though.

But so regardless of how sort of much epistemic humility we want to adopt about our models of the situation, it does seem it does seem that well, you've raised the second major kind of question I wanted to address,

which we'll see how long we want to spend on it. I think we're each of us in this panel is sort of immersed in the, you know, thermodynamic descriptions of things to different degrees. But like,

the second question I wanted to ask is, well, what it's probably there's no single simple relationship, although I would like there to be. But what's the relationship between these formalisms,

which describe after all, they're describing sort of objectives for agents, and just entropy,

constrained entry, maximization, things like that in the physical sense. So I don't know that just

asking the question in such a flat way is going to be a good conversation starter. But I mean, Dalton,

you brought up active inference agents look as though they are seeking to minimize free energy,

expected free energy in particular. So and actually, so let me just say something here. So I think,

like, in a sense, if you think of agents, or sort of inferential processes like minimizing

instantaneously, minimizing variational free energy, that's an easy, that's, it's a very easy, very clear

illustration of this sort of, you know, minimize energy while maximizing entropy, because like,

in the Helmholtz free energy, you have the energy, which is the average internal energy of the system,

minus the entropy. And so to minimize that free energy, you want to maximize the entropy, right? Or minimize the energy, or both at once. But so, but in a lot of these frameworks, we're not really talking about just that we're talking about policy selection. And like, I know, I mean, also, I think this applies as well to empowerment, like Daniel, so you're, right, you think of empowerment as it's not just, you want to choose, it's an objective for choosing states, right, so that you maximize the sort of the empowerment that you'll be in once you have once you're in that state. And so, so in either of these things, policy selection under expected free energy as an objective or empowerment, bears a direct relationship, I guess, to, you know, maximizing entropy in this Helmholtz free energy sense, like just maximizing the entropy of states under under a constraint directly in states. So I'm wondering if there's anything to say, okay, the overarching question, I mean, empowerment is, is can be used as a surrogate

[illegible]

Exploration is incidental.

It's not the main thing that empowerment does.

So yes, sometimes it can lead to exploration,

but that's not, it's a spinoff, if you like.

It's not its main agenda.

There are tools which are explicitly exploratory.

Right, right.

So one question I had here is sort of, of the various things that you might pursue as an agent, exploration is one of them.

Empowerment, when I try to reduce it to a bullet point,

I think of it as, based on things you've said,

I think of it as like,

sort of maximizing the variety of futures
you can realize or something.

So that's obviously different from, conceptually from exploration.

Yes.

In terms of trying to get information about things.

Exploration comes into play
if you need to develop your model.
If you have your model,
there's no point in exploring anything.
Right, so when you don't have a model,
exploration can increase your epistemic empowerment.
So empowerment can change by what forward model you have.
So if you have a very detailed forward model,
there's nothing to explore,
and you just maximize empowerment straight.
If you have a bad epistemic model,
where you don't know exactly what your actions will do,
there's an advantage for you to explore
what actually actions are going to do.
So empowerment will invite you into exploring the states,
but not just any states,
just the ones who are most promising
for maximizing your options in the future.
So as I say, exploration here is more side quest
rather than the main issue,
which I don't think it's necessarily bad.
We sometimes feel like if things are not primary,
that's a bad thing,
but I feel that evolution doesn't care.
Right, in evolution,
there are lots of these side quests
that turn to the main quest.
So for example, curiosity,
I would, for example, speculate,
and this pure speculation,
that curiosity might be a side quest
to make higher organisms ready for changes in the niche.
So, because curiosity is dangerous,
curiosity killed the cat.
The reason why they say that,
but maybe for higher organisms,
which cannot compete by larger numbers,
or that sensitive to environment,

it is a distinct survival advantage
to have curiosity in the population,
on population level.
On the individual level, probably not.
I don't think that on the individual level,
exploring Antarctica is good for your offspring,
but on population level, it might be.
So if I could add something just before you move on,
I do want to note that,
I think it's interesting,
Alex mentioned variability a little earlier on.
And it's interesting to note that,
variability is itself a way of increasing the capacity
for side quests.
You know, you introduce a whole bunch
of different mutations and see which ones persist,
see which ones increase fitness.
So it gives you a kind of test bed
for other beneficial things
that are not simply about increasing reproductive ability,
but are maybe unexpected ways
of increasing the duration
for which an organism persists,
and the amount which it can reproduce
and pass on its genes.
So there's kind of a link between this discussion
and the earlier discussion,
which I found quite compelling.
I also wanted to open, if that's okay,
just to note, perhaps,
I don't know if it's kind of technical more,
but something I've noticed from both empowerment
and active inference was that this idea that,
if there's, you know, if you have perfect knowledge
of the world, let's say, or your model is very certain
or whatever, and there might be ways to keep you exploring
so that you're never like converging,
but let's say that you do and that you're learned,

both of these two things kind of stay put forever, right?

And like, you could say, well, the environment is always changing, so you need to go back and like, you know, your internal states change, so you go back and fulfill your model or you minimize your surprise about your model.

Still, I feel like there's nothing fundamental about exploration.

It occurred me if I'm wrong.

In that sense of, as Daniel said, like it's a side quest to fulfill those things, but it's like if you converge, you stop, and then that's it.

You leave your organism there.

And so that's why I wanted to like kind of contrast that to what we've proposed, which is like, well, no, you keep doing stuff as if you're allowed to kind of thing.

And in that sense, that kind of can give you more robustness in your organism or in your artificial age and more in a fundamental way.

But yeah, that's kind of my question.

Yeah, Daniel, please go ahead.

I think this is a very good point.

The question is, when is it OK to, you know, dry out in a given niche or when not?

And my claim would be the following, and this is purely speculative here, that the reason you want to continue exploring when you have found a good maximum is because you have no idea what happens to your environment.

See, you sit on somewhere and everything is fine.

Everything is happy.

And you live in this perfect paradise, and you don't look around.

What happens is you lose information about the neighboring states.

And in so in other words, in the language of mathematics, you become only point oriented rather than neighborhood oriented.

And I believe that this makes it very vulnerable
if there is a major change.

So you will not be ready for any major.

I have some examples, but they're a bit too long to explain here.

But essentially, you're not ready for changes
where the local environment would actually tell you,
look, something's going to happen.

But because you said only on your point, you never notice.

And that's why I think we when we stand as humans,
we never stand still.

We are constantly by the ground constantly.

And anything we do has a certain variable,
even the most perfect sports people.

They should just reproduce exactly what they always do.

And they never do.

And it sounds like a like a bug,
but it seems to be a feature of nature to never do the same thing again.

Okay. Yeah, I like that.

Um, but it sounds to me though, that it's like,
um, it's an emergent thing, right?

So it is like, okay,
it's just because you live in a world where things change
and things have evolved to look for, for other things around.

Um, perhaps it's an even an empirical question.

I don't know just to say like, well,
if actually you have a very, very stable environment,
do you have, do you evolve beings that do not explore at all?

I actually, when you stop evolving and we have examples for that,
and, um, there are prox features, which are social specialized.
They require very specific set up to continue.

Um, you disrupt the environment.

It's finished, right?

Um, and you see this in nature, but mostly you don't because in large scale.

Echo webs, um, you don't have the stability.

There's always something that intrudes your, in your world, but you can see that in economy too, right?
An economy that is relied.

Oh, we always relied on the same things working fine.

And then suddenly something major, there's a major change.

Um, that happens.

You know, you now have the problem.

Um, you're not ready.

So, um, bubbling, um, as I think Adam said, bubbling is the right thing to do constantly to some level.

You don't want to waste too much on it, but you have to keep doing it.

There's an advantage to that.

Um, this, this is very interesting because my, one of my intuitions here is that it just sounds like it sounds as though there needs to be a reason for this, this babbling.

And of course it's very useful, but, but I guess I, I wonder whether just being alive, like that's, it is babbling.

Like that's, that's the fundamental thing.

So there's always, you know, if you plan and you can always, you can have these goals that pull you from the imagined future.

But then there's also something that just pushes from behind.

It would just sort of the essence of like being alive.

And I, I, I guess I want to link those two things, but they are quite different planning to stay alive.

Right.

Planning to maximize entropy on infinite time horizon is still quite a different thing from just doing stuff.

Um, but you don't.

You're in the bubble bubble, right?

I mean, one of the problems with, with many exploration strategies is they are, um, more undirected.

You don't care about knowing what the lint under your, your bed is doing.

You most of the time you're not interested in that specific things you're curious about.

This, this is true.

Um, so, well, okay.

That's sort of conveniently brings me to one question I had was this, this sounds, it's a funny thought, but like, is, is there any reason to think that?

So evolution is quite a general concept, of course, right?

It happens at different scales, but like evolution on the phylogenetic time scale.

It's how you usually think of it.

Um, is there any reason to think of it as having some kind of curiosity component?

Um, exploring state space, um, you know, in the similar sense to how an individual mind.

I mean, maybe that last thing makes it, uh, is, is not helpful.

You were nodding earlier, Daniel.

Um, so, I mean, you could think of that, you know, from the point of view of like multi-scale active inference, like, oh, nested agents all the way up and down.

But, um, that's a very thin sense of agent.

Um, but, um, I mean, so I, I think I was reading an article by, um, Terence Deacon, and he mentioned sort of, um, something about, uh, anyway, something about the fact that evolution is driven by, I mean, this is obvious. Evolution is driven by sort of variation.

Um, uh, so, you know, if you, if you're implementing a genetic algorithm, the obvious thing is you need the fitness function and you need the modification functions.

Um, so it's, it's, it's just, it's just sort of a question about whether, and I mean, Dalton, maybe you'd say, this is a matter of how you model things, but, um, if we're looking at processes of evolution on the, on, on the timescale of, you know, speciation and things like that,

is there anything that we could view as like a structured process of exploration?

Anyway, it's a question.

And, um, I don't have the answer, so.

I'll let that sit for a second while I dig for, um, well, if, sorry, is there any comments on that? Adam, thanks by the way for doing some chat jockey.

Uh, you've been quiet, but I really appreciate that what you're adding to the chat.

Um, if, I mean, if, I can make a comment, I just didn't want to talk to the panel.

Um, evolution, thinking of evolution as an agent is a very interesting thing.

It's not an agent in the usual sentence because the interface is very vague and, and, and this thing, but I do agree, uh, that yes, evolution has a way of learning.

As a way of, I think you just recently read a paper that evolution does have a ability to improve its learning mechanism.

I would be surprised if it doesn't do that.

Um, we know that Lensky's experiments are a nice example of, uh, pre structuring, um, evolve ability pathways.

Um, the longterm experiment that he did, uh, I don't know, 20 years ago.

I think it was this paper, uh, really, really nice paper, uh, fantastic experiment.

I do think that we, we can think of evolution in this, as long as we don't say, oh, it's an agent with clear membrane actuation or something.

We can think of it as a kind of, as a kind of, uh,

Curious learning mechanism.

Uh, but we also have to understand that evolution, unlike normal agents doesn't have that constraint.

It has to work on the constraints of physics and chemistry, but, um, energetically is quite unconstrained number wise.

It's quite unconstrained.

So it can be generous with that.

That, that makes sense to me.

I mean, well, thanks for, thanks for partly.

So I'm not, I'm not crazy or both, we're both crazy or to think this, that there's some, there's some, oh, that's all right.

Um, so, so there was one more kind of big picture thing that I wanted to get to.

So we have less than 10 minutes left.

Um, and that was just about rationality.

So again, we'll see if this sparse interest or not, but, um, I think of, I mean, so rationality again, has many senses.

But like, um, you know, often when you're thinking about agents and behavior and motivation, thinking about practical rationality, like I can reason my way to obtaining some goal, uh, which is a large part of what we've been talking about.

Um, so the simple question here is just, well, insofar as motivation, intrinsic motivation does not involve that. Insofar as we have like entropy seeking behavior and it's not just in the service of obtaining a specific goal.

Um, is it, what's it's a relationship to rationality?

Is it a rational, irrational?

Does it have to do with founded rationality?

Um, that's the question.

Um, I kind of like to think there's something super, super rational about this.

That it's not, it's not, it's not, not everything we do has to be in the, in the pursuit of some, of some.

Yeah.

I, I agree.

And yeah, I think from, I mean, I've been struggling with long-term illness and it's made me realize that the ends that you describe in rationality, the goals are like for healthy people.

Like for a sick person, it's just about expanding your action radius, gaining ground and just being pushed back and, and like expanding your world in a very literal sense and gaining some freedom, really.

And that's the basic thing.

that's the basic thing and of course you have to be eating or be well fed and but it's somehow natural for me to see that the freedom and the empowerment really is primary and only once you have like discovered the goals you can start to optimize so you have to form the goals and that's true I guess some form of maximum entropy or empowerment but yeah it resonates with me to think that it's very much primary yeah thanks that's interesting that resonates with how

I've been seeing it as well so yeah I'm not sure I'm not sure I have much more to add to that actually thanks for Sandra I was gonna ask you to chime in and I think that was a good moment to do

so um yeah just to get a bit back to sorry to I don't know whether I cannot articulate it but like somehow like this this primacy of like the maximum entropy it seems like I mean I've been reading Rovelli who's taking like this kind of more perspectival more interactionist view of of thermodynamics and maximum entropy and and this seems to map onto somehow onto environment for me that it's a relative measure a perspectival measure I cannot articulate it because yeah I'm not on top of the the physics of it but it seems to be that there's a link between the kind of like the blind entropy of course you have where heat is spreading where where gas is diffusing you have like the the structured entropy where or constraints are used to to better dissolve gradients and then you have like some kind of empowered entropy where where it somehow reflects on itself it it models its own the entropy prediction it it starts to um model that and optimize like the expansion of the of the paths the possibilities um with this agency then of course um but that seemed to be like like sim very

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

My name is William Gebhardt.

I am representing the Neuroadaptive Computing Laboratory from the Rochester Institute of Technology in New York.

I'm here today to talk to you about NGC Learn, specifically the V3, which is our upcoming release of NGC Learn.

It is a fast and modular computational neuroscience library.

And with the acquisition of V3, it just gives you a bunch of new tools to work with.

So first off, I just want to give a very brief overview on what computational neuroscience actually is.

So computational neuroscience at a glance is basically just the study and investigation of the brain functions using mathematical computational tools and theories.

And what this means is that instead of going to math first and then trying to claim that we are, you know, using things like the brain, we go to the brain first and then try and model the processes that happen in the brain with math.

Your brain is not actively doing a bunch of matrix multiplication.

Look at memory.

And these are all just the various things that we look at in computational neuroscience.

Now, what this kind of means is that we need tools to work on this because the brain is complex.

For example, if when our brain is interpreting the world, it is receiving lots of data, like gigabytes and gigabytes and gigabytes of data coming in constantly.

And what this means is that if we have a model, we need to obviously we're just going to be looking at a small pared down version of the amount of data that comes into our brain.

So we're always working with a slightly suboptimal picture.

But on top of that, your brain isn't ever idle.

So in traditional deep neural, like deep learning, you take an image, you feed it into your model and then you get an output.

So sometimes we do things what's known as recurrent neural networks where you give it an input, it goes and it processes and then you give it another input and it uses the output of the previous one and the new input and it processes.

That's closer to what your brain does.

But the problem there is all of the little internal parts of that network don't have values.

And so like we want to make sure that we're representing things with biology.

Sorry, was there somebody speaking?

Okay, sorry.

And so then this biological interpretation also lends itself to how we get inputs into our models.

So for instance, the focal point of your human eye can only really process things at roughly 60 frames a second.

Your periphery processes things a lot faster than that.

But the focal point doesn't process much faster than that.

But in between those frames, it doesn't sit idle.

And then obviously different sensors in our bodies sense things at different times.

For instance, the periphery, the rods in our eyes process things a lot faster than the cones in our eyes.

And so then we work inside of this biological framing that I was alluding to earlier, where when we model our neurons,

when we model our neurons, we want to make sure that they are stateful and that we are working in a more time aware system.

And we do this because your brain is a time aware system.

And so we still use a lot of the similar words that you're going to hear in traditional deep learning.

But for us, when we say we are talking about a synapse, we are talking about the actual modeled connection of the synapse.

When we're talking about a neuron, we are talking about voltages, currents, membrane potentials.

We are talking about the physical, biological aspects of a neuron.

And what this also means in our biological framing is that the way that information is transmitted from point A in our network to point B in our network,

it can't just send a floating point number because your brain doesn't send floating point numbers.

The way neurons communicate in our brains is with pulses.

For us, we call them spikes.

And so this in and of itself makes things a lot more complicated when it comes to modeling.

Because if we're sending discrete pulses over time, we have to do a full forward pass over our model to figure out where pulses happen,

send pulses, and then we have to do that again and again and again and again.

And basically what it looks like at the end of all of this is we have about a 200-layer recurrent network where it's just going down for like 200 time steps.

But really, it's not a recurrent network.

It's just constantly processing new information and constantly giving you some output, but that output might just be null.

So all of this comes together to basically say that everything works better as just one giant state machine.

So we have our model.

We're calling it S.

S.

There's something going.

Like there's always values that exist.

And the main thing to keep in mind with all of this is that these different initial conditions and initial starting state vastly change the model's output, even if it's shown the same information.

So that's kind of the overview of computational neuroscience, state for modeling, and kind of setting up our problem space.

Now, this is not a field that came about two years ago.

This is a field that people have been working and tinkering around in for a little bit of time now.

So that might break the question of, well, somebody probably has built something before, and they have.

I'm going.

I'm going.

But the thing about building previously built models is that they kind of generally operate as black boxes.

So because building these large stateful systems incurs a lot of overhead, most libraries that exist today hide a lot of what's going on.

And you have some very specific tools that you can use to modify the internals of your system.

But you don't necessarily have all the tools that you want.

If you want to come up with a new learning rule, maybe you don't have all of the values that you need for that learning rule.

And there's no good way of modifying the system internally.

And so testing new components and everything is not easy.

It's not trivial.

It's something that takes a fair amount of time.

And if we're testing models that take a long time to run, it means that you can just lose days, weeks to just trying to get a model to see if it'll do anything.

And you don't know whether or not it'll work.

And just as a reference point here, some of these alternative models take up to like eight hours to do an EPOCH over MNIST for a moderately sized spiking neural network.

Which is a lot.

And so this is kind of why we went to, like our lab, went to building our own computational neuroscience library.

And so that is called NGC Learn.

That is what I'm here to talk about today.

And basically, this is the little highlights reel and then we'll get into exactly how it's working.

But it is a public pip package.

You today, right now, could go and type in `pip install NGC Learn`.

And you will get a version of NGC Learn.

I will tell you right now, you're going to get 2.0, which is a perfectly functional version.

But it will not look exactly like what I'm talking about today.

NGC Learn at its core, though, is a state machine that sits next to JAX.

JAX, which is effectively the linear algebra library for NumPy that was the backbone of TensorFlow before it split.

And JAX allows us to do things that are known as just-in-time compilation, which allows us to take a bunch of Python code that we know has fixed sizes and fixed types and just produce effectively a C code snippet that

we can run on GPUs, TPUs, CPUs, whatever.

And in doing this, our models in some preliminary results, you can get from something that takes about 8 hours written in TensorFlow to something that takes about 15 minutes on the same hardware written in NGC Learn.

And then the advantage of this is that it also allows for the creation of new methods, new models, new components, whatever, incredibly easily.

Because as long as you follow a very simple coding paradigm, everything will interface with it perfectly.

So let's get into what some of that looks like at a high level.

So if we're going to build a model in NGC Learn, we do have to slightly change how we are thinking about modeling a network.

Specifically, for us, it is just one giant state machine.

But if we just treat it as one giant state machine, it gets very hard to keep everything organized and not end up putting values where they're not supposed to be or just, you know, moving all the data around.

So we use components inside of our state machine, which here we're calling a context.

We use components to kind of hold parts of the state together and modify only a sub part of our state and then be able to communicate with other components, which just also encompass a different part of our state.

And this whole component model just allows for us to keep easy track.

And it's basically bookkeeping that allows the end user to say, I'm going to take these same values and do something new with them.

Similarly, inside of a component, we have a compartment model, and this is where data is stored and is moved around.

And this is what allows us to very easily move data from one component into another component without a lot of overhead.

So more specifically, looking at components here, here we have one for what is known as the leaky integrate and fire neuron.

Now, specifically, the math that happens inside of this isn't particularly relevant.

But know that this is one of the most common neurons used in computational neuroscience for modeling.

What it is, is it is effectively a it's effectively a capacitor where it slowly builds up current, passes a threshold, discharges, emits a spike.

Now, what this means is that there are some fixed parameters.

If we look here, the number of unit, the number of neurons in our like layer of leaky integrate and fire neurons is fixed.

We're not growing new neurons magically as this is going.

That voltage threshold that it's slowly building up to, that's also fixed.

We have some time constants.

We have a resistance value and we have some voltage rest and reset values.

These are all fixed parameters.

These don't change in time.

You set them up once at the beginning of your model and then you just call run.

Now, they can vary from component to component.

But as far as time is concerned, they are fixed.

Then we get to the compartments, which are the variable parts in time of our component.

So here we have current as an input into our component at a various time.

We will have a current.

Similarly, if we pause at a different time, we will have a voltage.

And at that same voltage, maybe we have an output spike.

And we also have the last time that we spiked.

So we keep a track of all of these values that change in time.

And this allows us to think about our components in two steps.

We have here's every we have here's all of the math and the logic and the fixed parameters.

And then here we have a couple of values that we need to make sure are updated at the end of every time we are called.

So that way we keep up to date with our state.

Finally, how do we connect these together?

So if we have component A and we have its parameters, which are our fixed values, and we have our compartments, which are our variable values.

And then we have component B, which has different parameters and different compartment values.

But we want to take the output of component A, and we want to make sure that it gets put as the input to component B.

Oh, sorry.

That just means that you're just going to take the value here, and it just gets passed along.

And it's done, and it traverses along what is known as a cable.

And so inside your model, you can take any compartment value, and you can wire it with a cable into any other compartment value.

And what that means is that when component B looks for the value inside this compartment, it will just know that it actually should be looking at the value inside this compartment.

And so this is how that we can keep things nice and fast, is there's no duplicates of values.

But effectively, it traverses along it.

Now, with component X, Y, and Z down here, we're doing something a little special.

We have a summation.

So now it's entirely reasonable to assume that the inputs into our component need, like, we could have a variable number of inputs.

One of the best examples of this is if you have a recurrent connection inside of your model that are both feeding into, say, the current value of your LIF neuron.

What that means is that you would have going down your model, you would have the current coming in, and then you have current from somewhere up in your model also looping around.

And basically there, we might want to add these together.

We might want to subtract them together.

We might want to have them multiplied and have some basic arithmetic operation done to them.

And so we allow you to do that along these cables.

You can basically bundle these cables together and then just say, hey, add all of these together.

And you can do that.

And this compartment here knows how to do all of that.

So when you go and try and see the value of this compartment, it will automatically collect these values, add them together, and use it.

And you, as the end user, don't have to make sure that that happens.

And if, for instance, we unplug component Y from the model, it just runs.

If we add in a component alpha to our model that we want to add to this summation, we just wire it into the compartment like anything else.

It all just works together.

Now, this shows us how to set up our model, but we're a state machine.

So we actually have to be able to transform through states.

So transitioning our model through time.

So here we have our state transitions across a component.

We're going to look at these at specifically a component level, and then we'll look at them as a whole system, as our model.

Here we have component A.

It has our fixed parameters like before.

And here we have two different state transitions that we want to account for.

We have our reset transition, and we have our advanced transition.

Now, there's this little helpful key up here where anything that's a solid line means that that depends on those values.

And if it's a dashed line, that means it updates those values.

So we'll notice here inside of our reset function, it depends on this fixed parameter and this fixed parameter.

And then what it does is it just updates every compartment value inside the component.

Now, in the advanced state of our state transition, we'll notice that it takes this compartment value, this compartment value, this compartment value in, as well as these two fixed parameters.

And then it updates this compartment and this compartment over here.

And you can set these up in as complex of methods as you want.

Here, any amount of math can happen.

The only thing that we have to keep in mind is that these values are fixed and these values change in time.

That is the only thing that we need to keep in mind with this.

So if we want to do this across multiple components or what we call a process on our model, basically what would happen is we have some overarching process here.

We just have it called advance.

And what it's going to do is it's going to call advance here on component A.

It's going to compute all of its outputs.

Then since these are wired together, when we call advance over here on component B, it'll actually look at

the output of component A and use it for component B.

And then it will compute its output and put it inside its output compartment.

And if we were to keep chaining and have a component C over here that was wired out, it would just keep going.

And so this allows you to chain as many of these components together inside like a forward pass as you want.

It's also important to note that these do not have to be the same type of component.

They do not have to be the same type of transition.

So here it's all named advanced.

But as we'll see later, it doesn't always have to be named.

It doesn't all have to be named advanced.

Yeah.

And so then, yeah, and it's always deterministic, the order that all of this data gets transmitted in.

It will always go in the order you define the components in.

So you don't have to worry about race conditions and anything like that.

So finally, in order to actually get speed up from this, because right now we've just written basic Python code, we actually have to go through what is known as a compiling step of our process.

And this makes use of Python's abstract syntax trees, which I am not going to get into the complications behind getting syntax trees to work.

But basically, we look at your code that you have written in the generic Python classes, and we tweak it just a little bit.

And we'll see how it tweaks it just a little bit to produce code that is fast and efficient for JAX's just-in-time compilation.

This is the true source of NGC Learn's speedup, is that it can take code that is written in a very easy-to-read, effectively format, and then convert it into what is needed to be efficient for JAX's just-in-time compilations format.

And so basically, we just take that whole process and we send it through our compiler.

Now, the compiler as an end user is not something that you need to ever actually worry about, but there are advantages.

But with V3, and this is what we're getting into here with V3, there are some things that you gain from, that you gain that allow you to look and really inspect at what is our model doing?

What is our process actually doing? Which is very helpful for things like debugging.

So, what does this actually look like in code?

So, starting at the top, if we wanted to find a compartment on our class, right?

So, these all have selfs, so these are all inside of a class.

We can have inputs, and it's just compartment with some initial value.

Outputs, compartment with different value.

Weights, compartment with a third value.

This is how you make a compartment.

Now, there are a couple extra parameters that you can pass to compartments that are useful for things like plotting and whatnot, including, like, compartment names, units, all of that.

Then we have fixed parameters, which are just exactly like you're used to defining values in Python classes.

Just `self.shape` equals `shape`, `self.rscale` equals `resist scale`, whatever you want to call it.

You could just define them like regular values.

Then, when we want to define our advanced state for our transition here, we have our outputs, which is our compartment up here.

We're setting it to a matrix multiplication of our inputs and weights, multiplying it by some resistance scale value, and adding biases to it, which is not displayed here, but there's a bias compartment as well.

And what you'll notice is this is just a weight matrix.

This is just inputs, weight matrix, outputs.

And we'll notice that this looks pretty similar.

This `outputs.set` might look a little weird, but just if you think about this as just an equal sign.

This looks incredibly similar to regular Python code that you're used to writing.

The only thing that's a little bit new is that it has this little compilable flag above it.

Now, you might think that there's a lot of work that goes into that at compilable decorator wrapping around this.

And I'm going to give you a secret.

There's not.

All that is doing is just adding a singular flag to the method.

It is doing nothing beyond that.

So then we get to, okay, so now we have our component.

We have our state transition as an advanced state.

How do we use these, right?

Obviously, that's where the complications got to come in.

So not really.

So we have contexts, which if you remember are just our global state.

Basically, everything has to be inside of a context.

So we have with the context, your context name as some variable.

And then you just define them like you would regularly define Python objects.

So here we have our static synapse, which is a weight matrix from our inhibitory neurons to our excitatory neurons.

It has a shape.

It has a weight initializer, which we provide the ability to just get distributions generated for you.

So here we can use our distribution generator.

We can get a constant distribution.

And we can say that we want it to be hollow, which is the main diagonal is all zeros.

And then when we go back, we can say, nope, this one, we actually want it to be an identity matrix.

And there's uniform, like there's all the different, we have most of the various distributions that you're probably

looking for supported in this.

And all this is actually doing is just returning a method.

So these all get added to your model automatically, right?

And if you're looking at this, you might go, well, we still don't see where all of the compiling happens.

Where is this big step that gives you all of the speed up?

And I will tell you, it is just whenever you leave this with statement, you have to add no extra lines of code to your project.

It will automatically compile all of the methods.

So we will compile this advanced state method as soon as you leave this with block.

So it becomes incredibly easy to read.

You don't have to remember to call compile.

You don't have to do any of that.

It just happens automatically when you leave the with block.

Another thing that can be useful if you're building really large models and wanting to have nested functionality or repeated functionality,

you can make subcontexts or submodels inside of your main model.

So here we have our components.

Previously, we talked about cables and how we would connect the two things together.

This is the incredibly complicated syntax to do that.

So basically, here we have our Z1 excitatory neurons, and we have their spike output, or just S.

And then we wire it into the inputs of our weight matrix.

That is it.

There is nothing more that needs to be done with this.

It just is basically shifting it just directly into there.

Similarly, we can take the outputs of our weight matrix here, and we can shift it into Z1 inhibitory's current matrix, our current compartment.

And it just goes.

And if we want to do that fancy summation operation that we were talking about, it is just summation with your compartment values.

And you can just keep tacking them on here.

It's not two.

It is as many of them as you want.

We also support products.

We support negation.

We support a lot of other values here.

And you can also define your own bundling mechanisms, too.

So now we've got wires.

We've got components.

Now, the last thing that we talked about was a process.

So here's how we make a process.

This has a little bit more looking Python code, but it's incredibly repetitive.

So here, so there's two different types of processes that we have.

We have what are known as method processes, which are basically a process, whereas we are going to go over components and we are going to be calling transitions on them.

So here we are going through all of the components of our model and we are calling their advanced state method.

Now, theoretically, if we wanted to call something like, let's just say we were using somebody else's component and they didn't name an advanced state.

They just named it ADV because they wanted to shorthand it.

You could just type .ADV.

It will work.

These do not have to be the same name.

Similarly, the name of the component of the process does not actually mean anything relative to these values here.

Then what we might want to do is we have this advanced process, which does a forward pass over our model.

And then let's just say we want to add that we have an evolve process, which basically would be the equivalent of our update step over our model for all of our weights and so on.

And we call that evolve down here.

And then we want to join these together.

So we only have to call one method to do both of them.

And that is what a joint process is, is these are joining other processes together.

And you can nest these as deep as you want.

You can make multiple joint processes all in one.

You can make you can call an advanced process multiple times in one of these.

If you wanted to do advanced, advanced, evolve, advanced, advanced, evolve, it will all work and it'll just keep going.

So this is the processes set up.

And these also compile.

Right.

We talked about that.

These also compile.

And similarly to the width block of our model, as soon as we leave that width block, these are defined inside the width block.

As soon as we leave that width block, these also automatically compile.

You don't have to worry about getting things to compile in the correct order.

All of that is taken care of it automatically for you.

So what does this compiling actually look like under the hood?

So here again, we have our weight matrix for our forward for our forward step or advanced state.

And we're going to compile W1EI here.

Now, it's important that we remember that the spike output of Z1E is being wired with a cable into W1EI's inputs.

So if we look up here, we have advanced state.

The name changed a little bit.

It's now circuit underscore W1EI underscore advanced state.

That is because this is the name of our model.

This is the name of our component.

And this is the name of the method that is being compiled.

And notice that the input has changed.

The input is now CTX, or it stands for context.

So this is the global state being passed into this method.

Then what happens is this self.outputs gets changed to circuit W1EI's outputs is equal to.

And it's just this directly mapped over.

With we'll notice that the fixed value here becomes a global fixed value here.

And that self.inputs because circuit Z1E.s because it's wired here.

And so this just all goes through.

And so you give it a context.

It mutates the context where it needs to.

And then it gives you back the context.

So this is how we compile a method.

Now, for convenience sake of the presentation, this is...

I've got syntax highlighting on for this.

Practically speaking, you will never actually see what is down here.

This is all done dynamically and stored entirely in memory unless you get it to print it out yourself.

And then if you want syntax highlighting, you've got to copy-paste it into a Python file and get your IDE to syntax highlight it.

But practically speaking, you never need to actually do anything with this.

But it is, however, useful when you're debugging to look at this and go,

Oh, if this isn't updated, if this is still just W1EI's inputs, we know that it's not wired to anything.

And so then we could actually debug what's going on in our model by just looking at what it's compiling to.

Which is incredibly useful because if you just forget to add a step, you can see it because the output just doesn't have the step in it.

Similarly, if we have an evolve process...

So I did this one because it's got slightly less components in it...

Where we have just W1, right?

So this is just a Hebbian synapse that's doing an update.

What we'll notice here is that here is our compiled process where we have a context and loop arguments.

Now, we're not using loop arguments, but this is where you would actually give it parameters that you are passing into the forward step of your model.

So this is things like DT.

This is things like the current time.

This is things like the image you're looking at and whatnot.

And so then it goes through.

And I don't have the original code here, but you'll have to take my word for it that this compiles and looks correct.

But importantly, we'll notice here we have this W1 underscore compute update, right?

What this here is, this is another method that exists on the Hebbian synapse for doing its update.

Well, we also...

We successfully capture all of those two.

You can make as many submethods as you want as long as they all are just...

You know, they take in self, they take in values, it doesn't matter.

And you can call them.

And it all just builds together perfectly.

So how does this actually look when you want to, like, use the process, loop through the process?

What are you doing?

So for convenience, you can call your model.

You can call your forward process.getKeywords.

So for us, for this, that'll give us T and that gives us delta T, right?

You don't have to do this, but this will tell you all of the keywords that it is looking for to actually run your component or run your process.

And if you're missing any of them, it will complain at you.

And then we can pack all the inputs together, right?

Because, you know, we're going to be looping through all of these.

We need to actually make all of the inputs that we're going to call.

Well, luckily, that's really easy because you just call pack the rows of all of it and you tell it the number of iterations.

And then you can do one of two things.

You can give it a lambda function or just any function, really, that takes in X, which is your iteration number, and produces some value.

The next thing you can do is you can also just give it constants.

So dt equals one.

And so what this will do, if we say iteration is 10, this will give us basically a matrix that has T is equal to zero, one, two, three, four, five, six, seven, eight, nine.

And it'll have dt is equal to one at every single time step because it's a constant for the pass.

And then what we can do is, okay, so we have all of our inputs now.

Now we want to actually loop through all of those inputs.

Well, you can just call scan, which is very similar to Jax's scan if you've used it, if you've used any scan method.

I believe TensorFlow still has a scan method in it, where basically it'll just loop through all of the inputs that you give it in effectively a for loop.

But it is doing this all compiled in C, so we don't actually want to write our own for loop.

This is dramatically faster.

And the output of that will give you, hey, this is our spike output, which is something that we set when we compiled all of our method together,

and our final state of our model that we can do something with.

Because as it runs through, it doesn't necessarily actually change your model.

It just tells you this is how your model will change, and then you can choose what you want to do with that.

So there are some current papers being written with this, some current applications that have been written with this.

So we have a neural coding frameworks for learning generative models done by Arobia and Kiefer.

This one makes use of NGC Learn.

We have a reproduced Dillon Cook's model of unsupervised learning of digit reconstruction using spike timing dependent plasticity.

We have my own personal paper with my advisor called Time Integrated Spike Timing Dependent Plasticity.

That is also made in this model.

And then we have a contrastive signal dependent plasticity, self-supervised learning and spiking neural circuits, also done by my advisor, Dr. Arobia.

And you can find all of these models and more, actually, in a sister library known as the NGC Museum.

So this is a library that our lab works very hard to keep up to date for at least our projects for the latest version of NGC Learn.

And it houses all of the biomimetic and brain inspired computing.

And if you write a paper in NGC Learn and you want to add it to the museum, we are happy to add it to the museum.

The whole point of it is to be a public place to share code because there's nothing more annoying than reading a paper, looking at the coefficients that they claim to have used, reproducing their work and it not doing what it's supposed to.

So this is a place where we can store all of that.

It's also a place where we have some historical works.

So that way, if you want to get a baseline from some historical paper but don't want to implement it all yourself, we can do that as well.

And so with that, V3 should be releasing later this month.

I will say with Thanksgiving, it might get pushed back to early December, but we are making great strides and it should be up soon.

And with that, I will open the floor to questions.

Very cool.

So people in live chat can ask questions.

I wrote some down as well.

So just to stay close with the theme of the talk, what would you summarize are the major differences between NGC Learn 2 and 3?

So the main difference between NGC Learn, sorry, it's going to flash for a moment, is this right here.

So in NGC Learn 2, there was a very specific setup that you had to do when defining this advanced state method and you couldn't use self, which meant that you had to do a bunch of pattern matching yourself to make sure that all of the names were the same and that all of the inputs were correct.

Now, because we're making use of abstract syntax trees, you don't have to do any of that.

So you can just throw Compilable on a class method that has self all over it.

And we just extract that all away and turn it into the global context that you actually want to be referencing.

Yeah, awesome.

That's a big...

It's also faster.

I don't know if I mentioned that.

V2 is roughly 10 to 20% faster.

I'm sorry, V3 is roughly 10 to 20% faster than V2, which on the grand scale of things isn't that much faster, but it is faster.

Yes.

Yes.

Very cool.

So maybe just to kind of ground that, in terms of expert time or the size of models or networks that you could reliably render and execute, where does this move the frontier?

Like, was that a limiting step or was that preventing large systems from compiling or what?

So I think one of the things that it really stops is I think the time element stops people from getting into the field.

Pretty much everybody who has been working in the field for a while has their own little custom whatever that took them 5, 10 years to get, you know, to work and iterate on that gets them to something that's fast, right?

Because they've taken the time to figure out how to speed it all up and optimize it.

But if we have new students, if we have new PhD students, if we have students in a class, they don't have five years to optimize their code base in order to do preliminary tests.

What we allow for that is we just allow you to kind of be a little hands-off when it comes to how this is optimizing and becoming quick.

I will also note that this does not require fancy GPUs to actually run quickly.

That 8-hour and 15-minute benchmark is done on my laptop with no GPU.

That is the CPU speedup.

The GPU speedups are equally as fast, but you don't need it until you run out of CPU memory, which is always a constraint.

Thanks.

Thanks.

Okay.

Continuing with the questions.

Okay.

So does NGC Learn allow a different scheme or ontology other than the neuronal?

Like you had the block diagram with the voltage, current, spike variables.

So can you specify arbitrary stateful models in that block?

Yeah.

So here where we're defining this, this is just a class.

So this right here is a weight matrix, right?

This is our dense weight matrix where we have, it's just class weight matrix.

And then you do have to inherit from Jack's component, but like that's, that just gives you all of the backbone that you need, but doesn't actually require you to do anything.

You can define as many compartments as you want, as few compartments as you want.

You can define any number of fixed parameters that you want.

And you can define as many state transitions as you want with as many different names as you want.

It will just all find them and work.

The only thing I will say is that because this is a giant state machine, obviously if you have tens of thousands of changing values at every forward pass, it's going to get slow, but that's not a fault of the library.

Any library is going to get slow with that because just changing that many values in Python takes time.

So to those internal calls or nested calls have to remain within the NGC semantics or where would an external API call or another routine or program, how would that be brought into the design of what's that?

Here's Jack's matrix multiplication call.

This right here is just Jack's multiplication matrix multiplication call.

You can just call them.

It doesn't, it doesn't care.

If it's not a class method, it doesn't try and do anything to it.

So if it's not a class method, it'll just call it.

Now I will say that if it calls something that mutates things, that it probably, like, because this is all built on Jack's, Jack's generally speaking expects things not to be mutable.

Now this is a copy of what is passed out above just from the way Jack's works.

So like we can mutate it here, but like you don't, you want, you want your methods to

return values.

But that is a Jack's specific thing.

But practically speaking here, we have, this is Jack's numpy form matrix multiplication call of these two inputs.

Inside of NGC learn, we have an entire ordinary differential equation backend that you can just reference.

Like it's just NGC learn dot like ODE dot whatever ODE you want.

And it'll just work.

And like, that's not something that's like written specifically to work with the compiler.

It just allows it to function.

Like you, it just doesn't care if it's not a class method.

It's just, it just works.

Interesting.

But it might hang for a while.

If there was call local LLM method inside, would it spawn a whole Python thread?

Uh, so it's not necessarily going to spawn threads in that sense, but yeah, if, so for example, if I have, uh, if I'm doing a matrix multiplication of two gigantic matrices that is going to take like 30 seconds to do without any of NGC learn around it, just using Jack's optimized matrix multiplication with JIT compiling on a 30 by 30 mate, like a matrix that takes 30 seconds.

It's still going to take 30 seconds.

We are not speeding up the linear algebra.

Uh, we are purely speeding up all of the Python that sits on top of the linear algebra because it's really slow.

Writing values is really slow.

And so we are trying to make it as fast as possible without rewriting, uh, the accelerated linear algebra library or LAX, uh, which is what Jack's is built on top of.

It's the numpy bindings for all of LAX, which is what everything uses.

If you come up with a linear algebra method that is faster than LAX, I highly recommend you tell them because their entire mission is basically making it as fast as possible to do linear algebra on a computer.

Cool.

Interesting.

Okay.

A note in the chat, and then you could add a comment or I'll ask another question.

Note to audience members.

NGC learn does also support convolution, deconvolution, sparse synaptic transforms, et cetera.

In addition to a bunch of historical spiking and rate coded neuronal cell models.

NGC learn also supports generic iterative differential equation integration.

IE Euler midpoint RK4.

Yeah.

So I assume that's coming from my advisor in the chat.

Yes.

Yeah.

So yeah, I don't, I did not have space in this presentation to include all of the historical, like all of the components that we support.

But if you are used to using generic founded, uh, computational neuroscience components, there is a very, very good chance that we already have them in the library and you don't have to write them yourself.

So pretty much all of the, all of the, and that includes both rate coding and spiking.

Like we, you can do rate coded with this.

Um, I will say that it might be overkill for some rate coded models, but it would work.

Um, also technically speaking, you can do back propagation through this.

Uh, JAX does do back propagation, like, like gradients and whatnot.

Um, spiking neural nets themselves though, are not differentiable.

So I'm not going to get into covering it here.

And we don't truly have a tutorial on it because it's not something that we're trying to do.

Um, but it is possible to do it.

And I have gotten it to work fairly easily.

Okay.

I just have a few more questions.

So often in, in modeling, whether sort of a monolithic, monocomposite type model, or more of a synthetic intelligence approach, like nodes that have different kinds of programs running.

There's this continuum between taking a bit of a more problem agnostic architectural approach, like off the shelf neural network architectures or domain specific architectures, and then just kind of throwing in the training data going from there.

So that situation, the, the semantics of your model and what it does are being shaped kind of from a flat prior with the training data set.

And then on the other hand, sometimes there's, there's very situation driven modeling, even into the toy modeling space where people are looking to encode the semantics of like anxiety level as a parameter itself that might be undergoing some kind of Bayesian updates, but on the interpretable parameter itself, as opposed to, to that more machine learning driven approach without looking to constrain the.

How, how, how the topology of the initial network relates to the data set and the function.

So when you're designing these kinds of systems, where does it exist between like a super nuanced way to specify a neural network that's agnostic and trained sort of like a convolutional neural net, or do you encode in more direct fewer parameters of interest?

So, uh, I'm going to say that you can do both.

So we are specifically a library.

We're not, we're not gunning for a specific method here, right?

I'm not here to tell you that method A is better than method B.

Um, so with this sort of thing, with like my personal research and my use with this, um, a lot of the time is driven by, I am looking at the, so I've done a fair amount of work with reinforcement learning and similar spaces like that.

Um, in the context of computational neuroscience.

Um, and what that kind of leans to is very much thinking about, well, this is what my sensor data looks like.

So let's make sure that like the encoding scheme that we're using actually matches up with that.

Because like, sure, I could go take some generically large LLM and just throw data at it and hope that it works.

Um, but that's not necessarily going to be efficient.

Uh, computational neuroscience as a field of research also has a sister field of, uh, hardware research called neuromorphic computing, uh, which generally speaking, uh, imposes a lot more constraints on things.

Uh, so when I'm building models, I tend to work with a lot more constraints of physical constraints, uh, for like value types and, you know, magnitudes and the way information gets moved around.

Um, so you can definitely do it all the different ways here.

I would say one of the nice things about NGC learn, uh, though, from the standpoint of trying to come up with a new component, for instance, is the fact that you are just coming up with a new component.

So if let's just say I wanted to go out there and I wanted to make a new neuron, right?

Because what I was looking at advocates for me making a new neuron, um, you don't have to do a bunch of extra steps to get it to interface with everything else.

Um, you just have to define the logic basically and wire it up correctly and it will just integrate into your model.

I'm not going to tell you that it's going to work as far as like practical side goes, but as far as coding side goes, that's all you would have to do.

I hope that answers your question.

Yes.

And then here's my last question, um, or semi penultimate.

And then I'll see if there's any in the chat, but how do you see or go from the forward design phase and generating synthetic data and outputs from a priori specification?

How do you also work with the empirical parameter estimation, whether you were looking to design a sort of model level property like learning or, or are you fitting this to spike train data and looking to fit it at a more micro component level?

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

So some of that, so it's going to use the, the homework frame of it depends.

Um, so if we are in computational neuroscience, if we are looking at a toy problem that was not originally designed to be working with spiking, that's the biggest example of this is MNIST, right?

MNIST is a bunch of static images, uh, that have floating point values.

Uh, we don't have a bunch of, we don't have like, like we're not static models.

Our outputs aren't static.

Our outputs are just a bunch of spike trains.

So we do have to represent that somehow.

Um, and so we, uh, it is something that in NGC learn, we do try and offer some, uh, support for doing things like that, uh, for helping converting, collecting whatnot, um, in the models that we've built in the museum, uh, that can show like, yeah, here is one way to do like clustering with this.

This is one way to assign labels to output neurons.

So we do that sort of thing is where it's like, okay, like we have to make the synthetic data effectively work to something.

Um, but the, or I guess it's not quite synthetic, but like we have to basically make our own labels, um, that means something to us.

Um, and then on the follow-up side of that, when it comes from like, do we work forwards or backwards from either the real world and synthetic data or synthetic data towards the real world?

Um, I would say that good science does both, um, sometimes finding a real world problem that an idea fits to is challenging.

Um, and other times you have a real world problem that you're trying to solve.

Um, and therefore you definitely should be working backwards from it.

Very cool.

So now the actual last question, first you can add anything you'd like and just where are you at in your overall research journey?

What are you looking to have done by the end of this phase or just what's next for NGC learn and, and how you want to develop and do that and other things?

Yeah.

So I am in my fifth year of my PhD.

Uh, so I am slated to defend my dissertation fall of next year.

Um, I really like working in the space of research tools.

Uh, it is something that I have a huge passion for doing.

Um, and so research tools and more recently, specifically research tools for hardware applications.

So I mentioned a little bit earlier, uh, there's the whole sister field of putting spiking neural networks onto physical hardware.

Um, I have done some research in that recently and it is a lot of fun and I enjoy doing it.

Um, as far as more immediate plans of stuff, um, there's a survey paper that we are currently working on that should be coming out soon.

Uh, in relation to spiking neural networks and computational neuroscience.

Um, in addition to that, uh, coming out either December this year or January next year, uh, there should be a white paper actually out for NGC learn on its development, the thought process that went behind it and everything that was long as well as that.

Um, beyond that, um, I have writing my dissertation on the to-do list and probably one more, one or two more

research projects in there somewhere that we have not figured out yet.

Cool.

Great.

Yeah.

A lot of exciting reasons to explore NGC learn.

I have just cloned into it on another computer.

So I look forward to, uh, learning more about it.

If you clone into it right now, if you are using it through PIP, it will be V2.

If you clone it from the GitHub repo, if you just check out the V3 branch, uh, you will get all of what we're currently working on.

That is the dev branch for version three.

Um, but the release branch should be updated within the month.

All right.

Okay.

Thank you, William.

Good luck with finishing your PhD.

Thanks for having me.

All right.

Bye.

Bye.

Bye.

Thank you.

Thank you.

Thank you.

Thank you.

Interoceptive posture pictures for active inference informed neuromusculoskeletal NMS therapy.

43 minutes long.

Here it is.

Interoceptive posture pictures are a tool and a methodology for practical active inference informed neuromusculoskeletal therapy, as well as for people who want to move well as they age.

Interoceptive posture pictures, IPPs, are a tool and a methodology for active inference informed neuromusculoskeletal therapy, muscle, bones, joints, being able to move well, which is why it's also an amazing tool for anyone who wants to move well as they age.

I'm a neuromusculoskeletal clinician and educator.

And to me, active inference is a framework to connect and shape people's mindsets towards healthy behaviors in both body and mind.

Now, I understand that AIF is part of a broader understanding for PhD scientists, researchers, and others looking at how to model from a computer AI point of view, as well as understand from a personal point of view, how humans work in a whole new light.

I believe that for clinicians, patients, and the public, active inference can play a role in preventing the progression of, I hurt my back, back pain, I overdid it, neck pain, common stress.

Going from acute to chronic, and especially when chronic low back pain is the leading cause of disability globally, and neck pain and tech neck and normal daily stress is part of a somatic pain that a lot of people live within their lives.

Active inference can help clinicians and people help themselves.

In addition, it's also a way to mitigate full risk for aging boomers, as well as to address mild concussions, things called MTBI or PPCS, persistent post-concussive syndromes, after accidents or sports injuries.

And it's helping everyone, including you, to connect with your interoceptive sense, and your sense of balance, your sense of overall body alignment, to have the zones of postural mass balance, and have those zones of postural mass, those posture zones move in everything you do.

Balance, align, and move.

We call that BAM.

My name is Dr. Steven Weininger.

I am a doctor of chiropractic.

I've practiced for 26 years.

For the past 20 years, I've been lecturing around the world about teaching people to move well.

I'm a diplomate of the International Academy of Neuromuscular Thural Medicine.

And I'm very proud, last year, 2024, I published my first peer-reviewed paper in Frontiers of Neuroscience.

It really was a very cool thing, because I had wanted to, but never went into research, and I had gotten a whole new understanding of what it takes to be a PhD-level researcher.

I've written books, I've lectured literally around the world, and when I saw the word cloud for this conference, the first thing that popped out to me was energy principle and clinical. And the clinical application of this is something that I think is tremendous, and it's not yet been realized, especially in NMS care.

So, the title of our paper,

Introceptive Posture Awareness and Accuracy, A Novel Photographic Strategy Towards Making Posture Actionable,

came from my work teaching, which works with, well, this is some of the disclosures that I do need to talk about.

Bodyzone.com, which is public health awareness and ideas, largely focused on posture, and posturemonth.org, which is a public health awareness campaign.

And strongposture.com, which is a self-care campaign and tools for professionals mostly to give to patients to work along with those patients.

My main interest now is CESoup.com, which is online continuing education for professionals.

We teach doctors, therapists to incorporate the ideas of using a grid, the posture zone grid, which is a tool to objectify a simple posture picture.

And we certify people with this in posture practice.

Disclosure, these are things that I do have an economic interest in.

What you have an interest in is your posture and everyone does.

Posture isn't good or bad.

That framing is wrong.

It's old.

Posture is functional, which means it's strong or weak.

And the overlay with active inference is that maximizing the symmetry means reducing the prediction error, which saves energy.

Energy is the thing that life is all about.

Efficient movement that generates a fluid flow is not wasting energy.

And every unique body selects their movement strategies that match their internal models expectation of what's going to occur next, moment by moment.

And that's why good static posture and dynamic motion are not the same for every body, because not only is your body different from mine, but our experiences are different in life.

And what we've done and who we are.

I'd like to start to get you thinking about your body and how your body is basically working in what you do. So my question for you is, as I stand here right now on an overall biomechanical basis, and biomechanical means muscles and joints, how they're working together.

Right now, is my body balanced?

I've asked this question around the world to many audiences and people invariably say, no, he's not balanced, no.

And you're probably saying that right now.

But remember the question on an overall biomechanical basis, is this body balanced?

And the answer is, of course it is, because I'm not falling down.

Which brings me to the point, your posture is how you balance your body.

It's what you're doing to maintain verticality.

Humans are the only obligate biped on the planet that is a mammal.

Birds, also obligate bipeds, some other creatures in different places, but humans are the only mammals.

And mammals are unique because we've learned to do a number of things, one of which is maintain our internal energy at the same temperature.

The uniqueness of human bipedal moment to moment balance requires the inference to stay vertical.

Maintaining that takes a huge amount of the free energy that we have.

And conserving energy is a priority for anything that's alive.

This is a law of nature.

If there's an inferred prediction that has a higher energetic cost or a greater inefficiency, the organism is going to have a negative perception of that action and tend to shy away from it.

When that persists for a while, it can be felt as stress, fatigue, pain, or just feeling bad.

Human activities require accurate balance control from simple standing to complex sports movement.

Falling down is something that we don't want to do.

When you watch children learning how to walk, they fall down and they pass back up because they're made differently.

But as we develop, we become less resilient to falling down because we've learned how to stand tall and our bodies develop in that direction, which is also why.

Losing your balance in daily life is an early warning signal for problems breaking down with your health.

And you may say, well, I'm not old enough to have that problem yet.

But if your balance is weakening, you're getting a big flashing yellow light to do something differently.

And balanced assessment of how your body is standing both statically and dynamically is a sensitive indicator of neurologic health.

In 2012, maybe 2013, I saw a TED talk by Bud Craig and got me interested in interoception.

I picked up his book, read it, *How Do You Feel?*

And it sounds like an emotional, touchy-feely book, but the subtitle really got me.

It's an interoceptive moment with your neurobiological self.

The premise of this, based on a neuroanatomist that spent his time literally tracing where nerves in the body go,

is that the insulin, deep part of the brain, is where your somatic feelings of your body emerge from.

And there's a sense that drives this called interoception.

And the next question becomes, well, what is interoception?

I mentioned before that mammals are warm-blooded.

Mammals have to control their internal temperature.

And that began a process of controlling what's happening inside our body to different degrees for different mammals in different ways.

Humans do it differently.

And we've developed cognition in ways that have allowed us to socialize and to project our feelings from ourselves onto other people

and to have abstract thoughts about those in cognition.

And our whole progression over this past six or seven million years, we've developed a very unique cognition and very unique body and very unique abilities to connect with our body.

And focusing on interoception is what the NIH, the, actually the NCCIH division of NIH, in the five-year plan from 2021 to 2025 said,

we need more research about interoception because it is a huge and unexplored factor in promoting effective health and in addressing many syndromes and diseases of modern life.

The stress and the many problems developing from the incursion of technology into the activities of daily living.

You probably know what I'm talking about.

The definition of interoception from NIH is the representation of the internal world of an organism, which includes the processes by which the organism senses, interprets, integrates, and regulates signals from within itself.

Now, if you're on the biologic side of this, this is basic.

If not, I'm going to do some really fast anatomy to give you a sense of why this is so important from a processing point of view.

The insula develops in humans from the sylvian fissure, which is an infolding that's on the side of the brain.

That line that you've seen going left to right is the sylvian fissure.

When you were developing from an embryo, you went from one cell to a ball of cells to a tube of cells.

And at the top, there was an explosion of cells that started to fold in.

That stuff on the outside, the cells on the outside of the cortex, and that cortical infolding went down and got buried and became an anatomic structure called the insula.

And because it's been so deeply buried, it's been hard to study.

So anatomists have not really looked at it as we began mapping things.

Well, some of the newer technology has let amazing insights occur about this over the past 20, 25 years.

And these insights are exploding.

There's a ton of research going on about interoception.

Because when you look again, you'll notice that above that horizontal line, there's a groove going up and down called the central sulcus.

That is the line between the blue and the red.

And that line between the blue and the red is the line between the sensory cortex and the motor cortex.

And the sensory cortex is where information coming in from.

And the motor cortex is where information going out.

The coming in and the going out, there's what's called a homunculus.

Basically, it's a somatotopic representation of neural connections.

You may have seen pictures of a big nose and big, big lips, big eyes and small arms, which represent the point-to-point correlation of the nerves in that part of the body.

They're different for the information coming in and the motor control going out.

That is why there's a different sensory homunculus and a different motor homunculus.

They're roughly the same, two eyes, one nose.

But the proportions of the neural density is not the same.

When you look underneath, when you open up that Sylvian fissure and look down between the parietal lobe of the brain on top

and the temporal lobe of the brain on the bottom, you look at what's underneath it.

That's where you see the insula.

And the insula has a front and a back, an anterior and a posterior portion that controls things in the body.

It is the most uniquely complex part of the human brain, according to some people.

And it's a really good argument because it ties things together.

There's three different homunculi.

It's not just only the sensory cortex.

It's not just the motor cortex that has a representation or a point-to-point.

It's also the insula.

And that insula has homeostatic as well as autonomic, as well as allostatic input into the brain.

If you know what allostatic is, it's another big thread in what's occurring in life sciences.

And one way to look at it, which is conversation for a different day,

but if you know what homeostasis is, the ability to keep things level within the body,

allostasis is to homeostasis as prediction is to reaction.

Allostasis is predicting forward, which is one reason why active inference, in my mind,

is destined to reshape a lot of life sciences.

The low-hanging fruit, almost, is the biomechanical side of it.

Because when you start getting into some of the hormonal issues, it becomes far more complicated.

Oh, one more rather important thing from Bud Craig's book.

Emotional feelings emerge in the insula.

Craig writes that the insula optimizes homeostatic sensory motor integration and homeostatic behaviors.

Massio and Carvalho wrote a paper about interoception and the origin of feelings.

And they talk about how the interoceptive nervous system maps the homeostatic state of the body.

And that interoception is structurally and functionally different from extraoception.

Different pathways, different processing, everything's different.

It's a different aspect of your sensing your body.

It's a different way of reacting because of that different sensation coming in,
which is why a 2024 paper,

what is the role of interoception in the symptom experience of people with chronic pain?

A lot of words really well put together.

But the bottom line is, in common sense,

people who are more accurately connected to their body's inner feelings have less pain less often.

People who are inaccurately connected to their body's inner feelings tend to have more pain.

And this is why accuracy of somatic bodily prediction is so relevant from a pain point of view,
as well as from addressing the causes of the pain and empowering the person to work with and connect with
their body more effectively.

And to feel better about it.

And that emotional side of it is one reason why interoception has seen far more attention in the
psychological sciences working with emotions than in the physiologic sciences.

One of the directions within muscle and joint back pain care has been what's called biopsychosocial,
to work with the biological and the psychology and the social,

because these things affect how people deal with their pain on a simplistic level of

I'm under stress because of bad home life to a far more complex level that ties directly into active inference of
incorrect predictions

because of priors, things that have happened before that then make me look at how the world is going to act
towards me,

which then shapes how I act in the world.

And these things are not conscious, they might be, but they don't have to be.

In fact, it is a really strong argument to be made that active inference is how we feel.

Emotions are inferences about causes of your bodily state,

which is your body's best guess about, well, why you feel the way you do.

It's okay for me to feel this way.

And feelings in general are intraceptive predictions.

Your brain predicts the state of your body.

Now, pain can be a predictive error signal.

If something hurts, there's a torn tissue or a broken bone.

That's a prediction that if I move it's going to hurt.

And that's true.

When an expected and an actual body state doesn't match, you've got a prediction error.

Feelings aren't passive reactions.

They're active inferences on a moment by moment basis about what's happening inside your body.

I'm afraid to do this.

It's going to hurt.

Well, try it like this.

I thought it's going to hurt.

It's going to hurt.

Oh, it didn't hurt.

When you feel your posture more accurately, when you feel the muscle tension and the mechanics of what's occurring, or the pain more accurately.

When you're experiencing your brain's predictions about your body more accurately, then those direct sensations coming into the nervous system are more accurate to reality.

And you tend to have less pain.

And you're empowered to do things that can then help you have even less pain.

One of the biggest issues with musculoskeletal problem is people don't move because it hurts.

So they become less able to move.

So they move less.

And this is a problem that plays out over a long period of time.

But hopefully you've been living for a long period of time.

And I know many of us hope to live for a much longer period of time.

Interoception informs active inference.

Which is why looking at how someone's balancing their body on two feet, their posture, and balancing like this, that says something from a biomechanical as well as a social point of view.

I mean, our language says it.

He's kind of shifty.

He looks crooked.

He's a stand-up guy.

That two-leg balance symmetry and accuracy prediction has a dramatic effect.

In fact, it's a proxy, in my mind, for how your one-leg balance is expressed.

If I'm balancing like this on one side, or like this, or like this, whatever it happens to be, those motor patterns will be directly reflective of how I move in everything I do.

And this is one of the profound implications of active inference.

Biomechanics is more than the physics.

It's also the predictions and processing that's occurring within my body about how I'm going to move.

The old view is that muscles respond to muscle loads, and the nerves tell the muscles what to do, and that's all there is to it.

The active inference point of view is that muscles activate based upon the predicted loads.

And they also preload in that prediction, which means that, well, I'm just tight like this.

Because your body is preloading based upon actions that have occurred between your predictions and what's happened.

And the predictions may be stress, may be threat.

The effect is that people that live with anticipatory muscle bracing, have a protective guarding based upon

reactions, are moving in patterns that initially will compensate, would move differently, which is one of the concepts we talk about, that are functional. You can change it if you focus on it, but ultimately it adapts.

The structure changes.

When I use this muscle more and that muscle less, this muscle gets stronger, that muscle gets weaker.

Your brain distributes the load based upon the capacity of your body.

Is it the predicted capacity or the actual capacity?

It's the predicted capacity, not what's actually possible.

It's what you think you can do.

Even after the injury, the heel tissues may keep on being protected because you've learned that pattern.

That outdated prediction is carrying forward in the express pattern of how you're moving your body.

And that's why you end up with chronic load syndromes going on when you're putting more load on something.

I'm just like that way.

That's the way I move.

I don't know.

That's just me.

That asymmetry of function accelerates breakdown and asymmetrical development, creating a negative spiral.

That's the model that we look at.

And I believe that this is very much the active inference model that is creating a more accurate way of looking at the world.

George Box is a noted British statistician, has a great quote.

All models are wrong, but some are useful.

We came out with the idea of standardizing how to take a picture of posture that many questions have used, but adding something to it.

This is a one leg balance concept.

As a clinical framework that applies active inference principles to motor control assessment and exercise to enhance outcomes in neuromuscular skeletal rehab, as well as chronic pain management.

All you need is a camera.

Take a picture in four standardized views.

Looking at somebody from the front, from the side, and then from the front again, balancing on one foot, standing tall, and that's important, and standing on the other foot.

These ideas align very much with the word cloud that active inference is something that ties into the energy principle, and it is practical from a clinical point of view.

In addition, it's something that the public can do and that can be researched.

And this is why I'm talking here right now, because if you're doing any kind of research in the biologic sciences and you're working with subjects, adding a posture picture to what you're doing is a potential window into their body's prediction from an interoceptive sense, because that's where a lot of this information

coming from.

The sensors for your body for interoception are not the same sensors as the sensors for proprioception, how your body's moving.

The goals you tend in and the muscle spindles are dependent upon load and motion.

But just feeling something is a different kind of a feeling inside.

Interoception is thirst, hunger.

It's one of the internal body feeling.

Your body's sense of state is an internal body feeling, and the nerves for that go on different pathways up the spinal cord, and it's processed in the insula, which integrates this information with other parts of the body before going out to the motor cortex to guide the motions on a more gross level.

And the whole one leg balance idea is so important, because your walking begins with balancing on one foot.

And one foot, every step you take, you're balancing on one foot and all standing, which is why if you're doing that asymmetrically when you're standing statically, you've burned in that pattern every step you take.

That's why my book *Stand Tall Lives Longer*.

Our second posture principle was balance.

The first posture principle, motion.

Your body's designed to move.

Active inference drives those movement patterns.

Movement's about the whole body.

Motion's about one arc of motion within the body.

Prior beliefs about how your body should move within a global movement unfolds from all the little motions of the joints in that kinetic chain of your body.

And the predictions about the sensory consequences of the action drive what you do.

And the effort to avoid pain trumps minimizing prediction errors, minimizing energy use.

Well, it's harder to walk like this, but it doesn't hurt when I walk like this.

You'd be amazed to see how some people walk into the office when they're in an acute pain and what they manage to do to work out to have an affordance of some kind of a cane or a crutch in different directions.

Oh, as a sidebar.

I don't like canes.

I don't like crutches used indiscriminately.

I'd much prefer a staff to keep the body vertical than a cane to let the body fold forward because that tends to lead very much into, now you have a scooter to go around and now you have a wheelchair to go around in.

Effort keeps the body moving.

Compensatory patterns about how your body tends to move is your body's best inference about how not to have pain and how to move given your body's current biomechanical constraints about the muscles, what they've learned, and the nerves, how they've learned.

In other words, functional compensation drives structural adaptation and then structural adaptation constrains further changes in patterns of motion that change functional compensation, that change structural adaptation in how you move and how you balance in an ongoing spiral of those five concepts.

Taking a posture picture, seeing how somebody is standing with the instruction of stand tall, show me your best posture, take a step or two in place, just stand tall.

And then taking a picture of that can be powerful to engage them in making the change in the body.

At the end of this class, I'm going to show you two basic and simple exercises you can do to strengthen your balance and your body's alignment.

The protocol is to first have someone stand tall with best posture, take a step or two in place, look at them from the front, take a picture, face to the left, take a picture, face back to the front, have them stand tall, lift the left foot up in the air, and lift the right foot up in the air.

It's really easy.

One step at a time.

One step at a time.

And what you're looking at is where the head is, where the torso, where the pelvis, and where the feet are.

And we get into this in detail with clinicians.

This is not the place for that.

This is from the paper.

And looking at those four zones of postural mass, how they're balancing and the symmetry, especially on the front view of left and right leg coming up.

With something like this, a simple set of pictures.

This comes from a paper about changes and interception in mind-body therapies for chronic pain.

This is a systematic review and letter analysis.

Interception is the perception of physical sensations from inside the body.

When they're incorrect, there tends to be more pain.

Correcting them reduces the pain.

And when you're looking at an interoceptive posture picture, if someone's always leaning to the left, if they're lifting their left foot and their right foot, and they're always leaning to the left,

that's saying that their perception of where their body is in space isn't correct, especially if they're not hurting.

If they're hurting and it's compensatory, that's short term.

But when they're lying, they're shook the way that I am.

That's saying that I have changed my body so that I am incorrect about where my body is in space.

And that's why systematically focusing attention to strengthen that weak link, which is what we'll do at the end of this class for just a minute, with two exercises, can make such a big difference.

Not just in how your body looks, but in how your body feels.

Some examples, young lady standing tall, left and right leg coming up.

She looks pretty good.

Here's a young lady that's also standing tall.

You'll notice the pattern of how she's balancing her body on the left leg coming up and on the right leg coming up.

She's going way off to her left side with her body where her right knee is going out when she's raising her right leg.

Left side, she's keeping much more vertical.

The ability to maintain that verticality says something about how her body has learned to move at a very young age.

She may have had an injury. This came from a clinician. Hopefully the patient's followed through and she's done well.

But the fact that a child this young was brought to a doctor said something.

By the way, look at that first picture. Look at the child's feet.

How they're rotated off to one side when she's trying to stand tall.

That gives you a hint that there's been some asymmetry trained in that body.

And it's something that especially at a young age can be improved.

Here's a young man told to stand tall.

You'll notice he's internally rotating his hands so his palms are facing backwards.

That's not standing tall. That's not attention.

That's a child that's probably been using a whole lot of tech.

A clinical posture picture is an experiential perception of that expressed verticality of I'm standing tall.

This is why it's an expression of information coming in from the outside and the inside and your perceptions and prediction.

The predictions part of this is so important.

Correcting posture is not about telling someone to stand taller.

It's about making the body feel the predictions differently and reprogramming that step by step.

This is why active inference is such a powerful concept.

Those compensatory patterns, the things that can be changed, are inferences about what's going to happen in the body.

A lot of times this is called dysfunction.

The simple way of looking at it, if someone's hurting, the weighting of the inputs coming into the body are shifted.

You're not thinking about it.

Your body just reflexively changes.

Your brain is such an amazing tool because it can adapt to different situations.

Humans have a great adaptation potential.

There's an old saying, your greatest strength is your greatest weakness.

Our flexibility means that we can be trained unintentionally into moving in ways that we don't want to be doing when we think about it rationally.

Because that's what our body has been trained to do.

Our attention focusing on one link at a time, one piece of the puzzle at a time, can modulate biomechanical efficiency.

It's not a one shot, I fixed.

It's one thing that can be changed.

Which is why we talk about incorporating biobehavioral strategies, the NMS pros, to help people live well in a

tech world.

Don't want to let technology ruin your brain or your body because they're connected.

Before we go on to showing you exercises that I'd like you to do, I'd like to strongly encourage you.

Have someone take your posture picture because the emotional impact of, I thought I was standing tall from the front, from the side, and on one leg balance is powerful.

Clinicians know it.

I know it from working with thousands of people.

And what I'd like to ask you to consider once you've done this for yourself, because if you are a scientist, you're looking at this from a scientific point of view, as we all should be.

If there's any part of your research that you're using human subjects, I'd love to talk to you about helping you incorporate more systematic implementations of an interceptive posture picture.

As another way to get a window into how that body and individual is predicting actions for the body and how simple actions can change that by working with that person and engaging that person by educating them about their body and how it moves.

This is what we talk about in Stand Tall a little longer.

Your body is designed to move.

That's the first posture principle, motion.

The second is balance.

Posture is how you're balancing your body.

And it's also how you're balancing your mind.

A lot of the implications of the word posture is not physical.

It's your view on the world.

How are you looking at things?

Where do you stand?

And how you balance your body in space and over time is what we're retraining when we're focusing attention moment by moment on a portion of the body and how it relates and links to other parts of the body.

In order to balance because your body moves in patterns of motion that follow the path of least resistance of how you balance and align to move in patterns.

If it hurts to move this way, if it's mechanically constrained, if that's the habit that's caused muscles to get tighter and weaker in one direction and stronger in another direction, that's going to be the way that your body learns to move.

Initially compensating, you learn to move in the pattern that you teach it.

But then over time, when this becomes locked in, things become stiffer.

It doesn't move as well.

Ultimately, you can't move in that direction.

I just move like this.

Where changes in posture and changes in motion cause your body to physically change.

It becomes a spiral.

It keeps on building on itself over life.

My firm experience and belief and hope and efforts drive towards the knowledge that active inference informed education linked with individualized self-care is an approach that builds both cognitive understanding,

and that ties together things like pain science and cognitive behavior therapy on a base level.

Not therapy from a point of view of your childhood and things like that, but being able to control your body in an iterative, ongoing way, arc by arc.

Strengthen the strength and predictive accuracy to improve sensory perception, to have more accurate processing, and therefore prediction of the motor expression of how your body moves, and then creating a way to track that just by taking an interceptive apoptic picture.

We have a new course that I literally just finished.

Integrating interoception and biomechanics.

Prediction of where your body is, is part of what we're creating as a bio-behavioral approach to working with neuromuscular skeletal professionals.

It's also a way for psychological professionals that are working with chronic back pain to help and to be working with people who work with chronic back pain from a neuromuscular skeletal point of view.

That's the big focus as well as neck pain and concussion issues.

I'd like to show you two attentional focus exercises you can do yourself once you've taken a posture picture to see how your body is, to try to connect your perception of how your whole body is balancing with the outcome of if you can balance better.

If you feel your body is balancing better and you can stand at one foot and stay there for more than five seconds, more than 10 seconds.

And as quantified by an interceptive posture picture of observing the deviations of your body.

Connecting that balance, one, with two parts of your body alignment.

Where your head is and where your torso is.

We call it forward head posture.

You may call it tech neck.

But if you spend a lot of time in front of your computer, which most scientists and researchers do, you'll probably find that your head is more forward of your body than your parents' head was when they were your age.

In terms of some of the issues going on, if you take a picture of your kids, how they're standing and how their resting posture is as opposed to yours is now.

And if your kid's posture is worse than your posture is, it'll give you some insight on where the world is going.

This young lady is a model.

She had no back pain, no neck pain.

College student.

We did an event taking posture pictures.

I told her to stand tall, show me your best posture and just relax, stand tall.

And this is what she did.

Any way you measure where the head is versus the torso from the back margin, from the center of mass.

Her head is translating forward and you'll notice it's also going up.

It's translating up.

And that translation up is what happens because the muscles in the back part of the head become shorter.

They compensate shorter, ultimately adapting shorter to restrain and constrain how the head is balanced on the torso.

So looking at those four zones of postural mass, those four postural zones of head, torso, pelvis, lower extremity.

The very first alignment exercise is wall lean.

And it's really, really easy because all you're doing is having somebody lean against the wall with the feet a foot away from the wall so that their pelvis, their torso, and their head align with that vertical wall.

You can do this in all with any vertical wall, but you want to take your feet away because you want to reduce the confounding variables of feeling where your head is over your torso.

So first aligning the pelvis and the torso against the wall and keeping the head level, look straight ahead.

What you're dialing into is your perception of is your head truly level in space.

Have someone take a clear picture of your body from the side.

It's a way to discern functional compensation from structural adaptation.

So looking at her over here, you'll notice this, she was standing, looking straight ahead, thinking that her head is level.

If you looked at it from the back of her body, the margin from the back of her torso to the back of her head is quite a bit forward of it.

And if you look at the face, it's looking up in the air.

When I looked at her like this and I said, level your head and push it back towards the wall, that's what she did.

When I went up to her and said, pull your head more forward, level your head and externally put her head to level and let her push it back to the wall.

And I made her stop at the limit to explore the limit of that functionality, what the Markov blanket is of that prediction and stopping there.

This is where this body truly is.

Then told her to take five breaths.

She was able to do it with a lot of focus.

And that focus on keeping it level at the limit is what it takes to change bodies.

Having her do that for five slow breaths, three times a day will make a difference if she's being good about it.

If she's being correct about it.

And that's why working with someone to help you could be a family member to keep your head level as you're pressing back gently.

Not hard. Don't force it.

Because you want to remold tissue that's sensitive in the body, fascia.

Don't push it.

You want to be gentle.

Gentle works far better than hard in this instance.

You want to feel the tension.

And when you breathe into it, it'll give a little bit.

The imperative, what we call the must or the must versus try, is to keep the head level.

If you're looking at a side view, if your forehead is behind your chin, you're probably tilting up.

If you're away from the wall, putting something behind your head, a pillow, an inflatable ball, to get a feeling of where the wall is and pushing back against that very gently to build the awareness to retrain that control.

Awareness and control of your body and your environment is the center of the posture month.org ACE campaign, which is what we're behind.

To improve people's ability to connect with their body, beginning with the awareness of the body and then strengthening the balance.

Your posture is how you balance your body.

This is what we're looking at with an interoceptive posture picture.

Just looking at how someone's balancing.

How do you strengthen that?

Holding that for those same five breaths when you're balancing with your body aligned to your body's perception.

So finding a wall, standing tall and dialing into what that feels like.

And then locking in your knees, standing tall, engaging your core.

Open your chest up, look straight ahead.

Standing tall, kind of feel how that feels.

Touch the wall and keeping your outside leg locked, standing tall.

Simply lift the leg that's between you and the wall.

And you don't want to lean into the wall.

You don't want to lean away from the wall.

You want to focus on standing tall, but you do want to barely touch the wall.

You're going to want to say, okay, I can do it without it.

I can do it without it.

And the moment you do it without it, you've lost that feeling of inside your body.

The moment you go from, I'm maintaining my balance.

You're coming away from the feeling of standing tall.

Standing tall, touching the wall, lifting the leg and barely touch the wall.

How light can you touch the wall and stand tall?

And focus on the standing tall because the information coming in to your body from that little bit of touch is the link to reality that lets you strengthen how you're standing.

That will then help you strengthen your balance, tying those two links together.

Simply tying those two links together is a huge first step in strengthening balance and alignment of your posture.

And it's also an indicator that you can use when you take a posture picture, an intrceptive posture picture,

to validate the alignment of that objectively externally.

Doing that along with people's bodies internally is a way to create changes in how people use their body in everything that they do.

A lot of times people don't want to do this because I don't want to see it. It's too scary.

Herbert Spencer was a British philosopher and sociologist.

The profoundest of all infidelities is the fear that the truth will be bad.

I hope to hear from you. I hope to work with you. If you have any questions, drop me an email.

My name is Dr. Steven Weininger.

Thank you for having me here.

And I look forward to working with the active inference community with a clinical perspective to help people stand taller and move well.

Have a great day.

All right, we're back. Next is Observer Theory by Sam Seshal. This is a 31 minute pre-recorded video. Here goes.

Hi, a big thank you to the Active Influence Institute for having me.

Today I'll be talking about observant data in the context of all terms really adequate.

I'll start by outlining the history of these ideas, then explain how this extension provides a first stab at a general model for any observer sampling of computational universe.

This framework proposes a resolution to discrete versus continuous dynamic.

It handles cross-domain causation, i.e. mind-body problem, and begins to quantify the novelty of surprise of observer C. It also formalises the terminal object that stops infinite regress

in this model, and enables SARF mainly utilising the category-directed formulas developed by

Gorard and Arcewala. This extension of observer theory is built on the foundational work of

Stephen Wolfram, Jonathan Gorard, Cersei's Arcewala, D. Rikuls, Hatem al-Shabak, and all.

Start with a big, brief explanation of the Rulia.

Wolfram's physics project posits the underlying fabric of reality is a vast screen-space of all possible computations. Observers like us, sample reality through an enormous structure.

Recent formalisations are framing the synchronic theoretic terms. A key insight is that observers are not god-side views, i.e. there is no view from nowhere, and there is no true objectivity.

Wolfram has pointed out that observers leverage something called computational reducibility, which basically means they pick out regular lumps in Rulia, and equivalent relations between and the two-dimensional objects.

The same thing is that we can see what our brain can simplify. Wolfram also emphasised that we act as if we're stable.

We assume we persist through time and stay roughly in the same place, which lets us treat space and time as coherent backdrops or anything we say about observation.

These subconscious assumptions are exactly what we capture by positing this background object, the Rulia, and by formalising the observer in this framework.

In the first model, to get reality as we experience it, we need an observer to sample the Rulia.

Here, they formalised second-order cybernetic systems within the structure.

The Rulia updates state to state, and observers sample between state updates, creating a single thread of experience.

A coarse graining of the wandering amount of computational complexity to something manageable, which outputs everything you see around you.

Furthermore, observers hold an internal model that they update through their cycling, which helps them predict things within the Rulia, or in other words, find pockets of computational-induced ability.

The notion of a background object is a bit vague, so a background.

We can use a simple example of an observer looking at the sky.

That person will use fixed references, like the horizon, to interpret what they see.

In physics, it's usually background-independent.

But here, we introduce this object to capture what an observer takes as given.

This could be their internal world model.

It could be what they already produced in the Rulia.

This is a Rulia.

Wulfur emphasises that observers make strong assumptions.

For instance, we assume we persist through time and reserve for roughly the same things at any moment.

This is akin to having some background reference that doesn't disappear.

By explicitly modelling a background object, we codify the idea that observers always compare new inputs against unassumed stable contexts.

Historically, Wulfur's writings on observer theory note that observers correspond to patches of regularity in reality.

In other words, observers give unstable backwards in the grand computational universe you can guess.

Before introducing new ideas, I'll summarise a minimal observation.

Think of it like a very basic framework or a firm set.

It has three jobs.

1. Set something from the outside.
2. Store and update an internal record of what it says.
3. Act on the world based on that.

The authors, led by Tuxi Zasiwala, stressed that the simplest observer must have a backbone for distinguishing itself from its environment.

For example, the thermostat senses temperature, compares it to a set point in its memory, and turns a heat up on the ground.

This act of observation is reduced to a feedback room.

In their words, a minimum observer perceives external states, updates internal configurations, and generates an action in a closed feedback room.

This simple model captures the functional essence of observation without any assumptions of consciousness or scale.

It serves as a foundation for our larger model for generalised observers within the reading.

We now embed this minimal observer into the full Rudiad.

In the extension, we define an observer as a function.

SO takes R to R .

But what does that mean?

Well, we know that Rudiad is the full category of every possible computational universe.

But, observers have limitations.

They can only sample some parts of this universe.

The fundamental, SO, captures this sample.

It takes each state or transition in R and maps it to what the observer actually perceives.

In effect, it forgets or forces details the observer lacks the capacity to distinguish between.

The result is RO , the Rudiad for the observer.

And that's a function that takes the functor, SO, and applies it to the full Rudiad.

It implies the limit of the acceptable Rudiad for that observer.

Different observers can have different functors and thus experience different worlds.

In plain language, the functor encodes the observer's hardware and bias in our digital reality.

This makes explicit Wolfram's idea that our physics is a slice of the view we have determined by us as observers.

Next, we'll spell out the constraints to define the observable Rudiad.

Any observer has finite computational resources, which we call boundedness, so it can only process phenomena of limited complexity.

Formally, we say it's acceptable only if the boundedness exceeds a threshold, which could relate to a processing limit or an observer's memory.

Likewise, instantaneous or too brief events vanish.

We include a persistence threshold, meaning that the state must last long enough to register for the observer.

Finally, an observer doesn't care about absolutely everything.

They only notice relevant features.

We encode this via a Rudiad.

Only if the relevance is true is the information from that state integrated into the observer's internal model.

For a human, we can take the example of a bright flash.

It might be relevant to them in the dark, but random noise might not.

Together, boundedness, persistence and relevance filter the Rudiad and the observable Rudiad to a much smaller set of possible states that the observer can actually experience at a given time.

The field of observation is simply what the observer actually sees at a time of sign.

It's a subset of the observer will really add that passes all three filters.

In other words, this field contains exactly those states that are small enough, stable enough and meaningful enough for the observer to see them.

You can think of this field as the data in your current sensory body.

For a human, this field might consist of objects and colours you can see in your immediate field of you.

It's not everything in the world.

That would be the observer's total accessible reading.

It's what you're effectively attending to now.

This field is the basis for everything else that follows.

We define information content and information integration over this field.

The diagram on the page shows that in the context of the Rudiad.

The pink lines show the field and the red dotted lines show the full Rudiad that an observer could access.

By here there's a subset and it's the bit they're focusing on right now.

Now we quantify novelty or here surprise of each piece of observed data.

We define the information content as the total entropy of a state X less whatever the observer can already predict or has compressed.

We define the information content as the total entropy of a state X .

You can also say made computationally reducible from prior observations.

So the information content is essentially the net new information that sampling X offers.

If it's zero, the observer gains no new information.

If it's large, then it is novel or surprising.

I mean, this directly mirrors FEP but generalizes it to any observer.

In other words, our observer is trying to minimize the information content on average.

We tend to seek environments where surprise is limited.

Why is that?

It saves some of our bounded computation power.

We'll use this information content to build a measure of the overall information observer's size.

Getting a picture of a sense of the observer's experience, we define integrated information as the sum of the information content over everything in the observer's field.

The form that tells us how much new information the observer has coherently integrated at this moment and the moment being the same part of the time within the Wolfram model.

It's analogous to integrated information in consciousness studies, but again applies to any mass of observer.

If integrated information is high, the observer is experiencing a highly structured interrelated set of inputs.

If it's low, the inputs appear to be more random or destructive.

Integrated information aims for a concrete observer-centered measure of how complex and unified the observed information content is.

In practice, maximizing integrated information in this formalism is seen as an observer's general tell us, akin anthropomorphically to having a rich and meaningful experience of the world.

One of the biggest puzzles is how discrete computation can give rise to the continuous space and time that we experience day to day.

Our answer is that the continuity is observer constructed to save the computational boundedness of a given observer.

We formalize this by giving each observer a measure, indicated by the view zero over its field.

Intuitively, this weighs different samples according to the observer's sensitivity to them.

For example, humans' voice and your vision more heavily than the peripheral.

Consequently, the observer's perceived continuum is given by integrating the discrete information content against the measure.

In simple terms, the breaks filling in the gaps.

This aims to resolve the discrete continuous gap in the Hexman-Wulfram model.

The underlying Ruliat is all frames and the observer's mind interpolates to get a smooth movie.

Wulfram noted that when done properly, there's no content.

Spacetime could be a construct of an observer's court-screen of Ruliat.

So, with this, we can comfortably use continuous math in the observer's perspective, knowing it emerges from underlying discrete and spiral measure function.

Maybe the discrete continuous idea more commonly.

Let's look at a few analogies.

Imagine watching a movie.

An old movie.

The film is a sequence of static frames, but because they're shown rapidly, you, the watcher in the cinema, perceive fluid motion.

Similarly, the Ruliat provides discrete frames of reality in its hypergraph, and the observer's mind generates motion between them.

Another example is a blind screen.

Each eye has a spot with no photoreceptors where the optic nerve leaves the resting.

If you position an object there, it disappears.

You instead just see the back of the signal, nothing is missing.

Your brain literally fills in the gap for the computer loss.

That's the observer theory extension in action.

We don't perceive raw data.

We perceive an interpreted scene.

You could also evoke Newton's bucket.

We see water's surface flow under rotation because our perception assumes a fixed background.

All these examples illustrate the continuity and coherency of the observed and proposed.

In the formalism, this is captured by the measure integrating these three samples.

The theory posits the continuous reality is an observer construction to say the Q.

Let's move up from the mechanics of the observed and constructing continuity to what that continuity produces.

The layered domains of reality is experienced by our observers.

The paper proposes the observer's operationalised continuity not in a single domain, but across four nested layers, each expanding in degrees of representational freedom and coherence.

In short and simple language, this means that higher domains are more informationally dense and backed by less rules than lower domains.

It should be noted at this point that this decomposition can be done in any way a researcher chooses, even within the domains that I posit, or for entirely different sets of domains suitable for other classes of service.

The following decomposition utilises human's cognitive layers.

It's aphomorphic, but others could work.

I will draw the layers one by one.

The layer with the most rules is the pivot of the domain.

Here, computation appears tightly coupled with causality and energy conservation.

The main of classical mechanics deal with theory, balls and graphs.

Everything is closed and coherent.

And information of evolution aligns with notions like force of closure.

This is where standard physical computation lives.

Where our physics lives, at least until you get back to the quantum.

The next step is to move to what we might call proto-subjective space.

The valuation or domain into systems with observer-driven salience, such as attention, prioritisation or inference.

This is where concepts like active inference emerge.

Here, the observer doesn't just track visible space, but values its sex, creating meaning from a computational structure.

The third layer is the symbolic, and it's the world of form of abstraction.

The symbolic domain involves for rule-based transformations independent of their instantiation.

Things like language, form or logic, and mathematics.

Observers in this domain don't just respond to inputs, but create representational continuity.

Chaining symbols into expressions, proofs, and meaning structures that can be detached from any one physical or emotional, slash-valuational context.

For observers like us, this domain enables coherence across the physical evaluation domains.

It can describe physical systems or encode value structures.

It's to see with metatheoretic insight.

Lastly, we reach the edge.

A space of maximal modelling freedom that we've termed the minimally constrained domain.

This is a domain where the usual coherence rules of symbolic logic break down.

It's a place where we can talk about paradoxes, infinite regress, non-well-founded sets, things that we can imagine, but that we can't really instantiate or we can't totally describe in words.

It approaches the boundary of what we can access in the root yet.

From a categorical standpoint, this is a domain that gets closest to topological closure.

Observers are capable of traversing all paths and equivalences subject to identity, even those incompatible in lower domains.

This is also where our categorical terminal object, true infinity, enters.

It's a poetic mimic object that closes the space of coherent interpretation.

The hierarchical layer is crucial for modelling cross-domain observers.

Ones who, for example, infer value from physics or deploys symbolic logic to manage valuation hierarchies.

It also allows us to rigorously define how continuity and therefore computation means something different depending on the specific observer's focus.

Why do we talk about the hierarchies?

It's because one of the central aims of the extension is to show that mind to matter causation is natural in the general observed framework.

There's no magic.

Because all levels observers like us have access to are embedded in a single structure, we really add, a single causal chain, can thread through all of those domains.

We attempt a formalisation of cross-domain causation of rigorously.

For a state D in domain I and a state P in domain J , there is a causal relation if we can find a path of transitions in the observable region, connecting state T to state P .

The path is a sequence of states across domains.

In categorical language, we have a process category where projects, states and morphisms are processes.

A morphism from state T to state P spanning different domains is a cross-domain force influence.

So, as an example, a decision you make in the domain even with valuational or symbolic logic can amorphaously lead to a muscular contraction in the physical domain via the composition of factors between those domains.

Think about it as a map connecting to another map within one giant.

There's nothing a-causal about it.

It's just one connected multi-step morphism in the whole observer constructed reality.

In the observer theory, extension causation has two main factors.

Formal causation corresponds to a high-level pattern from a less constrained domain imposing constraints on a lower level.

For instance, the blueprint on an intentionally shaping physical act.

You can use the example of an architect's plans for a building that don't instantiate specific materials, but just the shape of the structure.

That limits what those builders in the physical domain can do to put it together, but it doesn't determine every piece of that puzzle.

Efficient causation is the usual physics-style cause effect, like neuron-inspiring.

Mathematically, we saw causal morphisms by how their constraint levels compare, i.e. how many rules are required to output said morphisms.

If a state has less constraint and goes to one with more, we'll call it four morphisms.

If it's the other way round, it's sufficient.

This picture shows clearly that there are just two directions of one unified structure.

Crucially, mind-body effects are simply morphisms in the observable rulead involving different domains.

We don't need new laws. We don't need any magic.

For example, deciding to pick up a car.

A mental domain morphism gets mapped via functor, through the symbolic and then the physical domains.

Eventually, we can add.

The laws of physics are untouched.

We're just following morphisms in the rulead under a particular observer's mapping.

Recently, Eric Wells formalised a mathematical model of the interactions of micro and macro scale causation that we feel supports this view.

Let me tie these ideas together.

The corny quantity we have is the information content an observer gains with, say, x .

This is explicitly tied to how an observer reduces uncertainty.

It generalises frictions and others ideas of minimising surprise.

An observer will naturally prefer observations that minimise aboriginal new information because they achieve computationally and enable more of their bounded resources to have availability to search for more new morphisms, potentially increasing global and local integrated information.

The sum of integrated information tells us how rich the experience is.

This makes observer dependent novelty precise for any class of observer.

What is new to one observer may be older to another because each has a different sampling factor, even if they share the same observable rulead.

For intuition, you can think of identical twins.

They have the same observable rulead and exactly the same bodies, but their divergent course of histories, meaning their sampling factor, works differently.

Thus, this photomplexus begins to quantify the audience's surprise in a way that depends on the individual observer's filters.

Finally, we tackle the question of photoshop.

Is there an infinite wall of mirrors of observers observing observers?

We answer by introducing a special object called true identity, which plays the role of the terminal object in our festival.

Concrete PTI has the property that every observer maps immediately to it.

In everyday language you can think of this as the ultimate perspective.

Our mind of God.

It's not reachable or observable by any finite observer.

But here, it's the source of all the rules that generate the rulead.

It provides a big extent of the lines so we're not stuck with an endless hierarchy.

This computationally intractable.

I.e. at some point, everything just makes sense.

As we summarise in the paper, TIS like an initial viewpoint or a unification or integration of all underlying information.

This ensures the framework is mathematically complete.

Every diagram has a limit.

And there's no paradox of what observes the observer.

True infinity is essentially the background, frame or ground truth that all observers implicitly assume, but can't prove or fully attain.

What does a deal like this, actually?

Here, it's a placeholder for ultimate computational completion.

Imagine that every computational process and every observer is eventually seen from some advantage.

True infinity is the advantage where everything comes together.

You could call it a first mover or even a God-like viewpoint, but in this formalism is strictly a mathematical object.

Because of it, our model doesn't have two sayings.

There is an absolute frame hidden in the background.

An unachievable fixed ground against which everything else is defined.

It shuts down the infinite regressive problem.

Nothing is left out of the.

And from the soft of the itself, there's an ultimate to the observant ladder.

So it serves as both a poetic condition and a precise mathematical capstone to the framework that aligns with the Gorat and R.C. Walla form of this on the construction of the Rudiad.

Simply put, categorical closure in the formal model enables the Rudiad to generate geometry and then, through a long chain, generate physics.

So, without it, we don't have a structure we can do things on.

It is no longer an infinite group, where it is an complete category.

Surprise!

The observatory extension of the author model can bridge domains that, in the author's view, are already obviously treated as a very much sterile.

By formalising observers as the symbols of a broad computational universe, and by introducing observer-dependent measures and information metrics, we can start to reconcile the subject in our way.

In our view, it doesn't introduce any contradictions with known physics.

Instead, it explains why our map-specific world looks like the advice from our vantage point.

I believe it gives us new tools to think about how consciousness and physics can play the same sandbox.

So, moving forward, this framework could guide us in neuroscience or in the design of AI about how observers build reality.

As just one example, if you increase observers' computational bounding methods, or increase it, we could use, say, the height of the speech camera.

We could expand its observatory area, but we might see a phenomenon.

Indeed, this is what we do with technology.

Without particle detectors, we can't attack the particles.

Another key takeaway is that observation is fundamental in computation.

By walking observers formally, we can better understand physics, cognition, and their intersection.

It still blows my mind.

100 years after the QM, there is still no general, formal, observable.

Shortly, I'm publishing a newspaper that extends the ideas of this paper of this observatory model to look at theology, ethics, telos, and human superstructures.

I'm seeing natural observers through this science.

Lastly, I want to thank James Files, Stephen Warden, Hanna Shatmari, Joseph Liddell, Andy O'SPAY, and Ben K, and Jed Solomon for all the discussions that helped with this paper.

Thank you for your time. Enjoy the pack.

All right. Sorry about the audio issues on that last one.

Thank you for that very interesting talk.

All right. One second before the next one.

Thank you.

My name is Jim.

Freda.

And I'm very, very pleased to be here for the 2025 Applied Active Inference Symposium.

All right.

Next up is Exteroceptive Bias, Belief Updating in the Global Model.

Jim Freda.

This is a 51-minute pre-recorded video.

My name is Jim Freda.

And I'm very, very pleased to be here for the 2025 Applied Active Inference Symposium.

I want to thank the organizers for being so open in their acceptance policy and letting someone like me present.

I am a deep tissue structural body worker, a manual therapist, an embodiment teacher, and a somatic historian and social theorist.

My interest is not in computation, computational neuroscience, mathematics, computer science, and physics. But I have a deep interest in a model of life.

And especially in the perceptual mechanics of a model of life from a somatic perspective.

My interest in active inference derives from my focus on what is called the deep core myofascial synergy.

Which is the anatomy, the synergistic and very new anatomy of our axial line, our postural core.

Sherrington describes this system as proprioception or self-awareness.

And when I came across Sherrington's marvelous text, the integrative action of the nervous system, and his neural architecture of the animal as a whole, it really blew me away.

It really changed everything about how I was working with people, working with myself, and understanding my own self-image and the world.

When I came across Carl Friston's version, updated version of the model, I was really blown away.

Very excited.

Very excited.

And I just love the work that the active inference community is doing.

I do not have computational, neuroscientific, computer scientific kinds of critiques and suggestions to offer.

But my field is anatomy.

And from the perspective of anatomy, and especially the self-organizing, self-aware, self-balancing, self-organizing, and self-directed generative anatomy of the human body, I do have a lot to say.

This is a new kind of anatomy.

For the purposes of today's talk, I'm calling it cybernetic anatomy in the cybernetic tradition, or we can call it synergistic myofascial anatomy.

And this is really the key to understanding the motoric element of the deep field in the model, the motoric element of the physical homeostatic function of active inference itself.

We're going to talk about a few major points here.

I'm going to go over, in addition to some guided body meditations that I hope that you will find novel and enlightening.

We're going to spend some time reviewing Sherrington's original model, the neural architecture on the animal as a whole.

I have a big section on that.

There's an awful lot of text.

I'm going to summarize it, but I put all the text there so that you can go back and do your research.

Because a lot of the belief updating that we're going to be doing in this talk today involves the major contours, the global outline of the superordinate model, which is three receptor fields, exteroception, interoception, and proprioception.

Now, the core node, proprioception, I argue, has been under-theorized.

And proprioception is the least well understood of the three different perceptual systems.

The model, we understand, contains two blankets.

But I want to give the inner blanket, the outer surface of the inner self, in the nucleus, what would be the, very importantly, mechanotransductive nuclear membrane, the lamina and membrane of the outer surface of the inner component of the self, as an intuitive example of a model, of the model, that is the inner self.

The model, that is, by analogy, we can also find in ourselves, in our body.

This motoric element has been under-theorized, and it's actually missing from the model.

Proprioception lacks a motoric element.

The deep field lacks motoric anatomy.

What we're going to find is exteroceptive bias.

And we're going to talk about that in terms of the blanket politics in the model, that if these two perceptual fields have agency and self-have identity, which it's clear that they do, and theoretically, which they must, then we can talk about their relationship in terms of blanket politics.

And I'm going to call forth both Carl Friston's Disconnection Hypothesis and Gregory Bateson's Double Bind as a way to discuss that.

I'd like to invite you into guided meditation on the grip, on how to get a grip on the reach.

The reach, the arm, the forelimb of any animal is something that we study to understand its unique adaptive potential in its environment.

The human upper extremity, the human forelimb, the arm and the hand, of course, are very, very special and allow us a tremendous amount of control and sensitivity within our environment.

And this has been studied hundreds and thousands of times by motor control theorists and neurologists and biomechanists.

And it's something that we should really be very familiar with.

Out here in the hand is the exteroceptive grip.

What I want us to focus on today is the proprioceptive prior, which is located in the armpit.

Now, that armpit is formed by a single large bone called a scapula.

Now, this scapula, you'll notice, is shaped exactly like a grip.

And your hand fits around it perfectly.

And indeed, there are five intrinsic muscles arrayed over this that literally grab onto it above and below with their tendinous insertions wrapping around the shoulder joint, forming the rotator cuff.

And this is the system that anchors the arm to the body, but also that manages that originary kind of joint, the shoulder joint that manages it and maintains centration and balance and stability in that very insecure joint.

So, we're going to reach out with our hand.

And if there's someone there, grasp their hand gently.

If there isn't someone there, if you have a cell phone or a coffee cup, grasp that and feel how your little finger is like a hook that secures that object in your hand.

How the thumb coming over the top is a kind of clamp and how you get a grip in the exteroceptive system reaching out into the world.

But even though this is the applied active inference conference, I'm going to suggest that we come back.

Not reach outward, but reach inward into the armpit, not the shoulder.

Now, this is again an exteroceptive knuckle, a superficial structure, and a rather insignificant, motorically speaking and perceptually speaking structure that is not well innervated.

We're going to come into this very sensitive armpit.

The armpit is a product of the scapula's attachment on the back of the ribs.

And those proprioceptive movements, postural movements, are long, slow, continuous, and mild.

They really are the expression of active inference in the mortal substrate.

So we have this grip in the hand with a parallel or a mirror synergy, an inverse homology in the scapula, in the armpit.

And see if you can feel that.

And just as an exercise, let's organize our arm with our shoulder.

Let's use the neck and the shoulder to lift the arm, which is very common and results in all kinds of pathologies like impingement syndrome and frozen shoulder.

Instead, let's come down deep within the armpit and bring that elbow in contact with the ribs and come up into the concavity, the sensitive pit of the gradient of the armpit.

And feel a hand.

Literally imagine that as you have a hand out in the world, you have a hand reaching inward in the body.

A literal hand grasping that scapula.

Now, that hand, that sense of a grip, wraps around the bony scapula with its fingertips, the tendinous fingertips of those muscles,

wrapping around the shoulder joint like fingertips around a doorknob, forming the rotator cuff, and giving it, giving that insecure glenohumeral joint the stability and centration that it needs for this amazing range of motion that we have.

That comes from deep within the proprioceptive prior of the armpit, of the concavity.

And indeed, the body has a system of major concavities on the axial line, all with this kind of a function.

And those five intrinsic muscles really create a synergy, a mirror synergy, an inverse homology that the brain finds very easy to work with.

Now, the last thing that I want to say is to come out of that meditation and to look here at this theory.

These are some recent, very interesting motor control studies of the reach and the grasp.

And you'll see here outlined in green, I've highlighted a graphic from each study.

Now, this is universal.

Every single motor control study originates in the shoulder, the knuckle of the shoulder on the one hand and on the other hand in the grip.

And attempts to calculate proprioception from this very false structure.

Instead, if we reach deeper within and understand the motoric anatomy of the proprioceptive, of the deep field of proprioception,

I wonder how that would change our studies and our research, our thinking about proprioceptive priors.

We're going to take a few minutes here to do a deep dive.

And I have a number of slides coming up with a lot of content.

We're going to do a deep dive into the model.

And we're going to go back to the origin, the 1906 origin, Charles Sherrington's neural architecture of the animal as a whole.

And it's Charles Sherrington.

Charles Sherrington was a great scholar, a tremendous scientist, and a man of letters.

And when he named something, he really thought about it.

And his names have weight.

And we've gone and really renamed or reinterpreted, redefined the neural architecture of the animal as a whole without considering the original model itself.

So we're going to spend some time going into that.

Exteroception, interoception, and proprioception were all words that Sherrington created with a great deal of intentionality and rationale.

Now, it's a perceptual model, but it's a spatial model, which is great for somatic thinkers like myself.

Because kinesthesia, the muscle sense somatics, is a spatial form of awareness.

Where we're not looking to ask why or how, but we're asking where.

Where do we go in our experience with our perceptual faculties in order to encounter those affordances in the environment and within ourselves that possess the information theoretic content that we need.

This is the nature of living systems and working with complex self-organizing and self-aware living systems.

Now, the model has two great separate fields.

There's an outside and an inside.

The debate really is what's the inside.

And what modern scholarship has done is taken interoception, exteroception, interoception, and made the very simplistic and naive assumption that, well, that's the outside and the inside.

Intero must mean inside.

And this isn't really true.

Sherrington referred to the deep field repeatedly.

And proprioception is the deep field or the core in the model.

Now, I'm going to go ahead and read this third paragraph.

The multicellular animals, regarded broadly throughout a vast range of animal forms, are cellular masses which present to the environment a surface sheet of cells.

And underneath that surface sheet, a bulk made up of cells more or less screened from the environment by the surface sheet overlying them.

Many of the agencies by which the environment acts on the organism do not penetrate to the mass of cells forming the organism's deeper part.

The various stimuli expend themselves as stimuli at the surface sheet and do not penetrate into the depth of the organism.

So, Sherrington is really trying to talk about a part of the body that is insulated most fully from the outside world, the deep field, the core, proprioception.

And he identified that as a mechanical system, a muscular system, the postural system, compressions and strains produced by muscles.

Not only that, but this inner receptor field, perceptual system, is not other aware.

It is most self-aware.

Let's take a look at Sherrington's arguments around interoception and the surface field.

Again, the surface field is broadly subdivisible.

Its subdivisions are two.

So, the external field is cutaneous.

But the inner surface, then, would be elementary.

Anything that passes through the boundary sheet of an organism has to be processed, for example, by the cytoplasm in the cell, by the alimentary canal, by the viscera, by the gut, what Sherrington called the kitchen.

This is not the deep field.

This is still something that, although shielded from the outside, still is basically working with processing, sensing stimuli that come from the outside.

Now, it's important to recognize, I think, and to take a moment with this bottom paragraph to just go ahead and look that exteroception is special and important in a way.

In that, and these words, I think, are really, really nice.

It's facing outward on the general environment.

And it receives freely, and it receives and has received for countless ages, the full stream, I love this language, the full stream of all the varied agencies, forever pouring on it freely from the outside world.

So, there's a bit of a basis with exteroceptive bias.

In terms of the perceptual fields, exteroception has to always be operative from the moment of birth onward.

And this is one good explanation for bias in the system.

But the real bias that we're facing is an accentuation of interoception and a diminishment of proprioception.

So, let's look at Sherrington.

The stimuli effective on the receptors of the deep field differ fundamentally from those on either subdivision of the surface field.

They are adapted for excitation by changes going forward in the organism itself.

Now, in interoception, in the gut, the excitations are coming from the red pepper and the garlic that you might have had with your spaghetti.

But in the deep field, it really is the contractions and relaxations of muscles of the postural system hugging the core, hugging the spine itself, which are the basis, the priors, the postural basis for the rest of the actions of the body.

Since in the deep field, the stimuli to the receptors are delivered by the organism itself, the deep receptors may be termed proprioceptors, meaning self-awareness.

And the deep field, a field of proprioception.

Now, this is a postural and a tonic system, and we've discussed that, but it's tightly interwoven, not with interoception, but with exteroception, and maintains tonic reactions in the skeletal musculature.

Sherrington describes these wonderfully steady, long, continued, untiring, and for the most part, mild contractions.

And this is known as their tone.

Tone, Sherrington spends a long time talking about tone.

The muscle sense was a major topic of 19th century explorations into perception, but also into the physics of life.

And I love what Sherrington says, and he says that tone serves as an adjuvant to so-called to the muscle sense.

Now, in the literature since Sherrington, these terms have gotten twisted around quite a bit.

And there's a real consensus in the literature.

These are a number of quotes here from different serious studies and scholars, making the point, for example, that proprioception predominates as the most misused term within the sensory motor system.

Jumping down, proprioception is one of the least understood senses, yet fundamental for control of movement and foot to the bottom.

Some major textbooks claiming to cover neuroscience barely mention it or mention it only in passing.

But the truth is that if you do a search online, you're not going to get the proprioception that Sherrington just described.

You're going to get a definition of it as positional awareness, typically, say, in the elbow or in the extremities.

That's a very different creature.

Positional awareness is a very different creature than self-awareness.

And then what of interoception?

Now, interoception has been redefined as comprehensive of all inner sense, which then includes proprioception or proprioception as positional awareness.

And proprioceptors might be considered interoceptors, Sarkiris says in a major study.

Interoception then refers to the perception of the physiological condition.

Now, what's happened is a very simple reinterpretation.

And none of these studies engage Sherrington more than for a few lines or a few paragraphs at the most.
But interoception has simply been redefined to mean inner sensation.
And we now understand from Sherrington that was not his intent.
That does not make sense.
The cytoplasm is not constitutive of the inner self of the cell.
That would be the nucleus.
So, interoception is not a source of self-awareness.
It's not realistic to expect from it a self-identity.
Exteroception, however, is, of course.
Our sense of, our outer sense of identity is quite rich and complete.
But that inner sense of identity that comes from the functional unity, the body-wide axial core, also functions at the level of a self-identity.
And has the neural mass to support that kind of a hypothesis in the cerebellum, the brainstem, and the spinal cord.
Now, I just want to take a moment here.
And I'm sorry if that was rough.
But I want to take a little bit of a moment, a few minutes of your time, to explore this in the body.
And we're just going to look at those three fields.
And let's start on the outside, just to rehearse the Markov blanket, the outer surface of the surface sheet.
With the five senses, with the extremities, with the skin, with our distance receptors.
And we can look around and we can hear the cars far off in the distance or children playing.
We can touch the things in our environment and touch ourselves on the outside.
We have so many ways of relating to the outer world that it can be overwhelming and all-consuming.
The importance of going deep within is so important.
So we're going to start that process at the inner surface of the outer boundary.
Naturally, we're going to start that process.
We can start in the mouth and in the nose.
Inhaling, imbibing, coming down the esophagus, feeling the blood, feeling the heart, feeling the lungs.
And all of the awareness that exists in the gut.
That's interoception.
Now that doesn't go all the way to the core.
But it does go all the way to the other end of our tubular vertebrate torso with its chordate body plan to the excretory system.
What comes in and comes out is a matter of interoception.
But is part of the surface field.
Next, we're going to start again here in the tongue.
And we're going to identify that muscular system of the tongue running back in the back of the throat.
In the back of the chest.

Behind the heart.
In the front of the spine.
Behind the lungs.
Then expanding suddenly here into the whole diaphragm.
And coming down through the gut.
Not in the viscera.
Behind the viscera.
In the motoric core.
The postural core.
The psoas muscles.
Running down through the pelvic basin.
And the groin.
Through the medial thighs.
And the adductors.
Down the legs.
Around the medial ankle.
And the medial arch.
To the big toe.
And the surface of the feet.
A body wide myofascial.
Motoric muscular.
Continuity.
This deep inner field.
Proprioception.
The postural system.
The deep core.
Doesn't contain.
These distance receptors.
It doesn't have.
Visual.
And verbal.
And auditory awareness.
It has haptic.
And muscular.
Awareness.
Which nevertheless.
Is tremendously rich.
And this is something.

That we're going to encounter.
All day.
Every day.
It is a daily part of our life.
But if you go into a yoga studio.
Then that's.
That's what you're going to experience.
Is that system.
As you move.
In those long.
Slow.
Continuous.
Tonic movements.
I have been.

I have been arguing.
That the core node.
Has been under theorized.
In the model.
And.
That the most important.
Aspect of that core node.
Is.
The motoric anatomy.
Of the deep field.
In the body.
And the deep field.
Is the postural system.
And is the source.
Of physical homeostasis.
And if we do not have.
A clear anatomical.
Understanding of that system.
Then I think it's.
Reasonable to say.
It's under theorized.
So.
It's only a recent discovery.

That the nucleus.
Has a nucleoskeleton.
And the importance.
Of the mechanotransductive elements.
In the nuclear.
Envelope.
Now.
It's also a recent discovery.
Of.
The.
The.
The discovery.
Of the myofascial.
Synergistic.
Nature of the body.
Is a recent discovery.
And we can see.
In these other pictures.
Here.
The core.
We can see.
This one.
Circular.
This cutaway diagram.
Of the abdomen.
Now.
We can see.
The outer.
Blanket.
Of the abdominal.
Muscles.
Of the core.
The traditional core.
That we think of.
On the outside.
Of the body.
Is complemented.
By.

An inner.

Core.

That.

Surrounds.

And supports.

And protects.

And expresses.

Its neural content.

The spinal cord.

This.

Inner.

Core.

System.

Is the myofascial.

Synergy.

That I've been.

Describing.

Now.

So.

I want to talk.

About.

Two.

Blankets.

A double.

Blanket.

Theory.

I want to propose.

That there are.

Two.

Blankets.

In the model.

And that they.

Have.

A relationship.

That is problematic.

We can call it.

Blanket.

Politics.

We can think about.

That problematic.

Relationship.

Between.

These.

Two.

Systems.

In the body.

In terms of.

Pristen's.

Disconnection.

Hypothesis.

Because.

Poor.

Integration.

Between.

These.

Two.

Major.

Perceptual.

Systems.

Is.

Going.

To.

Have.

Effects.

In terms of.

Neuromodulation.

And.

Sensory.

Integration.

Now.

I'm not.

Going.

To.

Discuss.

That.

But.

I am.

Going.

To.

Suggest.

That.

Bateson's.

Notion.

Of.

A double.

Bind.

Is.

Relevant.

Here.

Now.

Let's.

Look.

At.

A double.

Bind.

A double.

Bind.

Is.

A situation.

In.

Which.

There's.

A victim.

Who.

Is.

Exposed.

To.

Multiple.

Contradictory.

Injunctions.

Within.

A kind.

Of.

A power.

Dynamic.

A power.

Differential.

Where.

Escape.

Is.

Barred.

There's.

A dependency.

Now.

The result.

Of.

That.

Is.

Silence.

And.

Immobilization.

And.

And.

We.

Can.

See.

That.

In.

The body.

You know.

I.

Think.

It's.

Going.

To.

Take.

A lot.

Of.

Work.

To.

Really.

Flesh.

That.
Out.
So.
To.
Speak.
But.
The.
Double.
Mind.
We.
Can.
Think.
Of.
That.
As.
A.
Relationship.
Say.
Within.
A.
Family.
Between.
A.
Parent.
And.
A.
Child.
And.
Let's.
Say.
There's.
A.
Problematic.
Relationship.
Now.
The.
Child.
Is.

Sick.
And.
Is.
Has.
Recovered.
Is.
Getting.
Better.
And.
The.
Parent.
Visits.
In.
Hospital.
And.
The.
Child.
Gets.
Out.
Of.
Bed.
And.
Wants.
To.
Embrace.
Their.
Parent.
And.
Share.
Their.
Their.
Happiness.
And.
Their.
Their.
Gratitude.
But.
The.

Parent.

Backs.

Away.

And.

Signals.

No.

We're.

Not.

Going.

To.

Hug.

We.

Don't.

Touch.

And.

The.

Child.

Is.

Immediately.

Despondent.

And.

Perhaps.

A.

Little.

Bit.

Ashamed.

And.

Then.

The.

Parent.

Having.

Given.

One.

Contradictory.

Signal.

Gives.

The.

Next.

And.
Chides.
The.
Child.
Saying.
Something.
Like.
It's.
Perfectly.
What's.
Wrong.
With.
You.
It's.
Perfectly.
Natural.
To.
Love.
Your.
Parent.
It's.
Perfectly.
Natural.
To.
Want.
To.
Express.
Your.
Joy.
Why.
Are.
You.
Ashamed.
Of.
Yourself.
And.
So.
Then.

It.

Puts.

The.

Child.

This.

Dependent.

Being.

Who.

Can't.

Escape.

In.

A.

Situation.

Where.

They.

They.

Can't.

Do.

And.

They.

Can't.

Not.

Do.

They.

Can't.

Speak.

But.

They.

Can't.

Not.

Speak.

A.

Double.

Bind.

And.

This.

The.

Result.

Then.

Would.

Be.

Immobilization.

And.

Silence.

And.

I.

Think.

That's.

An.

Interesting.

Metaphor.

For.

This.

Outer.

Blanket.

The.

Exteral.

Septis.

System.

With.

All.

Of.

Its.

Power.

And.

Its.

Enveloping.

Presence.

And.

This.

More.

Dependent.

Inner.

Very.

Sensitive.

Perceptual.

System.

Which.

Nevertheless.

Has.

Its.

Own.

Impulses.

And.

Needs.

And.

Which.

Can.

Be.

At.

Odds.

They.

Can.

Be.

At.

Odds.

With.

Each.

Other.

And.

So.

A.

Lack.

Of.

Integration.

And.

We.

Would.

Call.

That.

The.

Double.

Bite.

Not.

Sure.

Really.

How.

To.

Work.

With.

That.

But.

I.

Find.

That.

To.

Be.

Productive.

In.

Terms.

Of.

My.

Own.

Working.

Out.

My.

Own.

Inner.

Blanket.

Politics.

So.

To.

Speak.

The.

Solution.

What's.

The.

Solution.

Well.

First.

To.

Identify.

That.
Dynamic.
To.
Be.
Able.
To.
Step.
Outside.
Of.
It.
At.
Least.
Conceptually.
But.
Also.
Empowerment.
To.
Give.
That.
Victim.
Agency.
And.
One.
Of.
The.
Problems.
About.
Under.
Theorization.
Of.
The.
Core.
Node.
Is.
That.
We.
Don't.
Have.

That.
We.
Don't.
Have.
That.
Empowerment.
And.
Agency.
At.
The.
Level.
That.
We.
Really.
Need.
It.
For.
Belief.
Updating.
In.
The.
Global.
Model.

Just.
To.
Be.
Really.
Simple.
About.
It.
There's.
A.
Lot.
Of.
Text.
Here.
But.

The.
Global.
Model.
Is.
Going.
To.
Be.
The.
Superordinate.
Level.
Of.
The.
Model.
Which.
Would.
Be.
The.
Model.
Itself.
As.
We've.
Already.
Discussed.
And.
It's.
Three.
Fields.
Rather.
Than.
Some.
Subsystem.
Within.
It.
It's.
Going.

Be.
The.

The.
Level.
Of.
The.
Individual.
Organism.
The.
Person.
The.
Self.
And.
Their.
Own.
Different.
Ways.
Their.
Their.
Own.
Three.
Different.
Fields.
Within.
The.
Individual.
Organism.
But.
Also.
The.
World.
Model.
That.
We're.
Working.
With.
And.
We.
Certainly.
Can.

In.
Terms.
Of.
Thinking.
About.
Surprisal.
We.
Certainly.
At.
The.
Level.
Of.
Global.
Phenomenon.
And.
The.
Anthropocene.
And.
Collapse.
And.
Technological.
Overshoot.
And.
Political.
Strife.
That.
We're.
Experiencing.
We.
Certainly.
Can.
Identify.
A.
Great.
Deal.
Of.
Surprisal.
Which.

Should.

Be.

Telling.

Us.

To.

Look.

At.

To.

Revisit.

The.

Global.

Model.

And.

I.

Think.

That.

This.

Interpretation.

Is.

A.

Salient.

One.

And.

A.

Relevant.

One.

Now.

A.

Little.

Bit.

More.

About.

Exteroceptive.

Bias.

Exteroceptive.

Bias.

Is.

Going.

To.

Be.

An.

Innate.

Tendency.

Within.

The.

System.

There.

Simply.

Are.

More.

Receptors.

There.

Simply.

Is.

A.

Lot.

More.

Going.

On.

Exteroceptively.

Than.

In.

Any.

Other.

Field.

Certainly.

In.

The.

Propriate.

Than.

In.

The.

Proprioceptive.

Field.

And.

The.

Animal.

When.

It's.

Born.

Immediately.

Has.

To.

Look.

Into.

The.

World.

Around.

It.

To.

Find.

Warmth.

Shelter.

Food.

Kin.

And.

That.

Need.

Doesn't.

Stop.

That.

Continues.

So.

There.

Is.

An.

Exteroceptive.

Bias.

That.

Is.

An.

Innate.

Tendency.

Within.

The.

System.

Although.

It.

Is.

Not.

Meant.

To.

Be.

Overwhelming.

Not.

Meant.

To.

Be.

In.

An.

Imbalance.

And.

Repressive.

Instead.

And.

So.

What.

That.

What.

Happens.

Then.

Is.

That.

A.

Control.

Bias.

Emerges.

When.

The.

Balancing.

Proprioceptive.

Priors.

Are.
Not.
Present.
In.
The.
System.
Then.
There.
Is.
A.
A.
Feedback.
Loop.
A.
Positive.
Feedback.
Loop.
A.
Vicious.
Cycle.
That.
Occurs.
A.
Control.
Bias.
In.
Which.
Exacerbates.
That.
Imbalance.
Now.
We.
Can.
See.
That.
In.
Right.
Hemispheric.

Dominance.

And.

Hyper.

Intellectualization.

changes.

In.

Technology.

And.

The.

Built.

Environment.

Self.

Evidencing.

Then.

Is.

One.

Way.

Of.

Looking.

At.

How.

The.

Body.

Calibrates.

Precision.

Tracking.

And.

Self.

Evidencing.

Now.

Self.

Evidencing.

What.

Is.

Self.

Evidencing.

I.

Think.

That's.

A.

Great.

Question.

From.

An.

Exceroceptive.

Perspective.

Self.

Evidencing.

Is.

A.

Little.

Bit.

Abstruse.

Because.

Exceroception.

Is.

Other.

Evidencing.

The.

Five.

Senses.

Provide.

The.

Basis.

The.

Evidence.

Required.

By.

Science.

And.

Reason.

This.

Is.

Conventional.

There's.

Really.

Nothing.

Wrong.

With.

It.

But.

Self.

Evidencing.

Is.

A.

Property.

Of.

A.

Self.

Aware.

A.

Self.

Directed.

System.

And.

That.

Would.

Be.

The.

Core.

That.

Would.

Be.

!

The.

Core.

Node.

In.

The.

Model.

Self.

Evidencing.

Then.

Is.

A.
Natural.
Property.
Of.
The.
Functioning.
Of.
The.
Proprioceptive.
Core.
In.
Its.
Relationship.
The.
Body.
Is.
Deep.
Deeper.
Inner.
Hidden.
Relationship.
With.
Itself.
Now.
There's.
A lot.
Of.
Ways.
To.
Look.
At.
That.
But.
We're.
Going.
To.
Talk.
About.

Precision.

Tracking.

And.

First.

We're.

Going.

To.

Talk.

A.

Little.

Bit.

About.

Fascia.

And.

We're.

Going.

To.

Talk.

About.

Precision.

Tracking.

In.

The.

Model.

Now.

The.

Rediscovery.

Of.

Fascia.

Is.

A.

Big.

Deal.

I'm.

A.

Bodyworker.

So.

I.

Work.
With.
Fascia.
That.
Is.
A.
Recent.
Phenomenon.
And.
I.
Would.
Argue.
One.
Of.
The.
Most.
Important.
This.
In.
Anatomical.
Science.
Since.
Vesalius.
In.
1543.
That.
The.
Restoration.
Of.
The.
Connective.
Tissue.
Matrix.
To.
The.
Image.
Of.
The.

Body.

Is.

A.

Powerful.

Powerful.

Form.

Of.

Belief.

Updating.

Now.

We.

Had.

Removed.

The.

Fascia.

Because.

It.

Was.

In.

The.

Way.

We.

Thought.

We.

Believed.

It.

Was.

Simply.

A.

Kind.

Of.

White.

Wrapping.

Tissue.

That.

Needed.

To.

Be.

Discarded.

So.

That.

We.

Could.

See.

What.

Was.

Underneath.

The.

Individual.

Muscles.

With.

Their.

Tendinous.

Connections.

To.

The.

Bones.

A.

Bunch.

Of.

Mechanistic.

Little.

Parts.

Was.

Revealed.

Through.

The.

Process.

Not.

Of.

Peeling.

Away.

A.

Wrapping.

Paper.

But.

Of.
Cutting.
Away.
Powerful.
Chunks.
Of.
Flesh.
Because.
The.
Muscles.
This.
Image.
That.
The.
Muscle.
Is.
Connected.
To.
The.
Bone.
And.
The.
Bone.
Is.
Connected.
To.
Another.
Bone.
And.
To.
Other.
Muscles.
Is.
Artificial.
The.
Bones.
And.
Muscles.

Are.
Invested.
Within.
A.
Connective.
Tissue.
Matrix.
That.
Provides.
Thirty.
To.
Seventy.
Percent.
Of.
The.
Force.
Production.
Of.
The.
System.
And.
Which.
Is.
Itself.
An.
Inner.
Skin.
A.
Layer.
Complex.
Inner.
Skin.
It's.
It's.
An.
Elastic.
Source.
Of.

Force.

Production.

That's.

Very.

Important.

But.

It's.

Also.

A.

Layered.

Inner.

Skin.

So.

That.

The.

Intermuscular.

The.

Intramuscular.

Septa.

In.

The.

System.

The.

Interstitial.

Tissues.

The.

Interstitial.

Space.

Is.

Very.

Very.

Sensitive.

We.

Now.

Know.

So.

We.

By.

Restoring.

That.

More.

Integrative.

And.

Holistic.

Self.

Image.

Anatomical.

Image.

Of.

The.

Body.

Then.

That.

In.

Itself.

Is.

Very.

Empowering.

It.

Certainly.

Is.

Empowering.

For.

Me.

As.

A.

Therapist.

Because.

I.

Understand.

It.

But.

Also.

In.

Terms.

Of.

My.
Movements.
It.
Gives.
Me.
A.
Lot.
Of.
Insight.
Into.
The.
Feel.
That.
I.
Have.
For.
These.
Deeper.
More.
Subtle.
Movements.
That.
We.
Say.
Experience.
In.
The.
Yoga.
Studio.
Or.
In.
Meditation.
In.
Tai Chi.
Or.
Rehabilitation.
Etc.
Myofascial.

Synergies.

Then.

Are.

And.

This.

Is.

The.

One.

That.

I.

Been.

Talking.

About.

Are.

The.

Basis.

Or.

A.

Kind.

Of.

Biomechanical.

Belief.

Updating.

In.

The.

Model.

And.

My.

Work.

At least.

Comes.

From.

This.

Wonderful.

Text.

By.

Tom.

Myers.

And.

Where.

He.

Has.

Positive.

Twelve.

Different.

Longitudinal.

Myofascial.

Continuities.

This.

Is.

Not.

The.

Only.

Example.

That.

Is.

The.

Preeminent.

Example.

But.

There.

Are.

Others.

In.

History.

And.

Others.

Available.

Today.

And.

An.

Interesting.

Hypothesis.

A.

Paper.

Was.

Published.

In.

The.

International.

Journal.

Of.

Molecular.

Science.

Arguing.

That.

The.

Myofascial.

Unit.

Not.

The.

Individual.

Muscle.

Is.

The.

Physiological.

Basic.

Element.

For.

Peripheral.

Motor.

Control.

That.

The.

The.

Basic.

Unit.

Of.

Motor.

Control.

Is.

Myofascial.

Synergy.

Now.

Again.

Some different pictures of the deep core.

And.

Then.

Getting.

A.

Little bit.

More.

Deeply.

Into.

My.

System.

What.

I've.

Tried.

To.

Do.

Is.

To.

Unpack.

This.

Axial.

Skeleton.

In.

Terms.

Of.

That.

Axial.

Line.

And.

What.

We.

See.

Here.

Are.

How.

That.

Axial.

Line.

Is.

Intersected.

At.

Each.

Major.

Juncture.

At.

The.

One.

End.

Of.

The.

Spine.

Where.

The.

Ground.

Connects.

With.

Where.

The.

In.

The.

Feet.

Where.

Where.

The.

Body.

Connects.

With.

The.

Ground.

And.

Each.

One.

Of.

Those.

Junctures.

Is.

Represented.

By.

A.

Circle.

Because.

Those.

Are.

Highly.

Sensitive.

Arrays.

Concavities.

Gradients.

If.

You.

Will.

Pits.

In.

The.

Body.

Full.

Of.

Sensitive.

Organs.

And.

Tissues.

Full.

Complex.

Junctures.

That.

Come.

Together.

With.

At.

The.

Center.

A.

A.

Bony.

A.

Bony.

Central.

Bony.

Servo.

Mechanism.

Now.

Um.

I.

Want.

To.

Apply.

That.

These.

Ideas.

To.

Movement.

And.

We're.

Going.

To.

Talk.

About.

Kinesiology.

So.

Conventional.

Kinesiology.

Of.

The.

Extero.

Subtero.

System.

Is.

Adequately.

Represented.

By.

The.

Very.
Very.
Primal.
Primordial.
Simple.
Stick.
Figure.
That.
We.
See.
On.
Cave.
Walls.
And.
That.
With.
Just.
A.
Few.
Simple.
Lines.
Can.
Represent.
The.
Expressions.
And.
The.
Movements.
Of.
The.
Body.
On.
The.
Outside.
The.
Extremities.
But.
It's.

Interesting.

To.

Note.

That.

Stick.

Figure.

Has.

Nothing.

To.

Say.

About.

What.

Is.

Obviously.

What.

Is.

Clearly.

The.

Most.

Important.

Part.

Of.

The.

Body.

Its.

Most.

Sensitive.

Contents.

The.

Axial.

Line.

The.

Core.

Now.

There.

Isn't.

A.

Stick.

We.

Even.

The.

Transmission.

Of.

A.

Car.

Has.

Linkages.

And.

Articulations.

So.

Getting.

Back.

To.

That.

Stick.

Figure.

In.

Conventional.

Kinesiology.

The.

Kinesiology.

The.

Movements.

Are.

Oppositional.

Election.

Coming.

Together.

Extension.

Coming.

Apart.

Adduction.

Bringing.

The.

Limbs.

To.

The.

Body.

Abduction.

Taking.

The.

Limbs.

Away.

From.

The.

Body.

They're.

Oppositional.

And.

Then.

Internal.

Rotation.

External.

Rotation.

Oppositional.

Movements.

Now.

Side.

Of.

The.

Offense.

In.

Proprioception.

In.

The.

Axial.

Line.

In.

The.

Core.

In.

The.

Synergistic.

System.

These.

Kinesiology.

This.

Kinesiology.

Is.

Not.

Oppositional.

It's.

Complementary.

It's.

Interdependent.

And.

The.

Movements.

Of.

The.

Axial.

Structure.

Are.

Also.

Diverse.

And.

Significant.

But.

We.

Can.

Describe.

Them.

As.

As.

The.

Grip.

We.

Talked.

A.

Little.

Bit.
About.
The.
Grip.
That.
We.
Can.
Get.
In.
The.
Armpit.
The.
Scapula.
Doesn't.
Just.
Grab.
The.
Arm.
It.
Reaches.
Around.
The.
Ribs.
In.
In.
In.
A.
In.
A.
Grip.
Like.
Structure.
And.
Contributes.
To.
Centration.
And.
Pressurization.

In.
The.
Torso.
Centration.
Coming.
Together.
Pressurization.
A.
Little.
Bit.
Of.
Expansion.
Work.
Together.
To.
Create.
Balance.
And.
Stability.
In.
The.
Body.
On.
A.
Longitudinal.
Level.
We.
Push.
Down.
Through.
The.
Posterior.
Line.
And.
Experience.
Lift.
Coming.
Up.

Through.
The.
Front.
This.
Is.
Something.
That's.
Experienced.
In.
Yoga.
All.
The.
Time.
Gravity.
In.
Fact.
Translated.
Through.
The.
Tensegrity.
Structure.
Of.
The.
Body.
Does.
Not.
Pull.
Living.
Organisms.
Down.
But.
In.
Fact.
Lifts.
Them.
Up.
It.
Provides.

A.
Vertical.
Kind.
Of.
A.
Gradient.
That.
My.
Physics.
Is.
Getting.
Way.
Ahead.
Of.
Me.
But.
Provides.
Lift.
And.
This.
Is.
Something.
That.
We.
Can.
See.
Commonly.
In.
Life.
And.
That.
We.
Can.
Experience.
In.
Ourselves.
As.
We.

Bring.

Our.

Posture.

Together.

That.

Poise.

Is.

A.

Form.

Of.

Lift.

And.

Then.

The.

Third.

Set.

I.

Think.

Of.

Of.

Complementary.

Cooperative.

Sort.

Of.

Movements.

Is.

Some.

Sort.

Of.

A.

Spiral.

Helical.

Pattern.

In.

The.

Body.

Now.

We.

Talked.

A.

Little.

Bit.

About.

Homologies.

And.

How.

The.

In.

The.

Exteroceptive.

System.

In.

The.

Body.

In.

General.

There.

Are.

There.

Is.

Segmental.

Homologies.

And.

That.

Helps.

Us.

To.

Organize.

The.

Body.

The.

Body.

Movements.

But.

When.

We.

Shift.

From.

Between.

Perceptual.

Fields.

From.

An.

Exteroceptive.

To.

A.

Propioceptive.

Field.

Then.

We.

Also.

Experience.

What.

I.

Think.

Is.

An.

Inverse.

Homology.

And.

That.

Helps.

The.

Brain.

To.

Sort.

Of.

Get.

A.

Grip.

From.

Within.

I.

Won't.

Say.

Any.

More.

About.

That.

We.

Talked.

A.

Lot.

About.

The.

Bony.

Serval.

Mechanisms.

Here.

We.

Go.

Into.

Them.

A.

Little.

Bit.

More.

deeply. The different bony cerebral mechanisms on the axial line, there are no other alternatives, no other options. It's very clear what they are. We can come into each one of these sensitive areas in each one of these major bones in order to organize that area. Now, here they are laid out in the skeleton. And we can see the circles around those bones and on the right hand side and highlighted in a kind of a green blue is each one of those prominent bony cerebral mechanisms.

Each one is at the center of a sensitive kind of hidden concavity, the groin, the throat, the armpit, in the soles of the feet. Now, these are clearly areas in the body where we're meant to know ourselves. But what happens when we get confused? So this is a little diagram, and I'm going to finish with this diagram today about how we actually, what Eric Starvo Galt means, as a control bias means that we wear the body backwards. And the green areas are the axial postural areas, the bony affordances, largely in the front, with the exception of the scapula.

And these are the areas, if you think about your body now, the most sensitive areas of the body, but are these the areas where we experience the most problems? These are the areas where I think that we experience the most repression and sometimes even shame. And when we see postural collapse, we see a collapse in these areas with a tremendous amount of stress and tension and compensation

required in their opposites. And those are the red areas. So when we have a head forward position, when we have collapse in this anterior structure, the back of the neck is carrying this heavy, imbalanced lobe, we get tension in the occiput, in the neck, and in the shoulders. When we fail to attend to the proprioceptive prior in the armpit, we make the shoulder do all the work. And of course, that's the third most common site of musculoskeletal complaints. It is not where we're meant to control the body. We're meant to find coordinative awareness in these concavities. And there are bony cerebral mechanisms of a low back. In relationship to the groin, the pubic bone in the front, the pelvis very much is shaped like a wheelbarrow. And those two big bones in the back, they're like the two big wheels or handles of a wheelbarrow. And we try our best to force that load. And we believe that we can control that load with brute strength until the front wheel of the wheelbarrow hits a pebble and the entire thing tips. So if we don't attend to the little front wheel to the low front, then we're going to have dysfunction in the low back. The feet, what I'm going to say about the feet is that that big toe is very clearly the power structure of the entire foot. And what's the first thing we do when we put it in modern shoes? Hallux, valgus, we collapse, we crush that structure. So that's my talk for today. And that's an effort to unpack belief updating and exteroceptive bias in a somatic way with reference to Sherrington, with reference to anatomy, myofascial anatomy. It's been a real pleasure. I want to ask you if you've made it this far in the talk, that you give me some comments, give me some feedback, hopefully some positive feedback. I've worked very hard on this. Again, I'm very excited to make this unusual somatic and atomical contribution to the field of active inference. Thank you very much.

All right, the next talk is by Sid. 10 minutes long.

Greetings, everyone. My name is Siddhant. And I'll walk you through our recent work on enhancing Byzantine-tolerant formation control for maritime buoy swarms using, you guessed it right, active inference. What are maritime buoy swarms? As you can see on my screen, maritime buoy swarms are a swarm of buoys deployed in the ocean in order to achieve certain tasks. For example, if there is an oil spill or like an incident in the ocean, these buoys help in monitoring that. They also help support missions like surveillance, search and rescue, and other kinds of environmental monitoring in hostile or contested waters. The threats that we focus on in this particular work rely on the configuration that you see in front of you. All of these buoys can be imagined as decentralized robots floating in the ocean with minimal motor control, and they can communicate among each other and a centralized hub. But because of this, they are also prone to cyber physical attacks. So we will focus on three kinds of attacks which can impact their operation and mission success. So for example, if the task is area coverage, like taking down one or more of these buoys is equivalent to a Byzantine problem where we cannot trust certain buoys anymore. And now we have to measure how to measure the mission success in this particular scenario so that the rest of the buoys can reorganize around this. And in order to do that, they need to detect the cyber physical attack at play. So the threats that we will focus on are GPS spoofing, communication jamming, and actuator hijacking. Why these three? Because these three push the resilience limits of existing control architectures which may not use artificial

intelligence or any active inference approach. Our goal in this work is to maintain formation integrity and task coverage while detecting and mitigating malicious agents. I'll start with our baseline approach and then explain how we integrated active inference, where it shines, where it still struggles, what alternatives we considered, and how we plan to push the results further. The original methodology without active inference uses a decentralized neighbor-to-neighbor approach in order to facilitate communication. Each buoy assessed trust using handcrafted manual heuristics, rules that downweighted agents exhibiting inconsistent trajectories, communication dropouts, or mission throughput anomalies. We combined that

multi-dimensional trust score with an attack-aware formation controller which essentially helped re-weight the neighbor influence based on trust estimates while enforcing formation geometry and coverage. This approach was practical and interpretable and it handled many different kinds of anomalies well,

as you can see in the graph in front of you. However, it lacked formal uncertainty quantification which is very important for this kind of a task.

It struggled with subtle or staged attacks that we inflicted upon the system and sometimes even over penalized honest agents during complicated maneuvers.

In our improvement methodology, like we see, we see that we introduced an active inference based trust assessor. The core idea here is to move from heuristic rules to a probabilistic generative model and variational Bayesian inference.

Each buoy in this new model maintains an internal state predicting neighbor trajectory's position, velocity, and acceleration, and even jerk to measure change in momentum while taking into account estimated formation forces.

When reality deviates from predictions, we compute a mahalanobis distance across those dimensions. High prediction errors suggest potential tampering. We then update our trust beliefs using Bayesian filtering, explicitly tracking both the mean trust score and its uncertainty. The formation controller uses those uncertainty-aware trust estimates to moderate neighbor influence.

Agent communication stays decentralized. The key change is that the trust updates now come from predictive modeling rather than fixed thresholds.

And this, as you can see, improves the detection accuracy a bit across all three metrics.

Where active inference is useful can be highlighted in the following ways.

Like, the graphs are pretty clear that active inference over a period of time can help in improving the results that we are seeking.

But to do that, it helps mostly in three areas.

First is early detection.

Predictive errors surface anomalies before they manifest as gross deviation.

We see this particularly with GPS spoofing position when they diverge sharply from expected trajectories.

Secondly, uncertainty quantification helps us in carrying variances along with trust means so that we can protect the honest agents when the model is unsure.

This reduces false positives during tight maneuvers.

Thirdly, it helps with attack specificity.

Different attack modes produce distinct error signatures which are important for us to quantify and measure. GPS spoofing elevates the position residuals.

Communication jamming shows up in missing observations.

Actuator hijacking disrupts velocity and jerk.

We calibrate these responses accordingly.

In our V2 results, this translated into an 8.4% average improvement in detection accuracy, a 1.6% boost in resilience, and 6.6% gain in task coverage with actuator hijacking coverage jumping nearly 20%.

And the results are pretty clear.

We do not claim that active inference is always the better approach.

We can see here that active inference does a lot better at task coverage and resilience score, whereas it is nearly as performant in other metrics or sometimes less performant in average trust score over time.

Where active inference still falters is that the generative model still simplifies the hydrodynamics and control inputs.

When cooperative agents execute aggressive maneuvers, prediction errors spike, and the system can result in false alarms.

Moreover, communication jamming creates data gaps that challenge the inference updates.

Even with tempered thresholds, detection improvements for actuator hijacking are minuscule.

So what do we do?

We see that detection improvements are modest with only 3.5% because actuation dynamics are hard to model at the granularity we need.

So active inference is computationally heavier for these decentralized swarms with lesser processing budgets.

So we need reliable processing power in order to ensure real-time updates, which may not always be a solution in a maritime environment.

The alternative approaches that we have considered was a reinforcement learning based trust estimation, where agents learn policies from simulated attacks.

It offers adaptability but lacks the interpretability and uncertainty guarantees that we wanted.

Another was a consensus-based filtering with fault-tolerant observers, classic control theory techniques.

These have strong stability proofs but require centralized coordination or strict communication assumptions.

Another thing that we could look at is a hybrid approach which combines Kalman filters with heuristic gating, which has the potential to show some promise, yet it also has the potential to lack explicit uncertainty handling.

What active inference gives us is a balance between theory, interpretability and decentralized operation.

The key finding for us is that active inference can outperform heuristic baselines across detection, resilience and coverage,

while providing principled uncertainty estimates.

GPS spoofing detection jumped over 15%.

Our communication jamming detection improved by 7%.

And actuator hijacking coverage improved almost 20%.

These are not basic numbers. These are quite significant.

So how do we go about it in a practical setting?

We confirm that these protective floors and attack-specific calibrations are essential.

Without them, active inference overreacts.

Finally, the predictive modeling gives the commanders richer diagnostic signals, not just what went wrong, but how confident the system is in that assessment.

This is where active inference really shines.

We can improve upon a few things as next steps.

Our next steps focus on richer generative models incorporating better hydrodynamic parameters for this environment,

external disturbance introduction, and learned control inputs.

We plan to fuse communication reliability signals more tightly with behavioral predictions to handle jamming gaps.

We also want to explore hierarchical active inference, where local agents share compressed summaries of their beliefs to enhance collective awareness without breaking decentralization.

Finally, we plan to benchmark against adaptive consensus and reinforcement learning hybrids to ensure that we are harnessing the best aspects of each approach.

In summary, active inference has pushed our maritime swarm resilience forward in a simulation environment, giving us predictive confidence and measurable improvements against Byzantine threats.

The work is not done.

We need to do a lot more to move this from a Python-based simulation environment to the real world.

But the path is clearer.

We need to sharpen the models, strengthen the communication inference loop, and validate under more diverse operational profiles.

Thank you.

Please feel free to reach out, and I'm happy to dive deeper into any section.

And it has been an honor presenting here at the symposium, and I wish you luck.

Thank you.

All right.

Next talk is by Nicholas Heinrichs, Hyperscanning Under Active Inference for Mental Health.

This is going to be an 11 minute recorded video.

recorded video. Hi, my name is Dr. Nikolas Hinries. I'm a neuroscientist working at the

Max Planck Institute for Human Cognitive and Brain Sciences. And hi, my name is Dr. Nikolas

Hinries. I'm a neuroscientist working at the Max Planck Institute for Human Cognitive and Brain

Sciences and at the Okinawa Institute for Science and Technology. I study how people come to feel

together. Today, I want to talk about a problem that's obvious to anyone who has ever been in a

redundant clusters while negative curvature marks bridging edges that connect different modules

this edge-centered measure captures local flow in ways that node-based synchrony can't our method which we've dubbed geometric hyperscanning computes the forman ricky curvature on time resolved inter-brain networks

derived from dual eeg f nurse or f mri recordings we treat each cortical region as a node and functional connectivity as weighted edges for each edge connecting nodes i and j the forman ricky curvature is computed

over the weight of that edge multiplied by the sum of one divided by the weight of node i and one divided by the weight of node j then you subtract the correction term which is the sum over all edges that share a node

with the edge under consideration of the ratio of the root of the product of that edge's weight and its neighbors

weight in this context the function w assigns weights to nodes and edges

to track how the overall geometry evolves we compute the entropy of the curvature distribution within the sliding window curvature entropy at time t is calculated by taking the negative of the sum over all edges present at that time of the probability of each edge multiplied by the natural log of that probability

here the probability of an edge is the curvature of that edge divided by the sum of curvatures across all edges this resulting scalar curvature entropy measures the volatility of interbrain topology

abrupt peaks or discontinuities in curvature entropy indicate topological phase transitions in the shared generative manifold when the entropy spikes it often corresponds to effective rupture when it declines the repair and reattunement are underway how do we how do we use these insights in therapy each partner in therapy so the therapist and the client

is modeled as an agent whose self-model contains a generative model of the other prediction errors in this repersive model manifests as topological changes in their combined neural activity by combined by computing

former ricky curvature on the weighted interbrain network this third brain between therapist and client and tracking the entropy of its distribution we obtain a real-time readout of when the interaction is stable rupturing or repairing

here's where the idea of a digital twin becomes powerful a mental health digital twin is a virtual representation of an individual's mental state that is continually updated with data collected over the person's interaction such a twin can guide professionals in diagnosing and treating patients using mechanistic models machine learning tools

the digital twin concept extends beyond simple simulations it spans the neurophysiological cycle of an individual and updated by real world data using physical and statistical models to answer what if questions

mental health digital twins could forecast deterioration predict treatment outcomes and recommend tailored interventions

implementing this framework requires robust data and careful modeling hyper scanning signals must be pre-processed to

to construct reliable time resolved functional connectivity matrices the curvature calculation depends on appropriate weighting schemes positive curvature highlights redundant local clusters while negative curvature highlights bridges entropy should be computed within windows suitable for the modalities tempo resolution shorter windows for eg or longer ones for fmri in real clinical settings the digital twin must integrate behavioral cues gaze prosody gestures and self-report data with the neural one each of these streams can inform the posterior over latent states ensuring the twin is grounded in multimodal evidence ethically digital twins raise issues of privacy and agency despite these challenges the potential benefits are considerable by embedding geometric hyper scanning within an active influence framework we can develop social markers real-time signatures of interpersonal coordination analogous to biomarkers in therapy these could alert practitioners to ruptures they might otherwise miss guide decisions about when to pause or reframe an intervention and help quantify progress over sessions in research curvature based metrics allows allow us to test hypotheses about how belief updating and effective revelation regulation unfold moment by moment in technology digital twins could inform adaptive interfaces and social robots that adjust their behavior based on the geometry of human interaction mental health interventions thrive on attunement yet quantifying attunement has remained elusive we propose a unified framework that models therapist client interactions as coupled generative systems formalizes affect as inference over self-model coherence and measures inter-brain dynamics using discrete geometry the form on ricky curvature and its entropy reveal dynamic phase transitions rupture correlation and reattunement which standard synchrony measures miss building on this insight we sketch a digital twin for psychotherapy a virtual entity that integrates real-time neural behavioral and emotional data to time stamp and anticipate shifts in relational state such a twin has a potential to assist therapists in guiding sessions towards the optimal outcome namely healing we invite collaborations to test this approach on real clinical data and further refine the models for broader use

thank you

thank you

thank you

okay

we're back

this is going to be

the session that would have been immediately following carl fristen's keynote address earlier today due to some technical issues the group with ian tenet recorded this during the time so here it is it is a 45 minute video

thank you

yep

yep

see you

great

okay

good

hello and welcome and thanks for joining

this presentation

this presentation for the fifth applied active inference symposium is titled an applied active inference group practice for transdisciplinary collaboration my name is ian tenet i'm based at anglia ruskin university

so i'll start with explaining why i'm interested in a particular type of group practice or communication style for transdisciplinary work and then think about how we're going to go about this today we'll do a live demonstration of this type of group practice and then i'll talk about the details of what it involves before we do the live demo

so at anglia ruskin university a year or so ago a new planetary health group was set up and um the idea was it brings together a new group of peers and um colleagues who are interested in planetary health with different backgrounds so some from engineering some from economics health different types of health and um law so we're in the stage now where we're still kind of storming forming and norming so understanding each other's interests and trying to figure out a shared way of looking at global planetary health issues and the idea of the transdisciplinary mindset appealed to me but i wasn't sure exactly what it means so i looked at the literature for some definitions and found a nice paper in 1998 titled complexity and human health the case for a transdisciplinary paradigm and this table sets out a spectrum of ways in which we can look at big problems so in this case health problems we can draw different boundaries around the problems so um if we considered something like a um a health problem like kidney disease a single discipline might be a kidney specialist thinking about the best medical medicine or therapeutic to treat the disease a multidisciplinary uh approach uh approach might be um uh kidney specialist teaming up with a engineer to develop a new device uh transdisciplinary

um according to this definition goes a step further and they widen the scope of the problem to look at the um um situation of kidney disease from multiple levels a kind of look at health as a dynamic system operating um even beyond the individual

uh so um staying with the example of kidney disease here we have a picture of uh dr william colf who was the first

um person to develop an artificial kidney these days we call it dialysis and he teamed up with engineers to build the first

um kidney dialysis machine these days we have transdisciplinary projects looking at things related to um uh the issue of something like end stage kidney disease so with kidney disease um some people can develop it after

having diabetes for a long time which damages the kidney so if we were going to go to the root cause of uh draw

expand the boundary of our problem and say why for example are there so many people in europe or in or the uk

developing diabetes which then reaches this stage of end stage kidney disease we might take a transdisciplinary

mindset and this project here called cities changing diabetes from 2014 it's um brought together people from lots of different disciplines including urban planners and uh maybe environmental toxicologists to look at what is it in the environment that might be causally related to people developing diabetes and the problems that they may come up with may be involved redesigning cities to make them healthier places to be in so often these are um it's a blending of disciplines for a long-term collaboration and it may end up in resulting with new frameworks or theories to work with however um when new people with special um interests and

working models come together there's often some challenges during the early stages and i'm going to use the johari window model here to um explain what i mean by this so with the johari window model um um it's it's called a window because it looks like four planes planes of glass and we have in the top left corner things that are known to um the self or one group of people but they're not known to other groups of people so we could say with people with some uh specialism or one discipline let's say a uh kidney specialist there are things that are known to the specialist in kidneys but not known to an engineer in the other corner we've got um things that are not known to the self but are known to the other group and then we've got in the bottom right corner things that are not known to either of the two groups and the aim really um of transdisciplinary research or teams will work better if we we expand the window where things are known to the self and the other and hopefully people familiar with um the active inference framework will recognize this as a form of reducing uncertainty or reducing surprise or minimizing error

so um free active inference involves reducing the discrepancy between what we uh the internal model and what we sense from the world and we can do this either through updating our model of the world or trying to change the world change what we sense from the world um a slide here from one of carl friston's papers where he um uses uh an example of songbirds singing and turn taking so the birds generate a shared model or shared generative model of their song by taking turns to listen and to sing and this is referred to as precision engineer engineered message passing and we all kind of do this sort of thing um every day we take turns to listen and speak but there are different um styles or ways we can do this the archetypal way of um thinking of um thinking of people coming together to share stories is around a campfire um but we do it in lots of different ways

um but we do it in lots of different ways the last month that was the sixth international workshop on active inference

so a conference is a style of um sharing generative models and um it may involve people presenting for long periods of time

um and then questions and answers and we're seeking to kind of understand each other and reduce our

uncertainty

about each other's models in the therapeutic world there's lots of um approaches to this on the right a picture of

uh counseling session so in some forms of counseling the counselor will not um necessarily uh try to uh coach or guide the person in counseling they'll literally listen and reflect um and allow the person to share but it's a very um structured approach another style of group chat maybe around uh less structured is around a social event around a table where people will take turns but there um there's no structure to say who has to speak first or who gets to speak or um who should be listening

so um it's a big spectrum of styles for sharing listening and um speaking and listening

and they range from having a lot of structure to little structure the approach we're going to use today that how we're going to go about this is um inspired some by some therapeutic traditions and one of these is um has been developed by john bavakey who's appeared on the inference uh institutes uh several several times

and he um calls this style dialogos and he says that it encourages a reciprocal flow state which is similar to um jazz or sparring so um this type of style has been developed by um the circling institute and for those of you who wish to find out more uh can visit the circling institute website but um it shares similarities to lots of therapeutic styles of communication or non-violent communication and um my partner has

um trained in this style of communication and has uh encouraged me to um adopt it and i'll just share a very brief example of how it's helped in our relationship so the idea is that if someone is speaking to you and um you rather than if it if something arises that maybe you disagree with or there's some uncertainty about you seek to understand the other person before you add your own opinion so this simple example was a day when we'd had a lovely day out and at the end of the day my partner said she wasn't feeling very loved and the part of me that might have been sensitive or um wanting to defend myself might have said um of course you're loved we've done all these lovely things today and so on but because she had um helped me understand the ideas of reflective practice um i sought to understand what she meant so i said okay

in what way are you not feeling loved um please help me understand what you're feeling and this led to a really interesting conversation and our relationship deepened as a result of it so today we're going to use um a structured form of communication and we're joined by susan andrew and leonardo to demonstrate this and there are um certain stages to it but the general principles are that when you're listening to someone speaking then you um seek to understand what they're sharing through reflecting what you've heard back to them and if there's an error it gives the person who spoke a chance to clarify or they may want to add further detail so the mindset we have to approach is to be comfortable to be comfortable with the the error and if you are sharing then be open to revealing your whole experience so that could be more the effective side of the experience so your emotional valence as well as just the facts the facts so i propose that this type of approach can help with generative model alignment in transdisciplinary research so through this turn taking in small groups we can continually update our

beliefs about others intentions and perspectives it can help with precision calibration so uh the practice can highlight which signals deserve more attention and which need revisiting can help with epistemic action and shared anticipations over time so hopefully the transdisciplinary group will develop a um kind of shared um expectations or uh beliefs about how they can approach the transdisciplinary problem um so a quick um bit of detail about what is involved so we imagine we're sitting in a virtual circle if this was face to face you could sit in a circle and it's turn taking between listening speaking so the first two people um person a and b person a will speak to the topic and person b will listen and then aim to reflect back what they've heard

and then um when person a has finished um yeah person b person b reflects and then when they've reached the position where person a feels understood person b turns to the third person and they share what they what they feel was surprising um or confusing or jarring about what they heard and then after then they add their own opinion

about the topic so their fresh opinion about the topic it doesn't have to be related to what the previous person said

then this continues around the circle so pers the third person then takes the role of the seeker so seeking to understand and they reflect what person two said and that continues so we're going to demonstrate this live now and um the topic we're going to look at is um about misinformation so a recent um piece of research says that of all the kind of um potential threats around the world that people are worried about such as natural disasters war um and so on disinformation is amongst the top so people are worried about um false information and the threats of that so what um i'd like the group to have in mind is how might our understanding of active inference help us notice and respond differently to the types of unhelpful information in society that could be ai generated or it could be through social media or it could just be everyday life so i'm going to stop sharing the screen and we'll return back to our small circle and um welcome back susan andrew and leonardo so you're welcome to unmute now and maybe before we start the group practice could you just introduce yourselves and say a few words about your interest in active inference

so we'll start with andrew okay that sounds good so um can you hear me okay yes great thanks um so yeah thanks ian for for inviting me to this session um i have been involved with the realm of active inference science for roughly three or so years um sort of my background has been a very broad combination of the social sciences economics but actually kind of a move towards complexity science uh neural imaging data analysis i work full-time as a data analyst so i'm just i'm a very curious person i'm very interested in many things and uh so i i interned and then uh became a member of the scientific advisory board with the uh the institute and some people will recognize me for teaching uh the the proverbial 2022 textbook uh by uh par and patsula and friston so thanks thank you andrew uh susan hi yeah i'm not a neuroscientist or data analysis uh i teach uh biology at uh university of california and i'm interested in active inference for the roles that it can help

us find better ways to help students learn and for the difference between how they perceive they're doing in a class and what actions they're taking

and how they're doing and how they actually are doing and i'm very curious about ways to break through that

And you say that these two things are not equal.

So information might not be viewed by the observer to be helpful.

So it might be viewed as unhelpful information, but it may actually be true in some sense.

It may actually reflect something accurate.

So it may not be misinformation in the sense of false information.

And you use the example of maybe something, a repeated message by a politician, for example, that may confirm, reaffirm someone's belief.

So even if it's misinformation, it may be used as it may be viewed by the receiver as helpful.

Even though it's not necessarily accurate.

So you were aiming to kind of show some examples of when information might be viewed as helpful or unhelpful or accurate or inaccurate.

And that those two things don't always, they're not always equivalent to each other.

Anything else you'd like to add or correct about my summary?

I think you did a pretty good job of the summary.

Yeah.

I think that I think it's useful to include the idea of what does it mean for information to be helpful or unhelpful and then kind of juxtapose that with what misinformation is.

Because I think that kind of by understanding the idea of information being helpful or not to someone, it actually can kind of help us potentially better understand the dynamics of the spread of misinformation.

Right.

That to kind of it's just reiterating what you said regarding it, reiterating what I said.

That is in our in our turn taking style here that, yeah, to for information to to make me feel good for me to self evidence for me to oh, I don't have to change my model of the world very much because these people are telling me what I think I already knew.

Right.

It's quite comfortable.

It's very low cognitive cost for me to process that information and update my model and perhaps the notes to us unbeknownst to us, it nonetheless could be misinformation.

So thank you.

you. So you were saying that you think it can be useful to include the notion of helpfulness or unhelpfulness when talking about misinformation. Anything else to add before we move around the circle. I think we should keep turning taking. Okay so I'm going to turn to Susan next and what I do now is briefly summarize what I found interesting about Andrew's sharing before I share myself. So yeah what I really liked about Andrew, what Andrew shared is that he gave an example of when something can be viewed as unhelpful but actually be accurate and then he gave an example of the opposite case. So for me that opened up something around transdisciplinary work when new teams come together and they you may have an expert who's got a special interest and they are hearing lots of information from someone else so it could be they're hearing some information from a finance team for example or a engineer but they don't view that information as helpful. Their interest might be on let's say the psychology of the issue.

So they may discard it as not helpful information but actually for the project as a whole

it may be useful. So for me that presents a real challenge. How do you help each member of a transdisciplinary

group see the value of information if it has got value? That's all I want to share.

Okay so reflecting back I think Ian was thinking about, I think Ian you were thinking about situations where you hear or one hears information

and one isn't necessarily sure in a group why that information is being presented. Is that correct?

Yes, yeah.

And then you have to try to figure out what to do about that presentation and did you come up with any, have you thought of ideas for what ways to manage a situation where information is coming up and you're not sure how to use it?

I guess what's coming to mind is there's a proactive and a reactive way. So the proactive way might be a group member might sort of put their hand up and say excuse me I'm not sure about the significance of that information or I'm not sure about a terminology, maybe I just don't understand the vocabulary being used. That requires the person to feel confident about their uncertainty.

Another way I can imagine is to maybe the facilitator or the organiser supporting or scaffolding that uncertainty in some way. So being prepared that people will not know the significance of each other's information and putting in place some kind of process, either background information or some kind of glossary or

or translation tool to help members to realise the significance for new groups.

So you're saying basically that if in a group where information is presented, someone's struggling to understand whether it's useful or not, they could ask the person themselves and say I'm not sure why you're saying that or maybe I just don't understand it. Or which puts pressure on them in a sort of social context to say that they don't understand which may be uncomfortable.

And then on the other side, it could be the role of the facilitator to realise that there are gaps in communication and step in and try to fill them. Is that sort of what you're suggesting?

That's it.

That's it. Yes. Yeah. Thank you.

Yeah.

So I don't think I have anything else to add at this point. So Susan, you're welcome to turn virtually towards Leonardo and share your thoughts on this topic.

Yeah, I think this topic is really interesting because what it seems often in transdisciplinary groups, it's very hard to understand whether you don't understand what they're saying or you don't understand why what they're saying is relevant or what they're saying isn't relevant.

And so you're always in this sort of internal quandary and trying to think, how do I present this in some way that, you know, yes, I may have to show my ignorance, but in the goal of moving the discussion forward, how do I react to this?

And so I think that that people, I think facilitators are often hired for reasons that they can try to bridge those gaps and help people understand them.

But I think sometimes facilitators either don't know the topics well enough to be facilitating those topics and that can create more trouble.

So I found some of these discussions to be kind of difficult for the reasons that both Andrew and Ian have mentioned.

And I wonder, Leonardo, what are your thoughts on this? And is anything I've said resonating?

Thank you, Susan. This is a very interesting topic. I'll just start by making sure I understand what you've explained to me.

So you're saying that you can see the importance of understanding unhelpful information in transdisciplinary groups because we, as a listener, we might develop internal troubles

because we might be afraid of looking ignorant if we don't understand something, but we have to balance it with the desire to progress the conversation.

And so, am I understanding? Is this relevant? Well, maybe a facilitator can help to understand, to make sure that the language is a bit more commonly understood.

But the problem here is that they might not be an expert and might actually make the discussion even muddier.

Is that the main topic or is there something else you'd like to mention?

No, I think that's the main topic. I think the other piece is trying to figure out whether what the person said really was on par for the discussion or was sort of something that was leading the whole group off of the discussion.

It's the main topic and the whole group off in the wrong direction and I really don't know how you deal with that.

I see. So, that is an interesting problem to tackle in problem-solving situations, yeah.

So, now, do I turn to someone or...

Yes, Leonardo. So, Susan, do you have anything else to add before...

add before. I do not. So now we would at the bare minimum we would close the circle so Leonardo you would turn towards Andrew and share a bit of a summary of what you've heard before and then add your own new insight. Fantastic. So the discussion Andrew so far has been about around really unhelpful information whether we can in transdisciplinary groups able to see what is really significant to the overall topic conversation with the group or what can we ignore and how can we address this problem of whether something is unhelpful given that we're not an expert in what the speaker is saying or even the facilitator is not an expert on what's being discussed. I found the role of facilitator actually very important in what I've been hearing in this discussion. It seems like it's a good solution could be sending some pre-meeting readings that aren't too long. Maybe all individuals should send out at least a few bullet points relating to the discussion before. Then so in the situation of misinformation and unhelpful the difference in that I mean I feel like if you're saying something as an expert it's maybe important to say that this is a true thing this is evidence-backed and maybe coming back in your explanations that well this relates in this way to the overall topic.

And I mentioned on the the fact of saying that what the statement you've expressed is true

to emphasize the truthfulness of your statement because if it's unhelpful

and false it won't bring to the conversation so it's just saying well I'm not trying to invent something.

So yes I think that was to emphasize the truthfulness of Usain and the relevance to the overall topic every time we express a statement and our pre-meeting readings. So what do you think of that Andrew?

Andrew? Sure thanks Leonardo. So to kind of recap my understanding of everything

so we have we're looking at the notion of unhelpful information in the context of transdisciplinary work

how do we kind of see the overall significance of particular pieces of information to the whole group

what can we ignore or perhaps what should we realize we shouldn't ignore and you laid emphasis upon the role of the facilitator

uh seems to be very important and they could perhaps help with organizing a bit through

sending out pre-meeting readings or bullet points in advance of the discussion

uh to maybe establish some kind of initial common ground between everyone and maybe what they're

thinking about but then that said uh being able to relate different pieces of information to various

topics or in the context of transdisciplinary research relating like the same topic to potentially

different disciplines I could also see coming in there and um so being able to keep those connections

um have I done a fair enough job at kind of kind of recapping there a bit definitely definitely um

there was just one point I wasn't too sure how to address it in the but it would be that to emphasize the

truthfulness of the statements so I guess as an expert use you want to you suppose you're saying something truthful but

you know I don't know if that's um it's just an idea I'm not sure if it's um practical but um in the in the um

definitely the relevance to the overall topic is the most important as you've uh repeated Andrew

i say ian do i should i can see you can you can maybe um

um so leonardo wanted to add an extra detail so if you're able to can you reflect that back to him

yeah uh i mean to to emphasize the truthfulness of the information you're sharing i mean i find that

quite significant as far as i understand bringing that up because if you know if you're working with a team of

people and you introduce you know say a new concept that's otherwise foreign to them um then if

they're not able to sort of relate it back to their own uh discipline and you know you could sort of

inject the the active inference framework here a little bit and say like if you say something that

simply does not sort of correspond in some meaningful way at least at a baseline at a micro scale as a

starting point with their their prior beliefs about you know whether it be the world at large or at

least like how their discipline works or their general topics and and foci that they focus on then

yeah i think i think well in that case you really want to like bridge that gap for them uh in a way that

you can you can justify by saying that there's some truthfulness here so um so like social science

music and clinical research um you know if you and i could establish some kind of gap or even if i was

sort of like proverbially selling active inference to you saying like oh you should use active inference

here's why here are various you know papers that have been published on like dynamics and clinical

therapy that relate back to the active inference framework and how it can be useful so being able to just

kind of provide that sort of additional information that can help establish that connection potentially

in a way that exceeds just me saying it to you right it's like no it's you know not to appeal to authority

here but here's some published papers you know that have much more detail if you want to pursue that

further

if it piques your interest so that's one way to deal with that great yeah yeah referring to the published evidence

it's a great way thank you

okay thank you so i think we'll close the the demonstration and close the circle there

what can be done with this type of um process is that you can keep going around the circle

again so you can see what else evolves with another another round but given the the time constraints

today um we'll draw towards the end i would like to hear um any views that you have and it can be kind of popcorn style so any any person can speak first um about how you found this process so you all kindly gave up some time before this session to have a practice with me and we practiced with the simple question of please tell me about your name and we went through some uh a round of reflection there so you you've experienced it twice now do you have any thoughts about um how this process of turn taking and reflection um could be useful or not or where it could not be useful

i can go first yeah i i think it is i think it is helpful because you uh get a chance to sort of

hear how other people perceived what you said which helps you clarify your own thinking uh you think oh is that how i put that okay and you get so so you have that role of self-reflection on your own thought

but then you also hear how the um how the thoughts of others dovetail and interact and i felt like we were moving towards uh conclusion if we had another couple of rounds we might have come up with a sort of

you know tiny version of how you should start organizing a certain kind of discussion so i thought

it was helpful thank you anything else to add andrew or leonardo i don't mind saying i agree with susan that it is helpful and we were getting closer to a conclusion for sure it still feels a bit

uncomfortable to make sure you're oh am i fully understanding am i adding am i repeating well but

i think it's a good process yeah yeah i uh i mean i think i think it's great and uh just so that i can

add a little bit more hopefully not too much i'm trying not to go into like teaching mode but you brought up the the bird song example in your introductory slides to this and what i love about that example

is through the turn taking between the the songbirds it's not just that they're kind of mutually uh

reducing prediction error but if you actually look and this is where it gets a little mathy i don't want

to get too abstract but if you actually look at sort of the state space that they collaboratively

explore um it starts to expand so they start to synchronize but explore new places in the state

space that neither one of them have individually explored before so through this kind of collective

emergent behavior they traverse new territory and so if you bring that back to us talking it's kind of

like well we all sort of have our own mutual our own respective explored spaces around these topics

you know we're our own degree of interest but by going around the circle i agree with with leonardo

and susan that it did seem like we're kind of reaching a new sort of conclusion about these things

and in some way it's kind of like where it's through this turn taking that we can kind of reach new places

and conversation and our thinking that wasn't may or may not uh i won't be overly optimistic but may or may

not have been immediately available to us before doing this so basically collaborative exploration

leads to new insights that weren't available before potentially yeah i think it's great very nice thank

you yeah that was really um useful to hear that view from an expert in the maths behind active inference so thank you three for helping uh to rescue this session with the technical difficulties and i hope it will be made available um to the symposium and i'm also interested in um working up this style of uh communication and group practice and translating it into the language of active inference for other people to use so thank you for joining you're welcome thank you thank you

so

all right next pre-recorded video is going to be sanjeev namjoji fundamentals of active inference a self-guided textbook for learning and applying active inference from first principles this is going to be a 20 minute video hello my name is sanjeev namjoshi and i'm a machine learning engineer working at versus if you've ever tried to learn active inference you probably know what it feels like to get lost in a maze of sub-disciplines neuroscience information theory bayesian statistics physics and more every paper seems to assume you'll already know three other fields with different mathematical notations equations seem to materialize out of nowhere with few good resources explaining where they came from or how they're even useful in the first place today i would like to talk to you about my progress toward providing greater accessibility visibility and knowledge of active inference and the free energy principle i'm excited to announce that my textbook fundamentals of active inference is complete and will be released in march 2026 from the mit press my aim for this textbook is to provide a unified systematic and rigorous foundation for students and researchers entering the field to reduce the steep learning curve however an important motivating question is why should you learn active inference in 2025 to answer this question i think it is helpful to turn the clock back 2006 and draw a comparison between the state of deep learning in its early days and active inference as we understand it today prior to the mid-2000s neural networks were just one of many possible models rather than the dominant choice in academia and industry this all changed in 2006 when hinton and his colleagues released the deep belief network paper which is generally understood as the start of deep learning as we understand it today after some various hardware innovations and the release of the well-known image net library we started to see more coverage and success of ai in the news but in 2012 deep learning truly provided its value with alex net and for the first time it achieved better than human performance on image detection tasks but this was just a start what followed was a proliferation of deep learning across industry and research i've added some well-known milestones just to highlight the explosion of progress in deep learning in the last decade but there is so much more that deserves mention and attention here and since i presented about my textbook at the 2023 active inference symposium the state of ai has dramatically changed once again today large language models dominate the ai landscape and have managed to overshadow many other fields and alternative approaches new and more powerful language models are

released

every quarter though the rate of advancement over previous models is beginning to slow so what about the state of active inference as a field from my perspective active inference lies in the same position as deep learning did in 2006 influential and on the brink of exploding in popularity and relevance this paper here from which this figure comes from is from contributors at the active inference institute and shows the current growth of publications in the active inference community in the last few years active inference as a field has seen a number of important milestones here i show just a few that broaden the scope and attention to the field we have the first international workshop on active inference in 2020 we have the first applied active inference symposium and the foundation of the active inference institute we have the release of the par pazulo and friston 2022 book as well as the pi mdp library and from 2023 to present there are so many pieces and changes here some of which include new julia packages rx infer and many many more papers in robotics and other industries as the international workshop of active inference has begun to grow and grow and grow i see active inference as perfectly poised for greater visibility and attention especially given the current interest in generative ai and deep reinforcement learning working alongside the active inference institute and other organizations my aim here is to provide some of the fundamental materials to capture the attention and notice of machine learning researchers and students to bridge this gap to bring active inference into its renaissance so why learn active inference in 2025 well active inference is highly interpretable the very nature of bayesian networks means that you can follow the chain of reasoning through a particular problem to see why a certain outcome was chosen over others for a particular model why a particular policy was chosen for example active inference comes equipped with a principled modeling system a way to model any system of interest using an elegant and all-encompassing compensating formalism based in information theory active inference also has a more robust interpretation of agency than current llms as we talk about agency in a lot of industry context today and includes an aspect of control of the external environment active inference is fundamentally compatible with the notion of world models that capture the cause-effect understanding of the external world and it's believed these world models are very important for preventing hallucinations that we see in large language models active inference can also be used to solve everyday practical problems in research and in industry and it may also expand the field of complex systems modeling which is central to fields such as physics climate science economics political theory psychology sociology and more active inference also provides a convenient way to univize unify many disparate fields that are exploring the same ideas in different languages or scientific conventions and although it's been yet to be fully demonstrated active inference does have the potential to be more cost and sample efficient at scale and more research needs to go into this area which is why we need a new generation of active inference researchers coming to contribute to the field due to the current state of active inference and its potential as a premier ai research framework my aim is to expand the reach of active inference beyond its current confines to this end i've aimed to

make my textbook as accessible for the widest audience possible with the least amount of required prerequisites the major focus is writing this book for an engineering audience

both machine learning engineers and traditional engineering disciplines neuroscience is largely out of scope for this book and there are many other resources available if you're interested in that particular aspect of the free energy principle and active inference many of the recommended and even optional prerequisites that you see here are typically known by undergraduate students in science and engineering and certainly by graduate students in these fields nevertheless the book is also written to be readable for those that desire to focus on everything the completionist that wants to know all the math the concepts and the code it's also suited for people who are more interested in the theory maybe just interested in the math and concepts to get a general sense of things so they can dive into the literature further and my hope is that ultimately those who are intuitionists who want to skim over the math and just get a sense of the ideas should still be able to read this book and ultimately i hope to convince you by reading this textbook that active inference may be challenging but it is not impossible to learn by any means part one of the book called fundamentals will set the stage by describing the statistical setting with an agent interacting with its environment we start as simple as possible with a single state and a single data point and then we begin to expand from there connections are made in chapter three to other classical parameter estimation techniques in statistics as well as a complete investigation into the basics of variational bayesian inference which happens in chapter four where variational free energy is also introduced the field of predictive coding is also explored in part one which we can see as a kind of precursor or one of the precursors to the modern formulation of active inference and this will also allow us to introduce the notion of prediction errors which are ubiquitous in the active inference and also the neuroscience literature part two the active inference core now introduces the core active inference methodologies chapters six seven and eight focusing on continuous state spaces

and chapters nine and ten focusing on discrete state spaces like part one we start simple in the simplest possible model we can imagine and then begin to grow and build up the model to the full vanilla active inference agent step by step

part three applications and extensions

the applications cover various developments that have occurred in the last eight to seven eight to ten years in the field which include modifications and extensions to base active inference and also a number of other topics included here like structure learning factor graphs robotic applications different types of free energies hybrid discrete and continuous models the free energy principle and so on there are also extensive appendices covering the history of the field and its connection to other fields a deep overview of all notation in the book a exhaustive summary of the different forms of variational free energy you might find in the literature and a least you need to know for all background mathematical topics in active inference you can see here that there's a lot of coverage that gives you the basic details so even if you're not an expert in all of these different background topics you have enough of a sense of how to actually understand why they're relevant in the context of active inference and

understand when they show up in various papers of interest

so now i'd like to give you a preview of what's contained in the textbook itself

typesetting and style is extremely important to successful learning so i've spent a lot of time attempting to make my work clear and readable margins are used for footnotes and supplemental figures that can aid reading comprehension there's a large focus on building intuitive understanding of concepts for example all algorithms are explained from scratch and in this book we start from a description of the agent environment modeling problem describe the univariate case extend to the multivariate case then we introduce variational inference add dynamics add generalized coordinates if applicable we look at hierarchical models and we also look at action building on it step by step and then looking also at learning and attention as well to see how they fit into the broader framework figures are designed to clearly walk the reader through the text and give detailed visualizations of important concepts there are also concept maps provided for most chapters which lay out the fundamental vocabulary that will be covered in the chapter and show how they're all interrelated to one another and this should hopefully form a useful guide as the different sections are also numbered and shown in the concept map so you can understand where that topic comes from and how it relates to something else that you may have read earlier in the text

the early part of the book focuses on basic concepts such as hidden state estimation that is estimating the conditional distribution of a latent variable given observed data and the aim here is to explain the modeling paradigm in the context of an agent attempting to infer the states of the environment this is an interaction between a generative model and a generative process and this is a perspective that differs from the way that bayesian inference is normally taught in chapter three and also in the code notebooks that i'll talk about in a moment we go through this perspective taking something as simple as linear regression and putting it into the context of an agent environment interaction to really make those concepts clear and understandable

typically introductory bayesian inference books focus on parameter estimation or what we call learning in this book and in part one i introduced the expectation maximization algorithm to make it clear what the relationship is between classical parameter learning and estimation techniques that you might see in a bayesian

context in certain textbooks and this notion of hidden state inference we deal with latent variable models to understand how they connect together within one algorithm which sets the stage for the active inference models that come later

variational inference is also explained in detail the book features a catalog of all the different forms of variational free energy

or expected free energy in the literature and how they can be derived from one another and how the different forms interrelate

the book also covers rao and bollard style predictive coding terms and ideas such as prediction error minimization

as well as a clear explanation of fundamental concepts such as surprisal and variational free energy

i'm also making extensive use of bayesian networks in this book so that all models can be seen graphically

and we also build up starting simple to more and more complex models throughout the book

factor graphs which are an alternative way of representing bayesian networks that have a number of other interesting and very useful properties

are explained in chapter 12 where i go through all the different models we've seen so far in the book and show how they can be represented as a factor graph to build that intuition and understanding of how these kinds of models work

are more than 230 custom figures that have been created for the book most of which are full color and figures are extremely detailed to give the reader a deep understanding of the content that i provide in the book and many of the models presented in the book in the book are also shown in pseudocode

which should aid in implementation my aim for readers is to go through these algorithms and be able to implement them in code yourselves although code will also be provided for those who also want to learn through reading the notebooks and running pre-existing code finally each chapter is filled with numerous examples and simulations these examples correspond to the jupyter notebooks that i will be talking about in a moment and try to show the application of a particular concept in a simulated environment

so now let's shift our attention to some of the supplemental materials that are going to be provided a few months after publication jupyter notebooks will be released on github and be fully reproducible using docker or other version version handling tools the notebooks will recapitulate every single figure that's shown in the book all simulations in the book and all background material with explanations for all this all these different items and it will also come with a library of modular functions that can be used to build custom active inference agents and environments fit for learning purposes these modular pieces can essentially be mixed and matched inside different environments or agents and that's how these jupyter notebooks often proceed they'll explain the basic concept first and then use those components or tools to build up an agent that does a particular task that corresponds to whatever's being presented in that particular chapter

there's a direct correspondence between the equations that you see in the text and the notebook jupyter notebook code as well so here's an example showing variational free energy in one particular simplified form that you can see here and i've tried to match the kinds of style of variables on the left here showing you how the matrix multiplication and addition and these different components fit together so as close as i possibly can i've tried to match them so it's very easy to see when you read through the notebooks exactly what that equation is telling you

and this is just another example here showing that there are detailed walkthroughs of different kinds of things here i'm showing you colored noise and how you can smooth out a noisy signal by passing over a gaussian filter and get smooth or colored noise which is very useful and active inference simulations and is present in a lot of raw data that sensors might pick up

there are also going to be full video presentations for each chapter this is going to be a project that i'll be working on in the late future after the book is released but this the idea would be to

eventually have an active inference course and you can see some of the details here of some slides that i've created along the route to this of this goal a good majority of proofs had to be cut from the book due to space concerns and i will collect all of those as well as other proofs in a document on archive that will contain all proofs for every equation in the active inference literature so the aim is if you ever see an equation you don't know where it comes from you should be able to look it up in this guide and be

able to find how it was derived and what assumptions were made in that derivation so finally let's talk about a few of the planned future resources i'd like to work on to further support the educational tooling around the book there are two potential side projects i'm interested in working on collaborators the first is providing a software suite for continuous state space active inference and abasion mechanics simulation suite these will both be very useful additions to the field and allow people from other disciplines to begin experimenting with active inference code and fill the gap right now in the software ecosystem i'm also very much interested in interactive learning the aim would be to have concepts explained in the textbook with various simulations interspersed in the forms of plots plot demos and other visualizations maybe sliders that you can move around you can tweak initial conditions and parameters and see how the resulting plots change and this would support everything from invasion mechanics to continuous and discrete space state space active inference as you can see a lot of this work will be very involved and take a lot of time so i'm very open to any volunteers that would like to help me with these projects so please contact me i have my contact information shown at the end if you're interested in contributing to these projects as i continue to expand this ecosystem i'm also working on a second book also to be published by mit press called fundamentals of bayesian mechanics if you're not familiar with bayesian mechanics it's an entirely separate field related to active inference in the free energy principle that talks about the separation between different kinds of stochastic processes separated by what's called a markov blanket rendering the two systems conditionally independent and this forms the background formalism and structure around which active inference models can be derived so it's kind of looking at the bigger picture looking at more thermodynamics and living systems and the mathematics of different kinds of interacting stochastic processes this book which is at the intersection of information theory and physics is currently under a work in progress and will be released probably sometime in late 2027 and i'll be giving more updates on that book as i proceed so just as another reminder again my textbook will be released in march next year and i'm very excited and looking forward to educating and collaborating with a whole new generation of active inference researchers

i would like to thank the active inference institute for hosting presentation and code during the time when the book was being written i got a lot of very useful feedback from everybody that joined in those sessions and i really appreciate all of your hard work you've all been acknowledged uh in my um book's preface as well for all that for all the contributions there and i'd also like to thank my my employer versus for their continued support of my book as well as the active inference community in general with the international workshops on active inference

and this is my contact information please feel free to contact me at any time and email is the easiest way but i'm also sometimes on the active inference institute discord as well if you'd like to see me on there thank you

all right that's the end of this stream next stream tomorrow part two bye

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